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На тему: Исследование возможности применения функциональных композитов с
модифицированным волокном в конструкции планера самолета

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_____ 20__ г.

Москва 2024 г.



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ABSTRACT

The need of weight reduction and fuel efficiency in the aviation industry calls for substituting traditional materials with ones that are lightweight and stronger. With newer advancements in material sciences, composite materials have been extrapolated to functional composites. One such functional composite that has garnered much liking in the aircraft industry is carbon fiber reinforced composite. CFRP's use in aircraft structures has been steadily increasing due to its high strength, stiffness and design flexibility. However, challenges remain in optimizing these properties for specific aircraft structural applications. Property optimization has been proven to be achieved by a fairly new method called fiber modification. This thesis investigates the possibility of using one such fiber modification method in the construction of an airframe component.

An IM7/8552 carbon fiber reinforced composite has been functionalized by single walled carbon nanotubes. The carbon nanotubes have been embedded in the interfacial region between the carbon fiber and epoxy matrix. The overall properties of a composite polymer depend upon the interface, thus, to improve the strength properties of the composite the CNTs have been embedded in this very region. The resulting composite was evaluated for its engineering constants and parameters like deformation in wing and stresses in the wing for a given loading.

A comparative study has been performed so as to better get a picture of the strength of such a functionalized composite. Finite element analysis was performed on micro and mesoscale to estimate the overall properties of the CNT functionalized CFRP. A unit cell representative volume element was developed to model the behaviour of the CNT functionalized interphase. Christensen's generalized self-consistent mathematical model was then employed to estimate the properties of the interphase. These results were then used to perform a laminate-level analysis to predict the overall elastic properties of the composite. The predicted properties of the composite were then used to model the wingbox of an A320 compare the induced stresses and deformations to that of a wingbox made of conventional CFRP.

It was noted that, the high strength CNTs improved the interphase's strength exponentially which led to less plastic strain in the fiber and the matrix. However, these extraordinary properties did not translate linearly to the overall properties of the composite. The engineering constants of the CNT functionalized composite were comparable to the conventional composite's. Thus, the values of deformation and maximum principal stresses of the wings made of the two materials were similar. Though, the overall result of the CNT functionalized CFRP was slightly better.

Further testing and simulation-based experimentation is required to warrant the use of such functionalized composites in airframe manufacturing. Fraction percentage of CNTs should be modified, even different fiber whiskers should be experimented with.

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1. INTRODUCTION

The aerospace industry's pursuit to optimize aircraft performance stimulates the development of high-performance structural materials. Composite materials are one of those categories of materials that play a significant role in present and future aerospace components. The implementation of composite materials as a major structural material was followed from the discovery of carbon fiber at the Royal Aircraft Establishment at Farnborough, UK, in 1964 [1]. Improved fiber and matrix materials (thermosets and thermoplastics) have led to enhanced mechanical characteristics in carbon fiber reinforced plastics (CFRP) composites, replacing conventional materials such as aluminium and titanium alloys in primary structures. Composite materials, also, provide the benefit of being able to develop complicated forms, as opposed to metallic materials. This decreases the amount of pieces in a component, as well as the necessity for fasteners and joints. Development of increasingly advanced composite presently continues and has brought an exciting era of lightweight commercial and military aircrafts. Some important examples of commercial aircrafts that are predominantly composed of composite materials and modern airlines are eager to acquire are Boeing 787, Airbus A350XWB and Airbus A220. Many defense forces in the current world also boast of aircrafts made extensively of composite materials, for example Lockheed Martin F-35 Lightning, Eurofighter Typhoon, Dassault Rafale and Saab JAS 39 Gripen. Composite materials are suitable for aerospace applications owing to their high specific elastic modulus and high specific strength and the production process results in a lighter product, the lower weight also leads to reduced fuel usage and emissions. For instance, by replacing aluminium alloy with CFRP, the weight can be reduced by 12%, this weight saving can be used to reduce fuel consumption or increase payload or speed. Additional benefits of CFRP include superior fatigue and corrosion resistance of composites and their vibration damping properties [2].

Functional composite materials can be defined as a type of material system containing two or more different phases in distinctive compositions and form on a macroscopic level including a continuous phase called the matrix such as polymer, and another phase of the functional agent in dispersed states such as fibers, whiskers or particles, both of which are called composite constituent materials [3]. Some advantages of functional composites are as follows:

- a. Designable functionalities and properties, they can be designed for different functional behaviors and degrees by selecting various types of constituents and compositions.
- b. Developing cycle for a functional composite is much shorter than for a composite structure for aerospace structures which usually takes up to 15 years.
- c. The performance to price ratio per unit mass material is higher showcasing a good prospect in a range of applications.
- d. Functional composites offer unique characteristics in certain applications that other materials cannot provide.

Functional composites, thus, play a vital role in aerospace structures due to their aforementioned advantages, making them an ideal choice for the development of advanced and efficient structures.

Advanced composites materials exhibit desirable strength and stiffness but they have drawbacks over traditional alloy materials due to their layered construction in which high aspect ratio fibers are held together by a weak matrix material. Structural composites have limited utility in aircrafts due to their low impact delamination, currently, most composites rely on thermosetting polymer matrices, which are brittle and susceptible to crack initiation and propagation. Engineers

sometimes overdesign structures for certain stress cases, such as impact and fatigue resulting in underutilized weight reductions from composite materials. Composite structures in aircrafts are more vulnerable to lightning damage as compared to metallic structures, as the composites are either not electrically conductive or much less conductive than metals. Composite materials in the aircraft exterior allow significant portions of the lightning current to flow into onboard systems and provide less shielding of onboard electronic systems from lightning electromagnetic fields [4].

Thus, to overcome these shortcomings of composite materials, fiber modification is introduced. Fiber modification in a composite polymer refers to the process of altering the surface properties or structure of the reinforcing fibers to enhance bonding and overall performance of the fiber which can give a composite polymer with improved mechanical, thermal and electrical properties. Fiber modification can be done by various techniques to induce chemical and physical changes on the fiber surface, namely plasma, chemical, nano-particle growth, coating and microwave irradiation [5]. Ansari [5] concludes that carbon nanotube (CNT) growth on the surface of a carbon fiber enhanced the wettability of the carbon fiber by increasing the surface area of the fiber, it helped in uniform distribution of the load from the fiber to the matrix. Moreover, the CNTs decreased the probability of stress concentration around the weakest fiber in the resulting composite.

When developing the safest and most fuel-efficient aircraft, structural materials with great strength and low weight have long been highly desired. Since single-walled carbon nanotubes have specific resistance in the order of 48.5×10^3 kN m/kg [6]; ultimate inherent tensile strengths of more than 100 GPa and an exceptional theoretical strength per weight, they have been tipped as potentially revolutionary structural materials. CNTs tend to be less dense than metals, and often provide improved strength and corrosion protection. The nanotubes exhibit about eighty times as much strength, six times as much Young's modulus, and one-sixth the density of high carbon steel. These properties of CNTs are desirable in airframe construction. It is of the utmost necessity that the airframe structure be lightweight, have high strength and showcase rigidity but at the same time, bend enough so as to absorb the disturbance energy and not break.

The interface is the area where the fiber and the matrix coincide. To have a successful, applicable composite, the interface must be strong and favorable towards maximum compatibility. A good interface is imperative for a material to endure stresses. The interface also plays a crucial role in determining the overall properties and performance of the composite. The interaction between fiber and matrix through, through the interface, helps distribute stresses efficiently, enhancing the overall mechanical performance of the composite. The anisotropic behaviour of composite polymer is influenced by the orientation and distribution of reinforcement fibers within matrix. This interface allows for design flexibility, enabling the customization of properties by selecting appropriate matrix, fiber and functionalization of the interface.

The addition of CNTs can improve the interfacial properties between the fiber and matrix, leading to enhanced mechanical, electrical and dielectric properties of the overall composite. Molecular dynamics simulations have shown that examining the stress-strain behaviour of CNT reinforced composites can help evaluate the elastic properties of the interfacial region. This understanding is critical for improving the manufacturing and performance of nanocomposites [7].

1.1. Goals & Objectives

The primary goal of this thesis is to determine the stress-strain states of the constituents of a composite polymer with embedded whiskered fibers and to determine the load-carrying capacity and failure modes of such a composite material when employed in an airframe structure. This work also summarizes an efficient mathematical methodology to evaluate the strength of a modified composite with whiskers and provides an analytical overview on the influence of dimensional and mechanical parameters of said whiskers.

1.2. Thesis Outline

This thesis work consists of 8 chapters and is organized as mentioned below.

Chapter 1: Introduction. The first section encompasses the outlining of the research topic and the scope of this thesis.

Chapter 2: Literature Review. This section gives a detailed overview of the themes pertaining to the topic and the developments made in the scientific community in those fields of nanoscience, material science and aircraft structural design.

Chapter 3: Analytical Formulation. This next section enlists the mathematical formulas required to determine the effective properties of the interphase and the stress-strain relations of materials.

Chapter 4: Computational Micromechanics Modelling. In the fourth section, the dimensional characteristics of the nanotubes used, as well as the dimensions, boundary conditions and loads imposed on a uni-directional RVE and a bi-directional laminate are established.

Chapter 5: Airframe Component Modelling. This section mentions the prototype aircraft chosen for wing box modelling, the construction of the wing box and its meshing for FEA.

Chapter 6: Results Appendix. In this section, the results of all the finite element analysis' is showcased and a comparative study is performed.

Chapter 7: Fabrication Evaluation

Chapter 8: Conclusion

Chapter 9: References

2. LITERATURE REVIEW

2.1. Prior Experience with Modified Fibers

Composite materials have a high strength-to-weight ratio and exceptional fatigue resistance, making them ideal for aeronautical applications. Polymer-based composite systems combine functional fillers with highly processable polymers, resulting in numerous functions and a wider application window. Composite materials for structural applications are often high-performance systems built of synthetic materials with high strength-to-weight ratios. However, optimal performance sometimes requires regulated production settings [8].

2.1.1. Fiber Composites

Fiber-reinforced composites, or simply fiber composites, have become the most important class of composite materials. They are composites consisting of fibers embedded in the matrix structure which combines the individual advantages of fiber and the polymer matrix. Fiber reinforcement contributes high strength and stiffness to the polymer composites while polymer matrix is responsible for light weight, ease of fabrication and corrosion resistance [9].

2.1.1.1. Composition of Fiber Composites

The microstructure of fiber composites materials is composed of the fiber and the matrix, which are fundamental constituents of the composite, and additionally the interphase layer [10]. Polymer matrices can be thermoplastic or thermoset. Thermosets are a family of plastics characterized by the fact that they are formed starting from a liquid solution that irreversibly leads to a solid material during a heating step [11]. While thermoplastics differ in that they can be made ‘plastic’ and malleable at high temperatures and are fully recyclable [12]. The choice of matrix depends on the required properties of the composite and the operating conditions. Fiber composites can be reinforced with multiple kinds of fibers such as glass, carbon, aramid and such. Fiber compositions vary, making it possible to choose the best one for a certain purpose. While interphase is defined as the region where the physical, mechanical and chemical properties differ from those of the original fiber and the matrix as a result of thermal, chemical and mechanical influences. Factors occurring within the interphase region can be listed as variable crosslink density and molecular weight, impurities, sizing, voids etc. [10] See Figure 1.

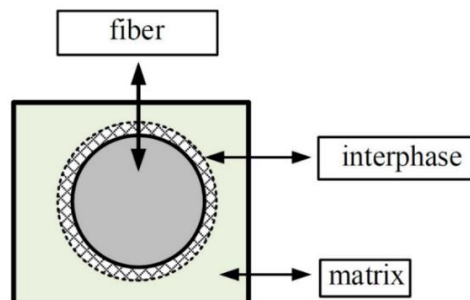


Figure 1: Morphology of fiber reinforced composite interface

2.1.1.2. Properties of Fiber Composites

Fiber reinforced composites offer high strength and elastic modulus that are either comparable to or better than many traditional metallic materials. As a result of their low density, the strength-to-weight ratios and modulus-weight ratios of these composite materials are superior. Fatigue strength and fatigue damage tolerance of many composite laminates are excellent. The properties of fiber-reinforced composite laminates depend on the direction of measurement, the tensile strength and modulus of a unidirectionally oriented fiber-reinforced polymer are maximum when they are measured in the longitudinal direction of fibers. Similar angular dependence is observed for other mechanical and thermal properties, such as impact strength, coefficient of thermal expansion and thermal conductivity [13].

2.1.1.3. Types of Reinforcements in Fiber Composites

The important types of fiber reinforcements are namely – glass fibers, carbon fibers and aramid. Glass fibers are one of the most widely used polymer reinforcements as it offers excellent strength and resistance to thermal, chemical and wear issues, however the machining of glass fiber-reinforced polymers (GFRPs) is slow and challenging [14]. Carbon fibers are high strength materials made of graphitic and non-crystalline regions, they have the highest specific modulus and strength. Carbon fibers also have the ability to retain their tensile strength at high temperatures, they are also chemically inert and electrically conductive.

2.1.2. Fiber Modifications

Fiber modification refers to the changing the properties of fibers in a composite laminate so as to improve the composite laminate's functionality. The modification can be through surface treatment, coating or integrating other materials into the fiber. Fiber modification frequently serves to improve the adhesion of the fibers to the polymer matrix, increase mechanical strength, improve durability. Incorporating whiskered fibers is the most effective technique to improve the composite laminate's desirable properties such as strength, durability, stiffness, and extended service life. Fiber modification with whiskered fibers involves embedding them into the interphase of the composite laminate, this can be achieved through various manufacturing processes as chemical vapor deposition, electrospinning etc. These whiskers improve mechanical strength, stiffness, thermal properties and electrical conductivity of the composite laminate.

2.1.2.1. Whiskered/Fuzzy Fibers

Whiskered or fuzzy fibers refers to fibers (base fibers) coated with fine crystalline structures called whiskers as shown in figures 2 and 3. These whiskers are made of crystalline materials as metal carbides, nitrides or borides. Whiskers improved the adhesion of composite constituents. The whiskers act as reinforcements within the composite, they enhance the composite's flexural strength and stiffness. The interphase layer between the whiskers is very strong and inhibits crack propagation. If the coefficient of thermal expansion of the whiskers and the matrix are comparable the thermal stresses are reduced and minimize the risk of delamination. They also make the

laminate wear resistant, thus, making the composite laminate suitable for industries that require lightweight material with high strength and rigidity.

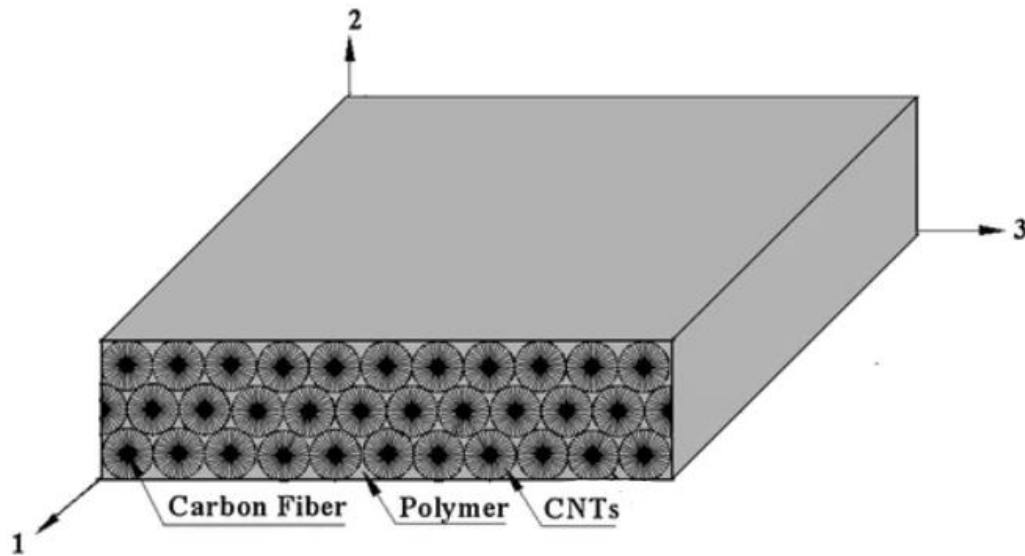


Figure 2: Schematic diagram of a lamina with CNTs (whiskered fibers)

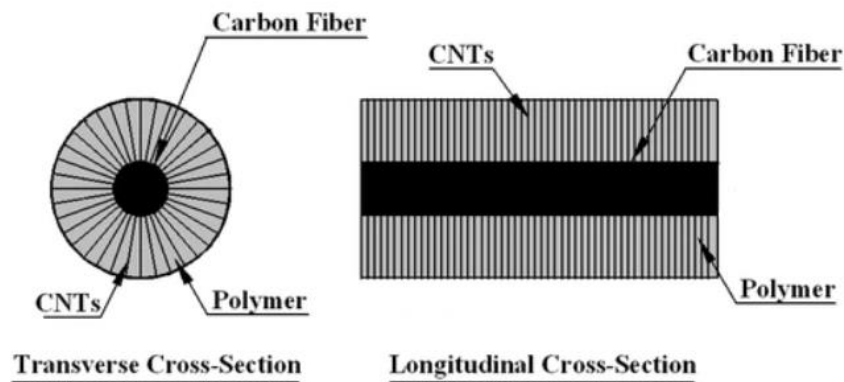


Figure 3: Transverse & Longitudinal cross sections of a whiskered fiber with CNTs

2.1.2.2. Methods for Enhancing the Mechanical Properties of Fiber Composites with Fiber Modification

In order to optimize the use of fiber composites it is important to improve their mechanical properties. Several studies have been conducted in the last few decades, both experiment and simulation based. Bose [15] was one of the first few to research and patent four different ways to incorporate whiskers in an intermetallic matrix to improve its strength and fracture toughness in order to use to composite at high temperatures with minimum creep. In their preferred method, they rolled SiC (Silicon Carbide) fibers and laid on a predetermined number of thin composite sheets to form the composition, this composite layer had a layer of MoSi₂ (Molybdenum Disilicide) reinforced with SiC whiskers followed by a MoSi₂ layer having continuous SiC fibers. The resulting composite in this case, showed improved properties for very high temperature applications.

In another attempt at improving the interfacial shear strength of a composite while investigating the change in tensile strength of the fiber, Li [16] coated ZnO (Zinc Oxide) nanoparticles with SiO₂ (Silicon Dioxide) via a sol-gel method. They formed a hybrid coating with the epoxy-resin

to form a core-shell structure on the fiber surface. The results showed no decrease in the fiber tensile strength and a 38.26% increase in the interfacial shear strength. It was found that there was a chemical bond formed between ZnO and SiO₂. In order to enhance the performance of ballistic composites, as they require high fiber matrix interfacial properties, Nasser [17] introduced ZnO nanoparticles and nanowires interphases on the surface of HNO₃ (Nitric Acid) functionalized carbon fibers to improve its interfacial shear strength under quasi-static loading conditions through a combination of mechanical interlocking and increased surface area. It was found that, under dynamic loading conditions, brittle failure of the ZnO interphases reduces the interfacial shear strength of carbon fiber reinforced composites, allowing for an enhanced ballistic performance. The reduced interfacial shear strength under dynamic loading, was found to be due to the stiffening of the polymer matrix at higher strain rates, limiting the effectiveness of the interphase's functional gradient.

Wang [18] employed three-point bending tests to research the damage behaviour of glass fiber epoxy composites repaired by patch embedding of ZnO whiskers. They performed acoustic emission and digital image correlation to monitor damage behaviors. Matrix cracking and delamination were found to be the main failure modes. It was found and concluded that ZnO whiskers and resin matrix have a good bonding property and failure load and matrix strength of the repaired specimens increases significantly.

There have been multiple such researches into carbon nanotubes being used as reinforcement whiskered fibers. One such research was shown by Zhou [19] carried out Short Beam Shearing tests with SEM observations of the damage evolution in a carbon fiber reinforced polymer composite with carbon nanotube (CNT) secondary reinforcement to assess its strength and fracture behavior. Different CNT distributions were developed to study the effect of secondary CNT reinforcement on a 3D multiscale computational model. It was reported that adding secondary CNT nano-reinforcement into the matrix and/or the sizing of carbon fiber reinforced composite ensures strong increase of the composite strength. The effect of the CNTs' reinforcements was strongest when small additions of CNTs in the polymer matrix were complemented by the fiber sizing with high content of CNTs.

2.1.2.3. Advantages and Disadvantages of Fiber Modification

While fiber modification of composite polymers offers numerous advantages, which comprise of high strength-to-weight ratio, fatigue resistance, corrosion resistance enhanced electrical properties and the ability to modify properties of materials for particular applications. Challenges include high cost, quality assurance, vulnerability to impact damage and interfacial compatibility issues. Balancing these benefits and drawbacks is critical for the effective integration of composite materials, ensuring that both performance and safety standards are satisfied.

- **Advantages of Fiber Modification**

Alves [20] showed that carbon fiber-reinforced composites display high strength-to-weight ratios suitable for certain structural applications where reducing weight while maintaining structural integrity is vital, such as aerospace operations. Turaka [21] added multi-walled carbon nanotubes and graphene nanoparticles to glass fiber-reinforced polymer to improve its mechanical performance, results showed in compressive strength and compressive modulus. Flexural strength and interlaminar shear strength increased by 62% and 298% compared to neat glass/epoxy resin

composite. Cheng [22] showed that the inclusion of carbon nanotubes in polymer composites led to substantial improvements in mechanical properties.

Fiber-reinforced polymer composites provide superior fatigue resistance compared to metallic materials and thus are important for aerospace applications where cyclic loading is a prominent stress during flight, Shankara [23] shows dependencies of fatigue of FRP composites, the fatigue tests to be considered and the challenges faced. Gaurav [24] compared the fatigue behavior of FRP composites and CNT-embedded FRP composites, it was concluded that addition of CNTs increased the static strength, enhanced the fatigue life as the load bearing capacity of the lamina increased, reduced the cryogenic fatigue failure of the composite.

Shi [25] showcased that the thermal conductivity and the in-pane electrical conductivity of a carbon fiber fabric and graphene reinforced segmented polyurethane composites (CFF-G/SPU), concluding that the composite has great advantages in phase transition and filling the interfacial gap in a short amount of time under the condition of electrical field with the interface temperature difference between two layers significantly reduced. Kareem [26] investigated the effectiveness of using hybrid filler in a polymer matrix and found that the interface bonding between the matrix and fiber got better and there was also a sharp increase in the dielectric constant and dissipation factor.

- **Disadvantages of Fiber Modification**

The unique manufacturing techniques required to make FRP composites can be substantially more costly than typical metallic materials, creating a problem for economical projects. Nunna [27] states that the high cost of carbon fiber composites has made the transition of various industries from metals to composites very challenging, they provide a model that presents the cost contributors by considering the entire process chain of manufacturing where it is revealed that the cost of the carbon fibers and the polyacrylonitrile precursor is dependent on the textile operating width and has a final contribution of 76.6% in the overall cost.

Composite materials require more complicated mechanical characterization than metal structures, they also require more complex repair methods. Composites may not necessarily outperform other materials in terms of corrosion resistance, cost, formability, joinability, strength, and toughness [28]. Due to the small size of nanotubes, they are usually curled and twisted, and therefore individual nanotubes embedded in a polymer only exhibit a fraction of their potential, their properties cannot be translated into high strength and stiffness finished products [29]. Spitalsky [29] shows multiple techniques of CNT chemical modification, grafting of macromolecules onto nanotube surface.

The general challenges fiber modification poses is achieving uniform dispersion and alignment of nanoparticles, if not carried out properly agglomeration of nanoparticles might occur which leads to localized areas of high stiffness and compromised mechanical properties. Matching fiber and polymer matrix properties is another major issue; it is important to ensure strong interfacial bonding between polymer matrix, fiber and embedded or functionalized particles in order to maintain effective stress transfer and adhesion between phases. There is a very restricted scope of fiber modification as some methods might only be effective for specific types of polymers limiting its versatility. There are also environmental concerns associated with the production of polymer composites and their modification, pertaining to the chemicals, certain hazardous substances and the industries involved in the manufacturing of the products.

2.1.3. Carbon Nanotubes (CNTs)

In 1991, Sumio Iijima [30] produced a new type of finite carbon structure consisting of needle-like tubes using arc-discharge evaporation method. He called these needles, helical microtubules of graphitic carbon and found that the needles grew at the negative end of the electrode. Through electron microscopy he found that each needle comprises of coaxial tubes of graphitic sheets, ranging in number from 2 up to 50. On each tube carbon atoms hexagons are arranged in a helical fashion about the needle axis and the helical pitch varies and aids the growth progress. Thus, we had the first known structures of carbon nanotubes in their double walled and multi walled forms. In 1993, Bethune [31] furthered this procedure and research and through coevaporizing carbon and cobalt in an arc-generator and formed single layered carbon nanotubes with a very small diameter of $1.2 \text{ nm} \pm 0.1 \text{ nm}$ comparable in size to fullerenes with 60-100 carbons, these were introduced as single walled carbon nanotubes.

2.1.3.1. Geometry of CNTs

The hexagon pattern is a major feature of a CNT, the pattern repeats itself periodically in space, because of this periodicity, each atom is bonded to three neighboring atoms. This structure is due to sp^2 hybridization in which one s-orbital and two p-orbitals combine to form three hybrid sp^2 - orbitals at 120° to each other within a plane [32]. The covalent bond (σ – bond) is a strong bond and plays a very important role in the mechanical properties of CNTs. Also, the π -bond which is relatively weak contributes to the interaction between the layers in MWCNTs and between SWCNTs in bundles. The curving of the graphene sheet re-hybridizes the σ and π orbitals, giving an admixture.

The geometry of CNTs is determined by the chiral vector of the hexagonal lattice [33]. The state of twisting of SWCNT can be defined by “n” and “m” chirality indices, these show the vector direction for rolling the hexagonal lattice. According to their chirality SWCNTs can be: armchair, zigzag or chiral. The chirality has a direct effect on the CNTs’ electrical properties. Saito [34] calls chirality a result of a defect in the cap region of a fullerene ball in order for it to grow into a cylindrical tubule. Confinement in the radial direction of the graphite tubules is provided by the monolayer thickness of the fiber. In the circumferential direction, periodic boundary conditions apply to the enlarged unit cell that is formed in real space and the subsequent zone folding that occurs in the reciprocal space.

2.1.3.2. Single Walled Carbon Nanotubes (SWCNTs)

Single walled carbon nanotubes are very versatile due to their promising electrical and mechanical properties. The material properties of small diameter nanotubes are predicted to exhibit quantum size effects based on their chirality and diameter [35]. The electronic and magnetic properties were found to be diameter dependent [36]. For some chirality, the stiffness of SWCNT decreases with decreasing diameter; but for a set diameter, the stiffness increases with increasing chiral angle.

Lourie [35] showed that diameter-selective resonance Raman scattering can be employed to find the elastic properties of armchair SWCNTs with different diameters. They reported a Young’s modulus of $\sim 2.8\text{-}3.6 \text{ TPa}$ for SWCNT using Raman spectroscopy. The mechanical response of SWCNT ropes under tensile load was measured [37] based on which the Young’s modulus range was determined to $320\text{-}1470 \text{ GPa}$. Molecular dynamics simulations were used by Lu [38] which

resulted in a Young's modulus of ~ 1 TPa and a shear modulus of ~ 0.5 TPa, he also stated that chirality, radius and number of walls have little effect on the value of Young's modulus. To avoid the difficulties with direct tensile loading, Wagner [39] transferred load to the CNT through matrix material, they were able to fragment embedded SWCNT and a tensile strength of ~ 55 GPa was reported. Walters [40] induced large elastic strains in ropes of SWCNTs using an atomic force microscope in lateral force mode, the freely suspended ropes were observed to deform as elastic strings with tension proportional to elongation. The maximum strain observed $\sim 5.8 \pm 0.9\%$ gave a lower bound of $\sim 45 \pm 7$ GPa of tensile strength (yield stress).

2.1.3.3. Dynamic Properties of CNTs

CNTs have remarkable capabilities, but are mostly found at the nanoscale. To use CNTs in real-world applications, we need to understand their characteristics on a macroscopic scale. Research is underway to develop standardized ways for using CNT characteristics at the macroscopic level. Previous research has used several techniques to incorporate CNTs into polymers, resulting in nanocomposites that have nanotube-like characteristics.

The conductivity, strength, elasticity, toughness and durability of formed composites may all be substantially improved by the addition of nanotubes forming CNT reinforced polymer composites [41]. The effective utilization of carbon nanotubes in composite applications depends strongly on the ability to homogeneously disperse them throughout the matrix without destroying their integrity. Furthermore, good interfacial bonding is required to achieve load transfer across the CNT-matrix interface, a necessary condition for improving the mechanical properties of the composite [42]. Bal [41] states that the load transfer from matrix to CNTs plays a key role in the mechanical properties of composite polymers and the most ideal mechanism of load transfer involving CNTs would be chemical bonding between nanotubes and the matrix, this will improve the interfacial interaction through ionic or covalent bond that enables a stress transfer.

In 1998 Schadler [43] stated that load transfer depends on the interfacial shear stress between matrix and fiber and a high interfacial shear stress will transfer the applied load to the fiber over a short distance and a low interfacial stress will require a long distance. Schadler then proceeded to disperse MWCNTs ultrasonically in Shell Epon 828 epoxy resin focusing on low loading ($\sim 5\%$ wt.) of CNTs for good dispersion in the matrix and used Raman Spectroscopy to measure the strain in the fibers. Measuring the Raman peak shift for tension and compression, it was found that the shift for compression was higher than tension indicating that the nanotubes carry more strain in compression than in tension. Two explanations were given for such behaviour (a) poor bonding between the matrix and outermost nanotube layer; (b) outermost nanotube layer is loaded but the load is not transferred to the inner layers due to weak bonding between nanotube layers. Lourie in 1998 [44] found that the compressive strength of thin- and thick- walled CNTs is more than two orders of magnitude higher than the compressive strength of any known material. Polyimide/carbon nanotube (PI/CNT) nanocomposites with different proportions of CNT were fabricated by in situ process [45]. The bending strength and microhardness of the PI/CNT nanocomposites were measured. The bending strength and microhardness of the PI/CNT nanocomposites were found to have increased with increasing CNT content and reached stable values at a certain content of CNT. CNT could effectively enhance the friction-reduction and anti-wear capacity of the nanocomposite because it increased the load capacity and mechanical strength of the CNT/PI. The variables such as applied load and sliding speed had a significant influence on friction and wear performance. Siochia in 2004 [46], found that the tensile modulus and yield

strength increased with the addition of SWNT loading in a polyamide SWNT composite. The increase was much higher than that observed for film samples, but much less than what was expected from an oriented discontinuous fibre reinforced polymer composite.

2.1.3.4. Functionalization of CNTs

The incorporation of chemically functionalized CNTs into polymers imparts enhancements in mechanical performance as it allows for chemical covalent bonding between CNTs and the polymer matrix. It has been established that the inherent brittle nature of epoxy can be improved by nanotubes in the matrix, but the toughness and mechanical strength can be further improved by functionalizing the CNTs with liquid rubber which consists of multiple functional groups. Konnola [47] achieved an improvement of ~117% in fracture toughness and ~25% in tensile strength. An increase in the storage modulus and glass transition temperature was also observed. Konnola surface modified MWCNTs with carboxyl-terminated poly (acrylonitrile -co-butadiene) by side-wall functionalism and dispersed homogenously throughout the epoxy resin system. Chen [48] states that functional modification allows increase in interfacial strength. This improvement enhances the load transfer mechanism of SWCNTs with the purpose of increasing the macroscopic and microscopic mechanical performance of composite systems. Jian [49] used molecular dynamics simulations on covalent functionalized CNTs in epoxy nanocomposite, it was revealed that the functionalization enhanced the interfacial shear stress and the Young's modulus of the CNT-epoxy nanocomposites. The functional groups also acted as sites of mechanical interlocking during deformation. Du [50] modified MWCNTs with PSS via a non-covalent functionalization technique, through electrical conductivity the electrical percolation threshold was found to be 0.3wt%, the mechanical properties of poly(ϵ -caprolactone) were significantly reinforced and an improvement in yielding strength, Young's modulus and dynamic storage modulus were observed. In another case, Mandal [51] fabricated nanocomposites using polythiophene (PTh)-wrapped MWCNTs via π - π stacking interactions. After the addition of 0.05 wt.% of functionalized MWCNTs, the Young's modulus, tensile strength, and fracture toughness of the poly (vinylidene fluoride) (PVDF) composites were two-to-three times higher than those of unfunctionalized MWCNTs.

2.1.3.5. Applications of CNTs

Aircraft design prefers materials with high specific strengths to minimize aircraft bulk while retaining structural integrity. Many modern aircraft constructions benefit from the particular strength advantages of aluminum alloys, such as 2024-T3 for the fuselage and graphite-epoxy composites for the empennage and control surfaces. A novel composite reinforced with CNTs may give aircraft designers with an additional structural material alternative for airframes.

O'Donnell [52] considered a notional 747-400 with carbon nanotube reinforced polymer as the primary structural material without including any modifications to the geometry or design of the airframe. The CNRP material made with 60-7% SWCNT showed some improvements, cruising at 39,000 ft consumed fuel 12-24% more efficiently. It was also observed that there was 5% reduction in vortex root circulation and the average mass savings of CNRP structure was about 10.07%. A similar result was found by O'Donnell in [53] where a significant airframe weight reduction was observed in four aircrafts. Owing to this weight reduction the performance, economy and efficiency of the aircrafts were seen to have improved. In yet another research, O'Donnell [54] reports notable reduction in mass at takeoff when CNRP was the primary structural material for

747-400, 757-200, Airbus A320 and Embraer E145. Each airframe modeled saw an average 17.32% weight reduction in the low initial takeoff mass category, with a minimum mass reduction in the high initial takeoff mass category of over 10%. Consequential benefits exist due to the mass reduction found in the notional CNRP-structured airframe analysis, including increases in aircraft flight range and cruise altitude, and a decrease in fuel consumption, which directly correlate to more economical flights.

Suresh [55] explores the usage of CNT-polymer composites in the aerospace field in order to enhance the electrical performance of composites with epoxy resin, CNT-polymer composites are essential due to their structural properties that could be applied to sch areas as in anti-radar protectors, anti-static materials etc. Kim [56] conducted tensile tests at 25% from their yield strength, and 4 wt.% CNT-reinforced polymer composites were discovered to break. In general, the 1 wt.% CNT composites doubled the Young's modulus and yield strength compared a pure epoxy laminate, proving that CNTs are emerging advanced materials that allow structure to be lightweight, have elevated temperature resistance and high strength-to-weight ratio.

2.1.3.6. Addition of CNTs in Polymer Composites

Iqbal [57] discusses the advanced applications of CNT/polymer nanocomposites in the aerospace industry. Noticeable improvements in thermal, mechanical and flame-retardant properties of the nanocomposites have been recorded with the addition of different nanofillers and diverse types of polymers. Srinivasan [58] highlights and reviews the development of metal matrix composites in aerospace and military applications, the importance of CNTs in the field of the aviation industry, challenges in aerospace and defense sectors and the importance of various processing techniques adopted in the development of CNT reinforced MMCs.

The incorporation of carbon nanotubes (CNTs) into natural fiber-based polymer composites has revolutionized their structural uses. Outstanding performance has led to a quick rise in publications and inspired efforts to produce technical products. Adding carbon nanotubes to polymer composites enhances their strength, stiffness, toughness, electrical conductivity, and thermal stability. Technological obstacles to overcome include repeatability of CNTs in large-scale applications, homogenous dispersion in composites, and interfacial interactions. The safety, dependability, and durability of these composites are still being investigated.

2.2. Mathematical Model of Three-Phase Composites

When nanoparticles are introduced to a polymer matrix, an interphase zone forms between the particles and the matrix. The composite comprises of three phases: polymer matrix, nanoparticle (filler), and interphase; as shown in figure 4. The interphase region has a direct effect to the overall material properties of the composite [59]. The essence of the three-phase model is that the entire periodic composite structure, with the exception of a single unit cell is replaced by the homogenous equivalent medium with some initially unknown effective characteristics. These unknown characteristics are determined from the relations based on the energy principle, i.e. the energies stored in the original composite medium and the equivalent homogenous medium must be equal [60]. A softer interphase than the surrounding polymer reduces stiffness and strength while increasing fracture resistance. On the other hand, a stiffer interphase than the surrounding polymer would reduce the composite's fracture resistance while increasing its strength and stiffness. The

nature of the interphase may vary depending on the specific composite system [61]. At the macro-scale, the interphase is a common boundary between the reinforcement and the polymer matrix, but at the micro-scale, the interphase is a transition zone where the chemical, physical, and mechanical characteristics of reinforcement and matrix materials change constantly or gradually.

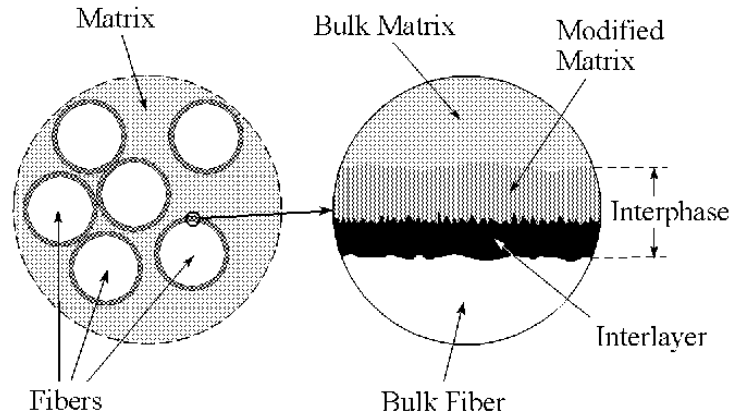


Figure 4: Schematic illustration of a composite interphase [62]

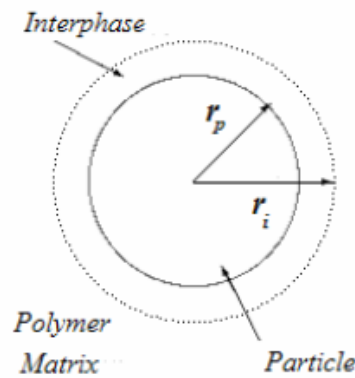


Figure 5: Three-phase composite model

The size of the interphase is very significant as compared to the nanoparticle size and must be taken into consideration. As mentioned previously, the properties of the interphase directly influence the properties of the composite itself. The idealization of a three-phase model of a composite involves replacing the investigated periodic structure with a homogenous homogenized structure with decreased parameters, with the exception of one distinctive cell. The solution involves deriving mathematical formulae for homogenized coefficients based on physical understanding [63].

The use of a three-phase model provides the following advantages:

- Ability to analyze composite materials with diverse layouts, such as fiber-reinforced composites with varying cross-sections, spatially inhomogeneous composites with spherical or cubic inclusions.
- The model's application technique is not affected by the geometric and physical properties of the composite material's inclusions or matrix.

- The model can be applied to a variety of problems, including inhomogeneous media with varying shapes, complex structures with non-periodic inclusion distributions, and variable dimensions.

2.2.1. Thickness and Volume Fraction of the Interphase

Williams [64] estimated the interphase thickness in a carbon fiber/epoxy system to be 500nm through a micro-pullout method. Mai [65] observed an interphase thickness of 1-3μm in a glass fiber/ epoxy composite determined through force modulation atomic force microscopy (AFM). Utilizing phase imaging AFM and nanoindentation Downing [61] found the interphase thickness to be in the range 2.4 – 2.9 μm. The principal method for determining interphase characteristics have been AFM force modulation and different types of nanoindentation.

The radius of the spherical particle is r_p which is added to the matrix, the annulus interphase region has the radius r_i , radius of the matrix is denoted as r_m . An increase in the volume fraction of particles results in an increment in the volume fraction of the interphase. The thickness of interphase, $\Delta r = r_m - r_i$, can also be changed based on the concept of glass transition without varying the volume fractions of particles [66].

The volume fraction of the interphase can be given by [67]:

$$v_i = 3v_p \frac{\Delta r}{r_p}$$

Where, v is the volume fraction and the indices p, i, m stands for particle, interphase and matrix respectively.

Hence, thickness of interphase can be found from:

$$\left(\frac{\Delta r + r_p}{r_p}\right)^2 - 1 = \lambda \frac{v_p}{1 - v_p}$$

Here, the parameter λ ($0 < \lambda < 1$), can be experimentally determined through differential scanning calorimetric measurement of these specific heat jumps.

2.2.2. Interphase Properties

In fiber-reinforced polymers, the interphase connects the fibers with the polymer matrix, allowing load to be transmitted based on strain compatibility. If the interphase fails prematurely, the strain of the matrix is incompatible with the that of the fiber, preventing load transfer from the weak to the strong component [68]. Therefore, the strain compatibility at the interphase is essential to the integrity of the fiber-reinforced composite. Stress concentrations are more likely at the interphase than in the fibers or matrix, leading to easy generation of microcracks at the fiber-matrix interphase. The load carrying capacity of the fiber-reinforced polymer reduces as the microcracks propagate. The fiber-matrix interphase is critical for long term mechanical properties of fiber-reinforced material under aggressive environmental conditions.

Zare [69] gives a method for characterizing the properties of the interphase in polymer nanocomposites reinforced with spherical nanoparticles. They used the Leider-Woodhams model to predict the tensile strength of the interphase using a modified stress concentration interfacial parameter from the Pukanszky model [70]. They also expressed the interphase thickness as a function of “ B ” and interphase tensile strength.

“ B ” as described by Pukanszky is an empirical parameter determined through experimentation. The parameter has been explained by and related to the interphase interactions.

$$B = (1 + 1A_f\rho_f) \ln\left(\frac{\sigma_{Ti}}{\sigma_{T0}}\right)$$

Here, A_f and ρ_f are specific surface area and density of the filler and l and σ_{Ti} are the thickness and strength of the interphase. The above relation for “ B ” shows that both the size of the interphase and strength of the interaction influence the parameter.

Leidner [71] proposed a model to predict the tensile strength of the interphase layer, the model assumes roles of matrix and nanoparticles in the ultimate strength of composites as:

$$\sigma_{uc} = (\sigma_a + 0.83\tau_m)V_b + \sigma_a S(1 - V_b)$$

Where, σ_{uc} – composite tensile strength, σ_a – interphase tensile strength (interfacial adhesion), τ_m – matrix shear strength, V_b – nanoparticle volume fraction, $(1 - V_b)$ – matrix volume fraction and S – stress concentration factor.

Kozlov [72] took the value τ_m equal to the shear yield strength of the polymer matrix τ_Y found from the macroscopic yield strength σ_Y as:

$$\tau_H = \frac{\sigma_Y}{\sqrt{3}}$$

Nanoindentation experiments conducted by Downing [61] on a glass fiber-reinforced epoxy matrix composite indicated that the relatively stiff fiber caused a gradient in modulus across the interphase region, with the modulus next to fiber approached that of the fiber and decreased to that of the bulk polymer as the distance from the fiber increased.

2.2.3. Interphase with Inclusions

The quality and strength of interphases are key parameters affecting the impact toughness of composites. If these regions are mechanically weak, they can limit the material stiffness and divert cracks. The introduction of reinforcing elements such as nanoparticles, nanotubes, nanowires, can effectively improve the material toughness. Studies have been done where CNTs were introduced as bulk reinforcement in the matrix of fiber-based composites, this proved to have improved the bulk strength but it left the fiber-matrix interfaces relatively unaffected. Livanov [73] created a thin layer of CNTs on the surface of ceramic layers which were separated by thin interlayers of polymeric adhesives. They employed an evaporation-driven self-assembly process, precisely controlling the thickness and uniformity of the CNT coating. They measured the hardness of polymer alumina interphase, with and without CNT-n by nanoindentation. They found that the specimens that had nanotubes were almost 100% harder than those with no CNT-n.

3. ANALYTICAL FORMULATION

Composite materials consist of two or more constituent materials bonded together so that the gross properties of the composite are superior to those of the constituents; e.g., the desirable properties – high strength, high stiffness, low weight are increased, while the undesirable properties – low ductility are suppressed [74]. The mechanical compatibility of the constituents of the composite polymers is important if the material is to be used for structural members. The desired gross properties can be achieved by selecting the proper constituent materials and putting them together in a proper geometrical arrangement.

The composite structure can be understood as a two-level system which includes a microstructure formed by the binder and a macrostructure characterizing the composite as a whole [75]. The microstructure is formed by combining the binder with filler and additives; the properties of the material at this level depend on the degree of filling, dispersity and surface activity of the filler, concentration and molecular weight of the binder. In a system united by a binder, there are aggregates, clusters - groups of particles glued together by a binder, and individual non-aggregated along with them - deformations and stresses (shrinkage, humidity, temperature, etc.) moistened with polymer and capsule the process of forming cluster structures introduces additional connections, and particles. The formation of the macrostructure of composite materials in the theoretical view is somewhat different. Here the foreground is the volumetric ratio of fillers and binders, the density of their contact, the difference or similarity of strength and deformation properties, the intensity of interaction between binders and fillers.

As the filler content increases, the matrix gradually transitions from bulk state to film state and eventually the film matrix becomes discontinuous and turns into thin islands on areas of the filler surface. The fillers are noted to enhance the properties to a certain extent, but beyond that, the properties of the composites were found to decrease due to their agglomeration [76]. This is purely due to the strong Vander Waals forces inducing the adhesion of filler particles with one another. This results in a decrease in mechanical properties and poor interfacial adhesion, which is directly governed by the geometry and concentration of the filler.

With increasing characteristics of the surface activity of the filler, flexural strength exhibits a decline compared to compressive strength. Due to the difference in the thermal expansion of filler and matrix materials during the formation of the polymer composite material structure and as a result of further heat treatment residual circumferential tensile stresses arise on the surface of the filler particles. Their growth during polymer composite material compression is limited due to the fact that part of the external load is absorbed by rigid filler particles. Bending is known to be more sensitive to stress concentrations, which is one of the main factors a general decrease in the strength of polymer composite material with increasing degree of filling.

The failure of composites has been investigated extensively from the micromechanical and macromechanical points of view. On the micromechanical scale, failure mechanisms and processes vary widely with type of loading and are intimately related to the properties of the constituent phases, i.e. matrix, reinforcement and interface–interphase [77]. Experimental observations have shown that unidirectional fiber composites exhibit five primary failure stresses, also referred to as limit stresses and simple strengths. These stresses result when the composites are subjected to failure under uniaxial loading in their plane. The failure stresses are identified individually as

follows: (1) longitudinal tensile (S_{11T}), (2) longitudinal compressive (S_{11C}), (3) transverse tensile (S_{22T}), (4) transverse compressive (S_{22C}), and (5) intralaminar (in plane) shear (S_{12S}) [78].

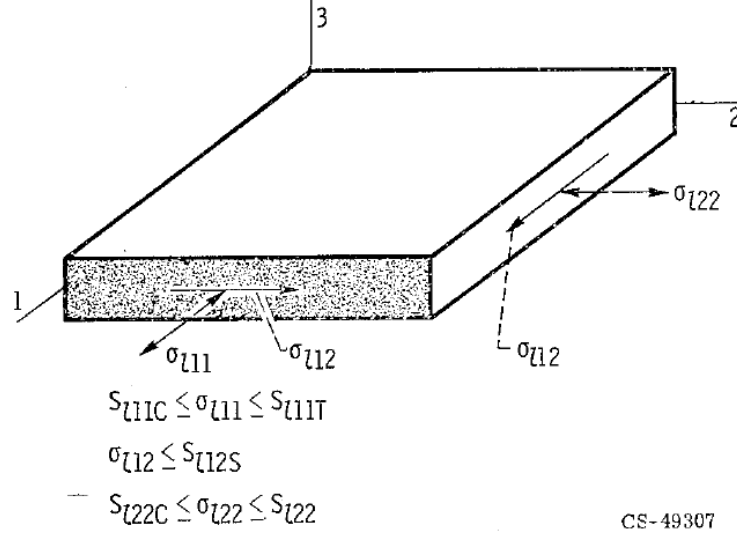


Figure 6: Unidirectional Filamentary Composite (geometry and simple strength definitions)

Experimental observations also show that unidirectional fiber composites under combined loading fail at stresses considerably different from those under simple loading. Therefore, five simple strengths and a combined-stress strength criterion are needed to describe the strength (limit) behavior of a unidirectional filamentary composite.

3.1. Structural Behavior of Composite Materials

A composite material is considered as a heterogeneous anisotropic continuum. Thus, the structural behavior of the composite material is described by the mechanical constitutive equation of the composite. The material coefficients of this equation describe the extent of mechanical response of the composite under the influence of external loads [74]. Materials and structures are two traditional areas of research linked by Hooke's Law. The constitutive equation for anisotropic laminated composites:

$$\begin{bmatrix} N \\ M \end{bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix} \begin{bmatrix} e^0 \\ k \end{bmatrix}$$

From classical plate theory, the variables used are:

$$N_i = \text{stress resultant} = \int_{-\frac{h}{2}}^{\frac{h}{2}} \sigma_i dz$$

$$M_i = \text{stress couple} = \int_{-\frac{h}{2}}^{\frac{h}{2}} \sigma_i z dz$$

$$e_i^0 = \text{in plane strain}$$

$$k_1 = \text{bending curvature}$$

Where the composite matrix is partitioned into four submatrices, so that [79]:

$$[A], [B], [D] = A_{ij}, B_{ij}, D_{ij}$$

$$= \int_{-\frac{h}{2}}^{\frac{h}{2}} (1, z, z^2) C_{ij} dz$$

The inverted constitutive equation would be:

$$N = Ae^0 + B_k$$

$$M = B_e^0 + D_k$$

Hence, $e^0 = A^{-1}N - A^{-1}B_k$, where the total strain is $e_i^0 = e_i + z_{ki}$

The material coefficients A, B, and D are expressed in terms of material and geometric parameters associated with the constituent materials and the method of lamination. This information provides a rational basis for the design of elastic stiffnesses of an anisotropic laminated composite.

Other strength theories of anisotropic materials include, using the Hill Yield Criterion, the quadratic function of the stress components:

$$2f(\sigma_{ij}) = F(\sigma_y - \sigma_z)^2 + G(\sigma_z - \sigma_x)^2 + H(\sigma_x - \sigma_y)^2 + 2L\tau_{yz}^2 + 2M\tau_{zx}^2 + 2N\tau_{xy}^2 = 1$$

Where F, G, H, L, M, N are material coefficients characteristic of the state of anisotropy, and x, y, z are the axes of material symmetry. This yield condition is a reduction of von Mises yield condition when the material coefficients are equal.

“Interaction Formula” in Hill’s notation takes the following form under the assumption of a state of plane stress in thin plates of composite material:

$$\left(\frac{\sigma_x}{X}\right)^2 - \frac{1}{r} \frac{\sigma_x \sigma_y}{XY} + \left(\frac{\sigma_y}{Y}\right)^2 + \left(\frac{\tau_{xy}}{S}\right)^2 = 1$$

Figure 2 shows the difference between yield condition of Hill yield criterion, interaction formula and von Mises, assuming tensile and compressive strengths of the material are equal. The Hill yield criterion can be applied to quasi-homogenous anisotropic composites subjected to combined stresses. It is assumed that there exist three mutually perpendicular planes of symmetry within the anisotropic body, this means that the body is really orthotropic rather than generally anisotropic from the point of view of strength.

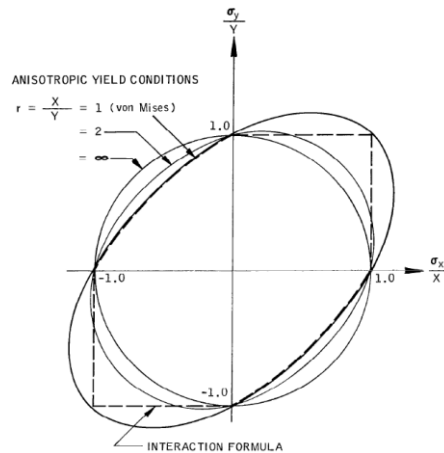


Figure 7: Comparative Yield Surfaces

The yield condition must be applied to the state of stress expressed in the coordinate system coincident with that of the material symmetry. Let, x-y (local stresses) be the material symmetry axes and 1-2 (global stresses) the reference coordinate axes of the applied stresses. The plane stress matrix, thus, would be:

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_s \end{bmatrix} = \begin{bmatrix} m^2 & n^2 & 2mn \\ n^2 & m^2 & -2mn \\ -mn & mn & m^2 - n^2 \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_6 \end{bmatrix}$$

Where, $m = \cos \theta$, $n = \sin \theta$

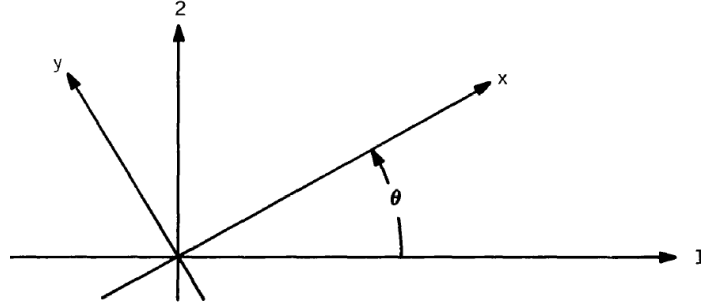


Figure 8: Coordinate Transformation of Stress (positive θ)

Stress is a measure of internal forces within a body. This together with strain are the key variables for the determination of stiffness and strength of a material. The mechanisms of deformation and failure are also interpreted in terms of the state of stress and strain. They are the fundamental variables for the mechanical behavior of materials similar to temperature and heat flux for heat conduction; or pressure, volume and temperature for gas [80]. Strain is simply the spatial variation of the displacements. There is no material property involved. Strain is related to displacement. Both are geometric quantities.

The constants in the stress-strain relation govern the stiffness of composites. Let, Δu = relative infinitesimal displacement along x-axis, and Δv = relative infinitesimal displacement along y-axis.

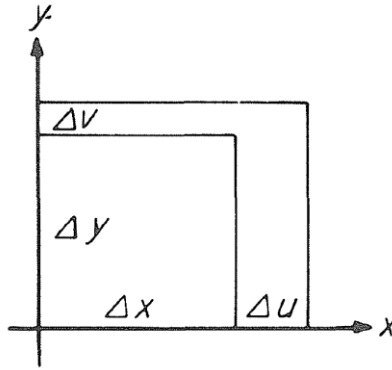


Figure 9: Normal strain and displacement relations

From Figure 4, we can define

$$\epsilon_x = \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} = \frac{\delta u}{\delta x}; \quad \epsilon_y = \lim_{\Delta y \rightarrow 0} \frac{\Delta v}{\Delta y} = \frac{\delta v}{\delta y}$$

Normal strain components are associated with changes in the lengths of an infinitesimal element. Distortion is measured by the change of angles. original rectangular element would be distorted

into a parallelogram. This combined action will produce distortion which is measured by shear strain. Figure 5 shows the combined action produced by the displacements.

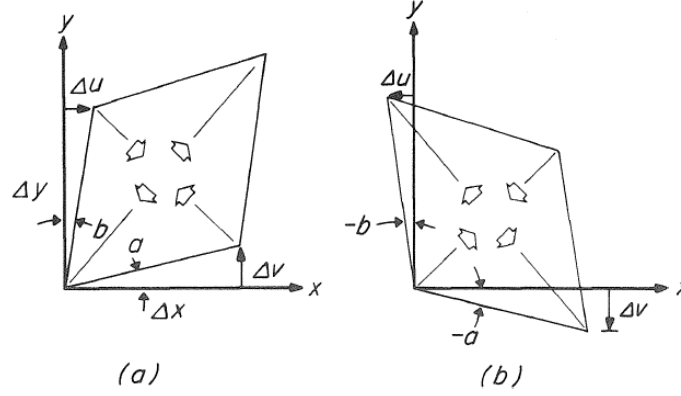


Figure 10: Strain-displacement relation for shear strain. (a) shear strain is positive; (b) shear strain is negative

The desired shear strain is: $\epsilon_s = a + b$

Where,

$$a = \tan a \cong \frac{\delta v}{\delta x}; b = \tan b \cong \frac{\delta u}{\delta y}$$

The resulting strain displacement relation is:

$$\epsilon_s = \frac{\delta v}{\delta x} + \frac{\delta u}{\delta y}$$

The stress-strain relation is, hence,

$$\epsilon = \frac{1}{E} \sigma$$

3.2. Stress-Strain Relations of Anisotropic Materials

An elastic body can be considered anisotropic if its elastic properties are different in different directions. Any components of the tensor stress σ_{ij} can lead to the emergence of all components of the tensor deformations [81].

Hooke's Law relation stresses to strains in a contracted notation can be written as:

$$\sigma_i = C_{ij} \epsilon_j$$

Where, σ_i = stress components; ϵ_j = engineering strain components; C_{ij} = composite stiffness matrix. Where, $i, j = 1, 2, \dots, 6$

The matrix notation of the aforementioned equation is:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{31} \\ \tau_{12} \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ C_{12} & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ C_{13} & C_{23} & C_{33} & C_{34} & C_{35} & C_{36} \\ C_{14} & C_{24} & C_{34} & C_{44} & C_{45} & C_{46} \\ C_{15} & C_{25} & C_{35} & C_{45} & C_{55} & C_{56} \\ C_{16} & C_{26} & C_{36} & C_{46} & C_{56} & C_{66} \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \gamma_{23} \\ \gamma_{31} \\ \gamma_{12} \end{bmatrix}$$

Therefore, the strains are [82]:

$$\varepsilon_1 = \frac{\delta u}{\delta x}; \varepsilon_2 = \frac{\delta v}{\delta y}; \varepsilon_3 = \frac{\delta w}{\delta z}$$

$$\gamma_{23} = \frac{\delta v}{\delta z} + \frac{\delta w}{\delta y}; \gamma_{31} = \frac{\delta w}{\delta x} + \frac{\delta u}{\delta z}; \gamma_{12} = \frac{\delta u}{\delta y} + \frac{\delta v}{\delta x}$$

Where, u, v and w are displacements in the x, y, z directions.

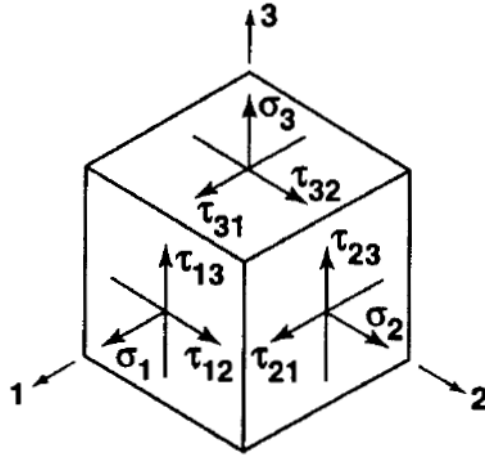


Figure 11: Stresses on an element

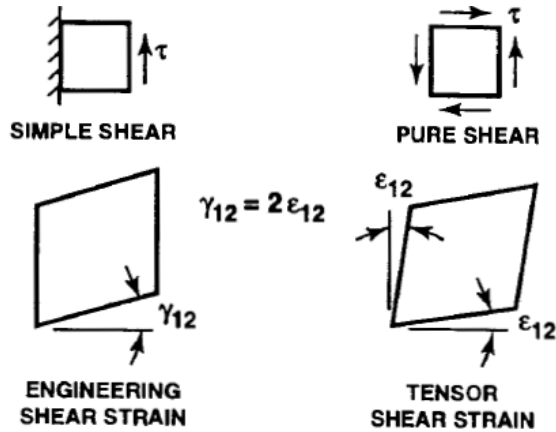


Figure 12: Engineering Shear Strain & Tensor Shear Strain

The engineering shear strain γ_{ij} is the total angle of shearing, whereas the tensor shear strain ε_{ij} is half of the total shearing angle.

The stiffness matrix C_{ij} , has 36 constants, but only a few are independent for elastic materials when characteristics of strain energy are considered. Elastic materials for which an elastic potential or strain energy density function exists have incremental work per unit volume $dW = \sigma_i d\varepsilon_i$, from stress-strain relation it becomes: $dW = C_{ij} \varepsilon_j d\varepsilon_i$

Integrating for all strains, the work per unit volume is: $W = \frac{1}{2} C_{ij} \epsilon_j \epsilon_j$; from Hooke's Law: $\frac{\delta W}{\delta \epsilon_j} = C_{ij} \epsilon_j$. The order of differentiation of work per unit volume is immaterial and remains the same on both sides. Hence, $C_{ij} = C_{ji}$, the stiffness matrix is symmetric. In similar manner, we can determine a compliance matrix S_{ij} , which is also symmetric $S_{ij} = S_{ji}$. The stress-strain relation is as follows:

$$\epsilon_j = S_{ij} \sigma_j \quad i, j = 1, \dots, 6$$

For an anisotropic body, the compliance coefficients depend on the choice of coordinate system. In the general case, they form a complete filled matrix $[S]$ of size 6x6. The reciprocity of the compliance matrix can be shown by virtue of the theorem given the condition $S_{ij} = S_{ji}$ is satisfied.

The matrix notation of Hooke's Law with compliance matrix is:

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \gamma_{23} \\ \gamma_{31} \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} & S_{15} & S_{16} \\ S_{12} & S_{22} & S_{23} & S_{24} & S_{25} & S_{26} \\ S_{13} & S_{23} & S_{33} & S_{34} & S_{35} & S_{36} \\ S_{14} & S_{24} & S_{34} & S_{44} & S_{45} & S_{46} \\ S_{15} & S_{25} & S_{35} & S_{45} & S_{55} & S_{56} \\ S_{16} & S_{26} & S_{36} & S_{46} & S_{56} & S_{66} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{31} \\ \tau_{12} \end{bmatrix}$$

Anisotropy of the most general form, when matrix $[S_{ij}]$ contains 21 independent coefficients, is a rare phenomenon in real materials. Usually, the structure of the material is such that its elastic properties in some directions are identical. In these cases, the number of independent coefficients in the matrix $[S_{ij}]$ (and $[C_{ij}]$) decreases, with suitable coordinate system Hooke's law can be simplified.

3.2.1. Plane of Elastic Symmetry

If in an anisotropic body its elastic properties are identical in any two directions with respect to some plane, then that plane is called the plane of elastic symmetry.

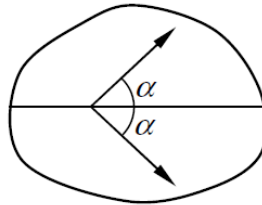


Figure 13: Plane of Elastic Symmetry

In such a case, the number of independent coefficients describing the elastic properties of the material is reduced to 13 and Hooke's law takes a simpler form when one of the coordinate planes is combined with the plane of elastic symmetry.

The type of matrices $[C_{ij}]$ and $[S_{ij}]$ are determined by the type of elastic symmetry and the choice of coordinate system. The type of symmetry of the material uniquely determines the number of

independent coefficients in these matrices. For any anisotropic body matrices $[C_{ij}]$ and $[S_{ij}]$ in an arbitrary coordinate system independent of the elastic symmetry will in most cases be fully filled.

3.3. Cylindrical Inclusions & Stress-Strain Relations

3.3.1. Composite in a Cylindrical coordinate system

Using a cylindrical coordinate system is preferable when estimating for cylindrical inclusions [83].

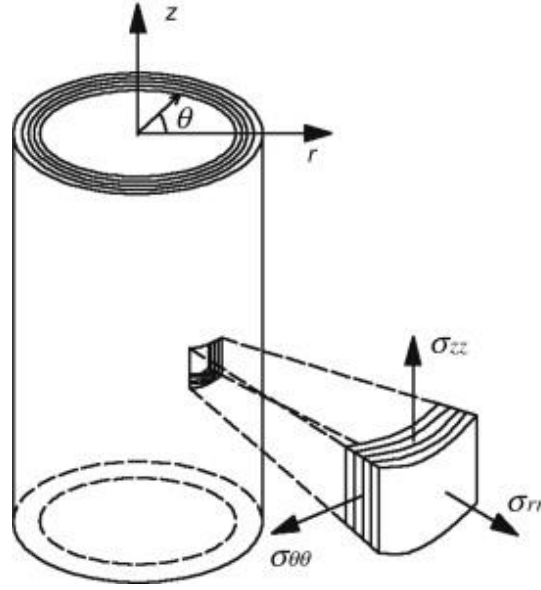


Figure 14: Interphase and coordinate system

To obtain the relevant strain functions, the allowable elastic deformations are substituted into Cauchy's differential relations [84]:

$$\varepsilon_{rr} = \frac{\delta u_r}{\delta r}, \varepsilon_{r\theta} = \frac{1}{2} \left(\frac{\delta u_\theta}{\delta r} + \frac{1}{r} \frac{\delta u_r}{\delta \theta} - \frac{u_\theta}{r} \right),$$

$$\varepsilon_{\theta\theta} = \frac{1}{r} \frac{\delta u_\theta}{\delta \theta} + \frac{u_r}{r}, \varepsilon_{\theta z} = \frac{1}{2} \left(\frac{\delta u_\theta}{\delta z} + \frac{1}{r} \frac{\delta u_z}{\delta \theta} \right)$$

$$\varepsilon_{zz} = \frac{\delta u_z}{\delta z}, \varepsilon_{rz} = \frac{1}{2} \left(\frac{\delta u_z}{\delta r} + \frac{\delta u_r}{\delta z} \right)$$

Using the equations of Hooke's Law, the corresponding stresses that are a function of displacements can be found for an orthotropic material:

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{13} & 0 & 0 & 0 \\ C_{13} & C_{13} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{C_{11}-C_{12}}{2} \end{bmatrix} \cdot \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \\ 2\varepsilon_{21} \end{bmatrix}$$

Here, C_{ij} and G_{ij} are components of the stiffness matrix.

The Equilibrium equations are given by:

$$\frac{\delta \sigma_{rr}}{\delta r} + \frac{1}{r} \frac{\delta \sigma_{r\theta}}{\delta \theta} + \frac{\delta \sigma_{rz}}{\delta z} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = 0$$

$$\frac{\delta \sigma_{\theta r}}{\delta r} + \frac{1}{r} \frac{\delta \sigma_{\theta\theta}}{\delta \theta} + \frac{\delta \sigma_{\theta z}}{\delta z} + \frac{2\sigma_{\theta r}}{r} = 0$$

$$\frac{\delta \sigma_{zr}}{\delta r} + \frac{1}{r} \frac{\delta \sigma_{z\theta}}{\delta \theta} + \frac{\delta \sigma_{zz}}{\delta z} + \frac{\sigma_{zr}}{r} = 0$$

Solving the above equations gives the required admissible displacement fields that satisfy the equilibrium equations for every orthotropic phase in the composite under a particular loading. Using Cauchy's relation and Hooke's law equations, the allowable strain and stress fields may be calculated based on allowable displacement fields.

3.3.2. Load Cases for Cylindrical Inclusion

Four types of loadings are considered [85] [86]:

- **Uniaxial loading along the axis of the cylindrical inclusion**
- **Pure shear along the axis of the cylindrical inclusion**
- **Pure shear in the direction transverse to the axis of the cylindrical inclusion**
- **Comprehensive load in the direction transverse to the axis of the cylindrical inclusion**

3.3.2.1. Case of Uniaxial Tension Along the Axis of the Inclusion

Displacements far from the inclusion in Cartesian coordinate system: $u_z = \varepsilon_0 z$, $u_x = u_y \neq 0$.

Rewriting in Cylindrical coordinate system: $u_r \neq 0$, $u_z = \varepsilon_0 z$, $u_\theta = 0$.

General solution for a heterogenous medium is given by: $u_r = u_r(r)$, $u_z = u_z(z)$, $u_\theta = 0$.

Where, $u_r(r)$, $u_z(z)$ are unknown functions.

Using Cauchy's differential relations:

$$\varepsilon_{rr} = \frac{\delta u_r}{\delta r}, \quad \varepsilon_{\theta\theta} = \frac{1}{r} \left(\frac{\delta u_\theta}{\delta \theta} + u_r \right) = \frac{u_r}{r}, \quad \varepsilon_{zz} = \frac{\delta u_z}{\delta z} = \varepsilon_0$$

Using Hooke's Law with the above equations to obtain the stress components in terms of displacements:

$$\sigma_{rr} = C_{rr}\varepsilon_{rr} + C_{r\theta}\varepsilon_{\theta\theta} + C_{rz}\varepsilon_{zz} = C_{rr}\left(\frac{\delta u_r}{\delta r}\right) + C_{r\theta}\left(\frac{u_r}{r}\right) + C_{rz}\varepsilon_0$$

$$\sigma_{\theta\theta} = C_{r\theta}\varepsilon_{rr} + C_{\theta\theta}\varepsilon_{\theta\theta} + C_{\theta z}\varepsilon_{zz} = C_{r\theta}\left(\frac{\delta u_r}{\delta r}\right) + C_{\theta\theta}\left(\frac{u_r}{r}\right) + C_{\theta z}\varepsilon_0$$

$$\sigma_{zz} = C_{rz}\varepsilon_{rr} + C_{\theta z}\varepsilon_{\theta\theta} + C_{zz}\varepsilon_{zz} = C_{rz}\left(\frac{\delta u_r}{\delta r}\right) + C_{\theta z}\left(\frac{u_r}{r}\right) + C_{zz}\varepsilon_0$$

Stresses σ_{rr} , $\sigma_{\theta\theta}$, σ_{zz} must satisfy one of the Equilibrium equations:

$$\frac{\delta\sigma_{rr}}{\delta r} + \frac{1}{r}\left(\frac{\delta\sigma_{r\theta}}{\delta\theta}\right) + \frac{\sigma_{rr}-\sigma_{\theta\theta}}{r} + \frac{\delta\sigma_{rz}}{\delta z} = 0$$

The other two Equilibrium equations are satisfied identically.

Rewriting the above shown equation in terms of displacements along with the mentioned stress components:

$$r^2 u_r''(r) + r u_r'(r) - \lambda^2 u_r = -\beta \varepsilon_0 r$$

$$\text{Where, } \lambda = \sqrt{C_{\theta\theta}/C_{rr}} \text{ and } \beta = \frac{C_{r\theta}-C_{\theta z}}{C_{rr}}$$

Solution to the above Equilibrium equation:

$$u_r(r) = u_r^h(r) + u_r^p(r)$$

Where, $u_r^{(h)}$ – is the general solution of homogenous equation and $u_r^{(p)}$ – is the particular solution to the inhomogeneous equation.

$$u_r(r) \text{ can be written as: } u_r(r) = D_1 r^\lambda + D_2 r^{-\lambda} + v \varepsilon_0 r^3$$

$$\text{Where, } v = \frac{C_{rz}-C_{\theta z}}{C_{\theta\theta}-9C_{rr}}$$

3.3.2.2. Case of Pure Shear Along the Axis of the Inclusion

Displacements far from the inclusion in Cartesian coordinate system:

$$u_y = \varepsilon_0 y, u_z = \varepsilon_0 z, u_x = 0 \text{ or } u_x = \varepsilon_0 x, u_z = \varepsilon_0 z, u_y = 0.$$

Rewriting in Cylindrical coordinate system:

$$u_r = \varepsilon_0 z \cos(\theta), u_\theta = \varepsilon_0 z \sin(\theta), u_z = 0.$$

In the case of pure shear along inclusion axis, cylindrical coordinates get rewritten as: $u_z = \varepsilon_0 r \cos(\theta), u_r = u_\theta = 0$.

General solution for a heterogenous medium is given by: $u_z = u_z(r, \theta) \cos(\theta), u_r = u_\theta = 0$.

Where, $u_z(r, \theta)$ is an unknown function.

Using Cauchy's differential relations:

$$\varepsilon_{zz} = \frac{\delta u_z}{\delta z}, 2\varepsilon_{\theta z} = \frac{\delta u_\theta}{\delta z} + \frac{1}{r} \left(\frac{\delta z}{\delta \theta} \right) = \frac{1}{r} \left(\frac{\delta u_z}{\delta \theta} \right), 2\varepsilon_{rz} = \frac{\delta u_z}{\delta r} + \frac{\delta u_r}{\delta z} = \frac{\delta u_z}{\delta r}$$

$$\varepsilon_{rr} = \varepsilon_{\theta\theta} = \varepsilon_{r\theta} = 0.$$

Using Hooke's Law with the above equations to obtain the stress components in terms of displacements:

$$\sigma_{\theta z} = G_{\theta z} 2\varepsilon_{\theta z} = G_{\theta z} \left(\frac{1}{2} \frac{\delta u_z}{\delta \theta} \right)$$

$$\sigma_{rz} = G_{rz} 2\varepsilon_{rz} = G_{rz} \left(\frac{1}{2} \frac{\delta u_z}{\delta r} \right)$$

Stresses $\sigma_{\theta z}$, σ_{rz} must satisfy one of the Equilibrium equations:

$$\frac{\delta \sigma_{rz}^i}{\delta r} + \frac{1}{r} \frac{\delta \sigma_{\theta z}^i}{\delta \theta} + \frac{\sigma_{rz}^i}{r} + \frac{\delta \sigma_{zz}^i}{\delta z} = 0$$

The other two Equilibrium equations are satisfied identically.

Rewriting the above shown equation in terms of displacements along with the mentioned stress components: $r^2 u_z''(r) + r u_z'(r) + \lambda^2 u_z'(\theta) = 0$

$$\text{Where, } \lambda = \sqrt{G_{\theta z}/G_{rz}}$$

Solution to the above differential equation:

$$u_z(r, \theta) = (D_1 r^\lambda + D_2 r^{-\lambda}) \cos \theta$$

$$\text{Where, } \lambda = \sqrt{G_{\theta z}/G_{rz}}; D_1; D_2 - \text{are constants}$$

3.3.2.3. Case of Pure Shear in the Direction Transverse to the Axis of the Inclusion

Displacements far from the inclusion in Cartesian coordinate system:

$$u_x = \varepsilon_0 x, u_y = -\varepsilon_0 y, u_z = 0.$$

Rotating the coordinate system relative to the z -axis by an angle θ , the displacements can be written as: $u_x = \varepsilon_0 r \cos(\theta), u_y = -\varepsilon_0 r \sin(\theta), u_z = 0.$

Rewriting in Cylindrical coordinate system: $u_r = \varepsilon_0 r \cos(2\theta), u_\theta = \varepsilon_0 r \sin(2\theta), u_z = 0.$

General solution for a heterogenous medium is given by:

$$u_r = u_r(r) \cos(2\theta), u_\theta = u_\theta(r) \sin(2\theta), u_z = 0.$$

Where, $u_r(r), u_\theta(r)$ are unknown functions; for simplification, a new notation is introduced: $u_r(r) = u(r), u_\theta(r) = w(r).$

Using Cauchy's differential relations:

$$\varepsilon_{rr} = \frac{\delta u_r}{\delta r} = u', \varepsilon_{\theta\theta} = \frac{1}{r} \left(\frac{\delta u_\theta}{\delta \theta} + u_r \right) = \frac{u - 2w}{r} \cos 2\theta$$

$$2\varepsilon_{r\theta} = \left(\frac{1}{r} \frac{\delta u_r}{\delta \theta} + \frac{\delta u_\theta}{\delta r} - \frac{u_\theta}{r} \right) = \left(-w' - \frac{2u + w}{r} \right) \sin 2\theta$$

Using Hooke's Law with the above equations to obtain the stress components in terms of displacements:

$$\sigma_{rr} = \left(\left(C_{rr} u' + C_{r\theta} \frac{u-2w}{r} \right) \cos 2\theta \right),$$

$$\sigma_{\theta\theta} = \left(\left(C_{r\theta} u' + C_{\theta\theta} \frac{u-2w}{r} \right) \cos 2\theta \right),$$

$$\sigma_{r\theta} = G_{r\theta} \left(-w' + \frac{w-2u}{r} \right) \sin 2\theta.$$

Stresses σ_{rr} , $\sigma_{\theta\theta}$, $\sigma_{r\theta}$ must satisfy two of the Equilibrium equations:

$$\frac{\delta \sigma_{rr}^i}{\delta r} + \frac{1}{r} \frac{\delta \sigma_{r\theta}^i}{\delta \theta} + \frac{\sigma_{rr}^i - \sigma_{\theta\theta}^i}{r} = 0,$$

$$\frac{\delta \sigma_{r\theta}^i}{\delta r} + \frac{1}{r} \frac{\delta \sigma_{\theta\theta}^i}{\delta \theta} + \frac{2\sigma_{r\theta}^i}{r} = 0$$

The other Equilibrium equations is satisfied identically.

Rewriting the above shown equation in terms of displacements along with the mentioned stress components:

$$C_{rr}(u''r^2 + u'r) - (4G_{r\theta} + C_{\theta\theta})u - 2(G_{r\theta} + C_{r\theta})w'r + 2(G_{r\theta} + C_{\theta\theta})w = 0$$

$$2(G_{r\theta} + C_{r\theta})u'r + 2(G_{r\theta} + C_{\theta\theta})u + (w''r^2 + w'r)G_{r\theta} - (G_{r\theta} + 4C_{\theta\theta})w = 0.$$

Solution to the above differential equation:

$$u(r) = Fr^\lambda,$$

$$w(r) = Dr^\lambda.$$

Where, F and D are unknown constants with a linear relationship:

$$F = 2 \frac{G_{r\theta}\lambda + C_{r\theta}\lambda - G_{r\theta} - C_{\theta\theta}}{C_{rr}\lambda^2 - 4G_{r\theta} - C_{\theta\theta}} D$$

The differential equations can be rewritten as:

$$u(r) = F_1 r^{\lambda_1} + F_2 r^{\lambda_2} + F_3 r^{\lambda_3} + F_4 r^{\lambda_4}$$

$$w(r) = D_1 r^{\lambda_1} + D_2 r^{\lambda_2} + D_3 r^{\lambda_3} + D_4 r^{\lambda_4}$$

Where, the roots λ^1, λ^3 – are positive values; and λ^2, λ^4 – are negative values.

3.3.2.4. Case of Comprehensive Load in the Direction Transverse to the Axis of the Inclusion

Displacements far from the inclusion in Cartesian coordinate system: $u_x(x) = \varepsilon_0 x, u_y(y) = \varepsilon_0 y$.

Rewriting in Cylindrical coordinate system: $u_r = \varepsilon_0 r, u_\theta = u_z = 0$.

General solution for a heterogenous medium is given by: $u_r = u_r(r), u_\theta = u_z = 0$.

Where, $u_r(r)$ is an unknown function.

Using Cauchy's differential relations:

$$\varepsilon_{rr} = \frac{\delta u_r}{\delta r}; \quad \varepsilon_{\theta\theta} = \frac{u_r}{r}$$

Using Hooke's Law with the above equations to obtain the stress components in terms of displacements:

$$\sigma_{rr} = C_{rr}\varepsilon_{rr} + C_{r\theta}\varepsilon_{\theta\theta} = C_{rr}\frac{\delta u_r}{\delta r} + C_{r\theta}\frac{u_r}{r}$$

$$\sigma_{\theta\theta} = C_{r\theta}\varepsilon_{rr} + C_{\theta\theta}\varepsilon_{\theta\theta} = C_{r\theta}\frac{\delta u_r}{\delta r} + C_{\theta\theta}\frac{u_r}{r}$$

Stresses σ_{rr} and $\sigma_{\theta\theta}$ must satisfy one of the Equilibrium equations:

$$\frac{\delta\sigma_{rr}^i}{\delta r} + \frac{1}{r}\frac{\delta\sigma_{r\theta}^i}{\delta\theta} + \frac{\sigma_{rr}^i - \sigma_{\theta\theta}^i}{r} + \frac{\delta\sigma_{rz}^i}{\delta z} = 0$$

The other two Equilibrium equations are satisfied identically.

Rewriting the above shown equation in terms of displacements along with the mentioned stress components:

$$r^2 u_r'' + r u_r' - \lambda^2 u_r = 0$$

$$\text{Where, } \lambda = \sqrt{C_{\theta\theta}/C_{rr}}$$

Solution to the above differential equation:

$$u_r(r) = D_1 r^\lambda + D_2 r^{-\lambda}$$

3.3.3. Stress-Strain Relations of a Composite Material with Cylindrical Inclusions

As determined earlier, for an anisotropic medium, the stress-strain relation for a Cartesian coordinate system is given by Hooke's Law:

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl}, \quad i, j = 1, 2, 3,$$

Where, σ_{ij} , ε_{kl} are the stress and strain tensors respectively. C_{ijkl} is the tensor of elastic moduli (stiffness tensor) of the fourth order.

A transversely isotropic medium is a medium that has symmetrical properties in a plane perpendicular to the direction of orientation of cylindrical inclusions. In the case when the packing of the cylindrical inclusions in the cross-section is uncertain, it can be assumed that the medium is essentially homogenous. For a transversely isotropic medium, the stiffness tensors depend upon the position of the central axis of the cylindrical inclusion relative to the coordinate system [87].

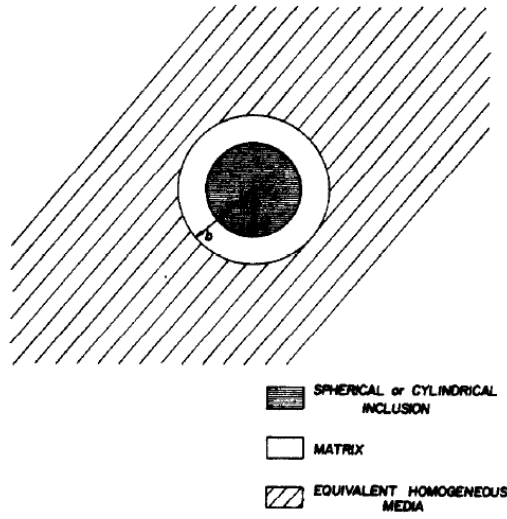


Figure 15: Model for transversely isotropic materials

The stress-strain relation for a transversely isotropic medium depending on the axis of symmetry is presented [86]. Here, the representation of the stiffness tensor C_{ijkl} has been altered, in which a pair of indices have been replaced by a single index $11 \rightarrow 1, 22 \rightarrow 2, 33 \rightarrow 3, 23 \rightarrow 4, 31 \rightarrow 5, 12 \rightarrow 6$. index

3.3.3.1. Assuming z -axis (3^{rd} axis) as Axis of Plane Isotropy

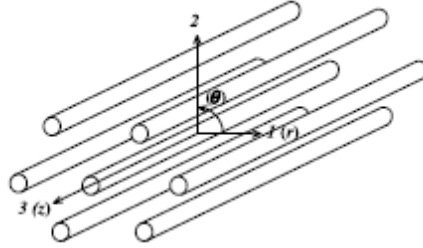


Figure 16: System of a transversally isotropic medium with axis 3 as the axis of symmetry

The appropriate components of the stiffness tensor in Hooke's Law take the form:

$$C_{11} = C_{22}, \quad C_{33}, \quad C_{12}, \quad C_{13} = C_{23}, \quad C_{44} = C_{55}, \quad C_{66} = \frac{1}{2} (C_{11} - C_{12})$$

The matrix notation is:

$$\begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{pmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{13} & 0 & 0 & 0 \\ C_{13} & C_{13} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{C_{11}-C_{12}}{2} \end{bmatrix} \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \\ 2\varepsilon_{12} \end{pmatrix}$$

The elastic constants C_{11} , C_{12} , C_{13} , C_{33} and C_{44} determine five independent effective properties of a transversally isotropic medium where the axis of the symmetry plane is directed along the z -axis. These constants are determined by the corresponding Young's modulus E_{33} ; shear moduli μ_{12} , $\mu_{23} = \mu_{13}$; bulk plane strain modulus K_{12} and Poisson's ratio ν_{31} using the equations:

$$C_{33} = E_{33} + 4\nu_{31}^2 K_{12}, \quad C_{13} = 2K_{12}\nu_{31}, \quad C_{11} = \mu_{12} + K_{12},$$

$$C_{12} = -\mu_{12} + K_{12}, \quad C_{44} = \mu_{13} = \mu_{23}.$$

Other equivalent equations can also be used to determine the elastic constants C_{11} , C_{12} , C_{13} , C_{33} and C_{44} :

$$E_{33} = E_L, \quad E_{11} = E_{22} = E_T, \quad \mu_{13} = \mu_{31} = \mu_{23} = \mu_{32} = \mu_L,$$

$$\mu_{12} = \mu_{21} = \mu_T, \quad \nu_{12} = \nu_{21} = \nu_T, \quad \nu_{31} = \nu_{32} = \nu_{LT},$$

$$\nu_{13} = \nu_{23} = \nu_{TL}.$$

The matrix notation of the compliance tensor S_{ij} takes the form:

$$\begin{bmatrix} 1/E_T & -\nu_T/E_T & -\nu_{LT}/E_L & 0 & 0 & 0 \\ -\nu_T/E_T & 1/E_T & -\nu_{LT}/E_L & 0 & 0 & 0 \\ -\nu_{TL}/E_T & -\nu_{TL}/E_T & 1/E_L & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/\mu_L & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/\mu_L & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/\mu_T \end{bmatrix}$$

Where,

$$\mu_T = E_T / 2(1 + \nu_T).$$

The relation between elastic stiffness, C_{ijkl} , and engineering constants can be written as:

$$C_{11} = C_{22} = E_T(1 - \nu_{TL}\nu_{LT})Y, \quad C_{33} = E_L(1 - \nu_T^2)Y,$$

$$C_{12} = E_T(\nu_T + \nu_{TL}\nu_{LT})Y, \quad C_{13} = C_{23} = E_T(\nu_{LT} + \nu_T\nu_{LT})Y =$$

$$= E_L(\nu_{TL} + \nu_T\nu_{TL})Y, \quad C_{44} = \mu_L, \quad C_{66} = \mu_T,$$

Where,

$$Y = \frac{1}{1 - \nu_T^2 - 2\nu_{TL}\nu_{LT} - 2\nu_T\nu_{TL}\nu_{LT}}.$$

3.3.3.2. Assuming x -axis (1^{st} axis) as Axis of Plane Isotropy

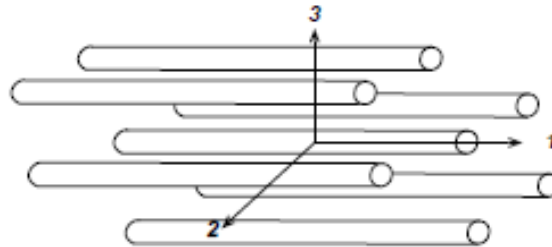


Figure 17: System of a transversally isotropic medium with axis 1 as the axis of symmetry

The appropriate components of the stiffness tensor in Hooke's Law take the form:

$$C_{11}, \quad C_{22} = C_{33}, \quad C_{12} = C_{13}, \quad C_{23}, \quad C_{44} = \frac{1}{2}(C_{22} - C_{23}), \quad C_{66} = C_{55}$$

The matrix notation is:

$$\begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{pmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{22} & C_{23} & 0 & 0 & 0 \\ C_{12} & C_{23} & C_{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{C_{22}-C_{23}}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{66} \end{bmatrix} \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \\ 2\varepsilon_{12} \end{pmatrix}$$

The elastic constants C_{11} , C_{22} , C_{12} , C_{23} and C_{66} determine five independent effective properties of a transversally isotropic medium where the axis of the symmetry plane is directed along the x -axis. These constants are determined by the corresponding Young's modulus E_{11} ; shear moduli μ_{23} , $\mu_{12} = \mu_{13}$; bulk plane strain modulus K_{23} and Poisson's ratio ν_{12} using the equations:

$$C_{11} = E_{11} + 4\nu_{12}^2 K_{23}, \quad C_{12} = 2K_{23}\nu_{12}, \quad C_{22} = \mu_{23} + K_{23}, \\ C_{23} = -\mu_{23} + K_{23}, \quad C_{66} = \mu_{12} = \mu_{13}.$$

Other equivalent equations can also be used to determine the elastic constants C_{11} , C_{22} , C_{12} , C_{23} and C_{66} :

$$C_{22} = C_{33} = E_T(1 - \nu_{TL}\nu_{LT})Y, \quad C_{11} = E_L(1 - \nu_T^2)Y, \\ C_{23} = E_T(\nu_T + \nu_{TL}\nu_{LT})Y, \quad C_{12} = C_{13} = E_T(\nu_{LT} + \nu_T\nu_{LT})Y = \\ = E_L(\nu_{TL} + \nu_T\nu_{TL})Y, \quad C_{66} = \mu_L, \quad C_{44} = \mu_T.$$

3.3.3.3. Case of Complete Isotropy

In such a case, two independent stiffness tensors remain, C_{12} and C_{11} .

$$C_{11} = C_{22} = C_{33}, \quad C_{12} = C_{13} = C_{23}, \quad C_{44} = C_{66} = C_{55} = \frac{1}{2}(C_{11} - C_{12})$$

The matrix notation is:

$$\begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{pmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{C_{11}-C_{12}}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{C_{11}-C_{12}}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{C_{11}-C_{12}}{2} \end{bmatrix} \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \\ 2\varepsilon_{12} \end{pmatrix}$$

With Lamé constants μ and λ the matrix notation of Hooke's Law turns into:

$$\begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{pmatrix} = \begin{bmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix} \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \\ 2\varepsilon_{12} \end{pmatrix}.$$

Comparing the matrix with stiffness tensors C_{ijkl} and matrix with Lamé constants μ and λ :

$$C_{11} = \lambda + 2\mu, \quad C_{12} = \lambda$$

It is known:

$$\mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{\nu E}{(1+\nu)(1-2\nu)}.$$

Comparing the known values of the Lamé constants and the relation between them and stiffness tensors:

$$C_{11} = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}, \quad C_{12} = \frac{E\nu}{(1+\nu)(1-2\nu)}.$$

3.4. Modeling the Effective Properties of Modified Composites with Whiskered Fibers

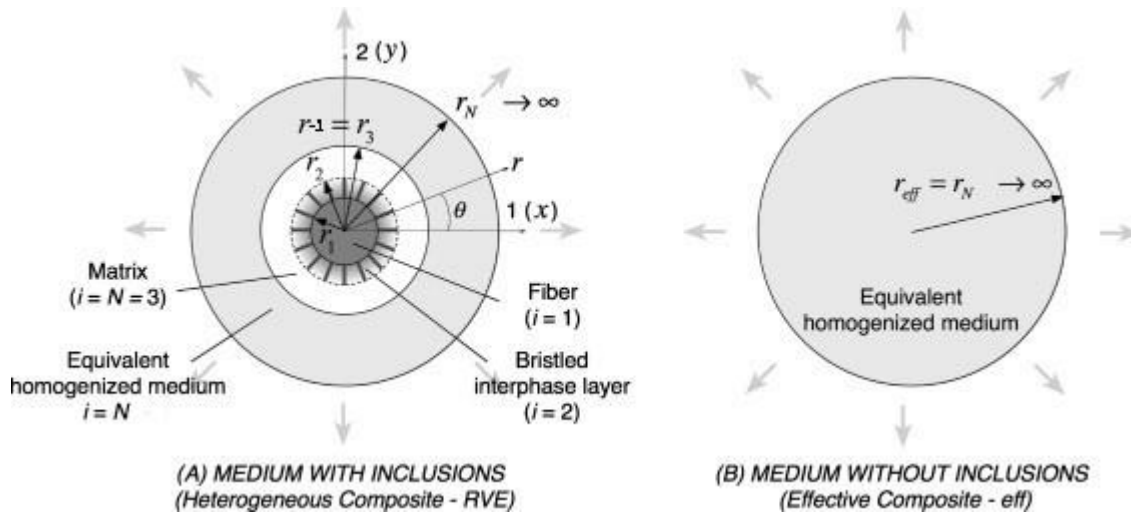


Figure 18: (a) Cell of a whiskered fiber composite; (b) Cell of an effective composite

For the phase N ($i=N$) in figure 18(a), where the equivalent homogenized medium assumes the transversely isotropic property with its plane of isotropy perpendicular to the principal axis of the fiber, its stiffness tensor's components are given as [88]:

$$\begin{aligned} C_{11}^N &= C_{22}^N, \quad C_{33}^N, \quad C_{12}^N, \quad C_{13}^N = C_{23}^N, \\ C_{44}^N &= C_{55}^N, \quad C_{66}^N = \frac{1}{2} (C_{11}^N - C_{12}^N). \end{aligned}$$

The specimen is referred to a Cartesian-coordinate system x - y - z (1-2-3), where z (3) is the direction of the fiber and x, y are transverse planes. Representative volume element (RVE) as shown in figure 18(a) is used for the composite cylinders method. This approach can be extended for cases where number of layers are $N \geq 3$.

The specimen is subjected to boundary conditions over its surface [89]:

$$\begin{aligned} u_i^0(S) &= \epsilon_{ij}^0 x_j \\ T_i^0(S) &= \sigma_{ij}^0 n_j \end{aligned}$$

Where, u_i^0 and T_i^0 are displacement and stress-vector components respectively. x_j and n_j are surface coordinates and components of the outward normal to the surface respectively.

For the boundary conditions, it can be shown that (1) the average strains over the specimen $\bar{\sigma}_{ij} = C_{ijkl}^* \bar{\epsilon}_{kl}$ are ϵ_{ij}^0 and for (2) the average stresses over the specimen are σ_{ij}^0 . The specimen is assumed to be macroscopically homogenous, whereby it is intended that for either one of boundary conditions (1) or (2) strain and stress averages taken over large enough RVE.

When (1) is prescribed, the average strains are ϵ_{ij}^0 and $\bar{\sigma}_{ij}$ have to be found. Conversely, when (2) is prescribed, the average stresses are known and the average strains are found. The strain energy W stored in an RVE is given as [90]:

$$W = \frac{1}{2} \bar{\sigma}_{ij} \bar{\epsilon}_{ij}$$

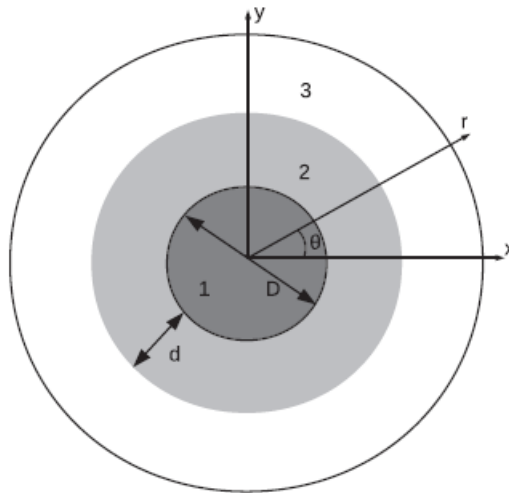


Figure 19: Simplified RVE

r_i is denoted as the external radius of each layer and the length of the cylinder is taken as $2L$. The volume average with V as the RVE volume [91]:

$$\langle \phi \rangle = \frac{1}{V} \int_V \phi(x, y, z) dx dy dz = \frac{1}{V} \int_V r \phi(r, \theta, z) dr d\theta dz$$

The volume fraction of inclusions is given in terms of the length and diameter of the whiskers:

$$c_b = \frac{M_b^2 d_b^2}{4\pi(l_b + D)D}$$

Where, M_b – number of whiskers on the fiber; d_b – diameter of the whisker; l_b – length of the whisker; D – diameter of the base fiber.

The packing density of an arrangement refers to the proportion of whiskers on a single base fiber. For calculation purposes, the packing density is the ratio of the number of whiskers on the fiber to the maximum allowable number of whiskers that fit along the circumference of the fiber:

$$\rho_b = (M_b / M_b^{\max}) 100\%$$

Where, $M_b^{\max} = \pi D / d_b$ – maximum permissible number of whiskers that can fit on the fiber circumference.

Five effective properties need to be found in order to adequately describe the overall behavior of the whiskered fiber composite, namely – plane strain bulk modulus K_{12}^{eff} , axial shear modulus μ_{23}^{eff} , axial Young's modulus E_3^{eff} , transverse shear modulus μ_{12}^{eff} , transverse Young's modulus E_1^{eff} and Poisson's ratio ν_{23}^{eff} , ν_{12}^{eff} .

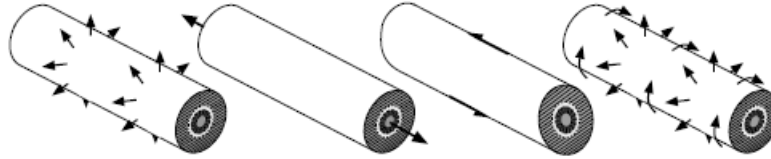


Figure 20: Load Cases

It is considered that the fiber⁽¹⁾ is transversally isotropic with symmetry axis along the fiber axis, the interphase layer⁽²⁾ is also a transversally isotropic medium with symmetry axis directed along the axis of the whiskers and the matrix⁽³⁾ is an isotropic medium. The stresses in the fiber, interphase and matrix are determined through:

$$\begin{pmatrix} \sigma_{rr}^{(1)} \\ \sigma_{\theta\theta}^{(1)} \\ \sigma_{zz}^{(1)} \\ \sigma_{\theta z}^{(1)} \\ \sigma_{rz}^{(1)} \\ \sigma_{r\theta}^{(1)} \end{pmatrix} = \begin{bmatrix} C_{11}^{(1)} & C_{12}^{(1)} & C_{13}^{(1)} & 0 & 0 & 0 \\ C_{12}^{(1)} & C_{11}^{(1)} & C_{13}^{(1)} & 0 & 0 & 0 \\ C_{13}^{(1)} & C_{13}^{(1)} & C_{33}^{(1)} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44}^{(1)} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44}^{(1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{C_{11}^{(1)} - C_{12}^{(1)}}{2} \end{bmatrix} \begin{pmatrix} \varepsilon_{rr}^{(1)} \\ \varepsilon_{\theta\theta}^{(1)} \\ \varepsilon_{zz}^{(1)} \\ 2\varepsilon_{\theta z}^{(1)} \\ 2\varepsilon_{rz}^{(1)} \\ 2\varepsilon_{r\theta}^{(1)} \end{pmatrix}$$

$$\begin{pmatrix} \sigma_{rr}^{(2)} \\ \sigma_{\theta\theta}^{(2)} \\ \sigma_{zz}^{(2)} \\ \sigma_{\theta z}^{(2)} \\ \sigma_{rz}^{(2)} \\ \sigma_{r\theta}^{(2)} \end{pmatrix} = \begin{bmatrix} C_{11}^{(2)} & C_{12}^{(2)} & C_{12}^{(2)} & 0 & 0 & 0 \\ C_{12}^{(2)} & C_{22}^{(2)} & C_{23}^{(2)} & 0 & 0 & 0 \\ C_{12}^{(2)} & C_{23}^{(2)} & C_{22}^{(2)} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{C_{12}^{(2)} - C_{23}^{(2)}}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{66}^{(2)} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{66}^{(2)} \end{bmatrix} \begin{pmatrix} \varepsilon_{rr}^{(2)} \\ \varepsilon_{\theta\theta}^{(2)} \\ \varepsilon_{zz}^{(2)} \\ 2\varepsilon_{\theta z}^{(2)} \\ 2\varepsilon_{rz}^{(2)} \\ 2\varepsilon_{r\theta}^{(2)} \end{pmatrix}$$

$$\begin{pmatrix} \sigma_{rr}^{(3)} \\ \sigma_{\theta\theta}^{(3)} \\ \sigma_{zz}^{(3)} \\ \sigma_{\theta z}^{(3)} \\ \sigma_{rz}^{(3)} \\ \sigma_{r\theta}^{(3)} \end{pmatrix} = \begin{bmatrix} C_{11}^{(3)} & C_{12}^{(3)} & C_{12}^{(3)} & 0 & 0 & 0 \\ C_{12}^{(3)} & C_{11}^{(3)} & C_{12}^{(3)} & 0 & 0 & 0 \\ C_{12}^{(3)} & C_{12}^{(3)} & C_{11}^{(3)} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{C_{11}^{(3)} - C_{12}^{(3)}}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{C_{11}^{(3)} - C_{12}^{(3)}}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{C_{11}^{(3)} - C_{12}^{(3)}}{2} \end{bmatrix} \begin{pmatrix} \varepsilon_{rr}^{(3)} \\ \varepsilon_{\theta\theta}^{(3)} \\ \varepsilon_{zz}^{(3)} \\ 2\varepsilon_{\theta z}^{(3)} \\ 2\varepsilon_{rz}^{(3)} \\ 2\varepsilon_{r\theta}^{(3)} \end{pmatrix}$$

3.4.1. Plane Strain Bulk Modulus; Uniform Load Across the Fiber

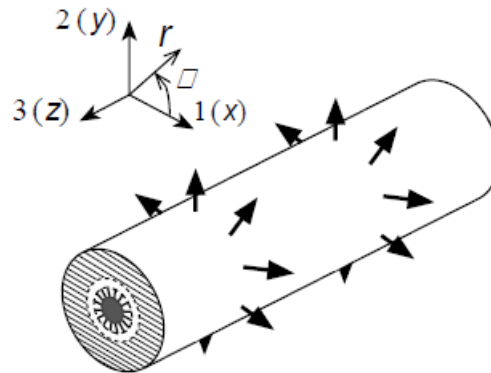


Figure 21: All-round loading

The effective plane strain bulk modulus, K_{12}^{eff} , through application of displacement field can be described by:

$$u_r^{(i)} = \sum_{j=1}^2 D_j^{(i)} r^{n_j^{(i)}}, \quad u_\theta^{(i)} = u_z^{(i)} = 0, \quad i = 1, \dots, N$$

$$\text{Where, } n_1^{(i)} = \sqrt{C_{22}^{(i)}/C_{11}^{(i)}} = -n_2^{(i)} \text{ or } = \sqrt{C_{\theta\theta}^{(i)}/C_{rr}^{(i)}} = \sqrt{C_{22}^{(i)}/C_{11}^{(i)}} = 1$$

The axis of symmetry is parallel to the fiber axis, $n_1^{(i)} = -n_2^{(i)} = 1$, for isotropic and transversely isotropic materials. The displacement field thus, reduces to the expression [92]:

$$\begin{aligned} u_r^i &= B_1^i r + \frac{B_2^i}{r} \\ u_\theta^i &= 0 \\ u_z^i &= 0 \end{aligned}$$

For, $r_{i-1} \leq r \leq r_i$. Where, B_1^i and B_2^i , are constants with $1 \geq i \geq N$. For this document B_1^i and B_2^i are denoted as $D_1^{(i)}$ and $D_2^{(i)}$ and can be determined from the boundary conditions shown in figure 22.

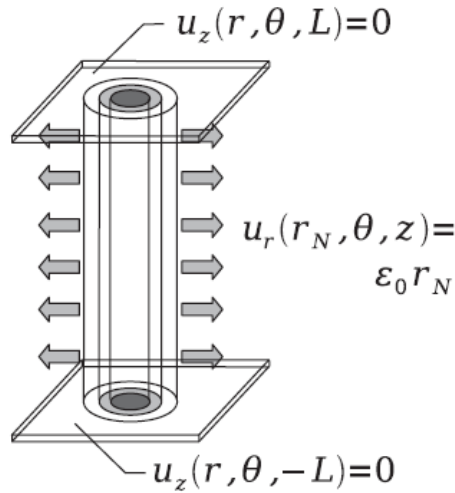


Figure 22: Boundary value problem for plane strain bulk modulus

The external boundary condition at infinity ($r_N \rightarrow \infty$):

$$u_r^{(N)}(r_N) = \epsilon_0 r_N$$

The condition for stress non-singularity at fiber center ($r=0$), it ensures that the displacements remain bounded at the origin: $D_2^{(1)} = 0$

The condition of continuity that enforces ideal bonding between phase layers:

$$u_r^{(i)}(r_i) = u_r^{(i+1)}(r_i), \quad \sigma_{rr}^{(i)}(r_i) = \sigma_{rr}^{(i+1)}(r_i), \quad \text{for } i = 1, \dots, N-1$$

The volume averaged strain energy for the composite cylinder assembly is given as:

$$W^{\text{RVE}} = \frac{1}{2} \left\langle \sigma_{rr}^{(i)} \epsilon_{rr}^{(i)} + \sigma_{\theta\theta}^{(i)} \epsilon_{\theta\theta}^{(i)} \right\rangle = \frac{1}{r_N^2} \sum_{i=1}^N \sum_{j=1}^2 \left(\frac{C_{22}^{(i)}}{n_j^{(i)}} + C_{12}^{(i)} \right) (D_j^{(i)})^2 \left(r_i^{2n_j^{(i)}} - r_{i-1}^{2n_j^{(i)}} \right)$$

For a homogenized cylinder, $n_{eff} = 1$ and for the effective medium, $D_1^{eff} = \epsilon_0$. Accordingly, the strain energy of the effective homogenous material is:

$$W^{\text{eff}} = \frac{1}{2} \langle \sigma_{rr}^{\text{eff}} \varepsilon_{rr}^{\text{eff}} + \sigma_{\theta\theta}^{\text{eff}} \varepsilon_{\theta\theta}^{\text{eff}} \rangle = 2K_{12}^{\text{eff}} \varepsilon_0^2$$

Since, $W^{RVE} = W^{\text{eff}}$:

$$K_{12}^{\text{eff}} = \frac{1}{2\varepsilon_0^2 r_N^2} \sum_{i=1}^N \sum_{j=1}^2 \left(\frac{C_{22}^{(i)}}{n_j^{(i)}} + C_{12}^{(i)} \right) \left(D_j^{(i)} \right)^2 \left(r_i^{2n_j^{(i)}} - r_{i-1}^{2n_j^{(i)}} \right)$$

The effective plane strain bulk modulus does not depend on the applied strain at the boundary as $D_j^{(i)}$ contains ε_0 .

3.4.2. Axial Shear Modulus; Pure Shear Along the Fiber

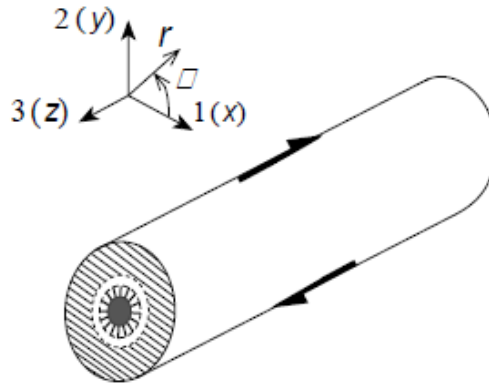


Figure 23: Shear along the fiber

The axial shear modulus, $\mu_{23}^{\text{eff}} = C_{44}^{\text{eff}}$, through application of displacement field can be described by:

$$u_z^{(i)} = \sum_{j=1}^2 D_j^{(i)} r^{n_j^{(i)}} \cos \theta, u_\theta^{(i)} = u_r^{(i)} = 0$$

$$\text{Where, } n_1^{(i)} = \sqrt{C_{44}^{(i)} / C_{55}^{(i)}} = -n_2^{(i)} \text{ or } = \sqrt{G_{\theta z}^{(i)} / G_{rz}^{(i)}} = \sqrt{C_{44}^{(i)} / C_{55}^{(i)}} = 1$$

The axis of symmetry is parallel to the fiber axis, $n_1^{(i)} = -n_2^{(i)} = 1$, for isotropic and transversely isotropic materials. The displacement field thus, reduces to the expression [92]:

$$u_r^i = 0$$

$$u_\theta^i = 0$$

$$u_z^i = \left(B_1^i r + \frac{B_2^i}{r} \right) \cos(\theta)$$

For, $r_{i-1} \leq r \leq r_i$. Where, B_1^i and B_2^i , are constants with $1 \geq i \geq N$. For this document B_1^i and B_2^i are denoted as $D_1^{(i)}$ and $D_2^{(i)}$ and can be determined from the boundary conditions shown in figure 24.

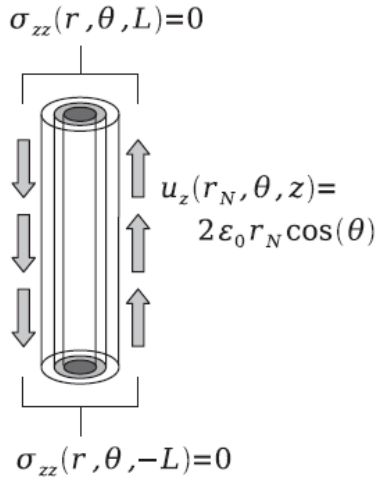


Figure 24: Boundary condition problem for axial shear modulus

The external boundary condition at infinity ($r_N \rightarrow \infty$):

$$u_z^{(N)}(r_N, \theta) = 2\varepsilon_0 r_N \cos \theta$$

The condition for stress non-singularity at fiber center ($r=0$), it ensures that the displacements remain bounded at the origin: $D_2^{(1)} = 0$

The condition of continuity that enforces ideal bonding between phase layers:

$$u_z^{(i)}(r_i, \theta) = u_z^{(i+1)}(r_i, \theta), \sigma_{rz}^{(i)}(r_i, \theta) = \sigma_{rz}^{(i+1)}(r_i, \theta) \text{ for } i = 1, \dots, N-1$$

The volume averaged strain energy for the composite cylinder assembly is given as:

$$W^{RVE} = \frac{1}{2} \langle 2\sigma_{z\theta}^{(i)} \varepsilon_{z\theta}^{(i)} + 2\sigma_{rz}^{(i)} \varepsilon_{rz}^{(i)} \rangle = \frac{1}{2r_N^2} \sum_{i=1}^N \sum_{j=1}^2 \frac{C_{44}^{(i)}}{n_j^{(i)}} (D_j^{(i)})^2 \left(r_i^{2n_j^{(i)}} - r_{i-1}^{2n_j^{(i)}} \right)$$

For a homogenized cylinder, $n_{eff} = 1$ and for the effective medium, $D_1^{eff} = 2\varepsilon_0$. Accordingly, the strain energy of the effective homogenous material is:

$$W^{eff} = \frac{1}{2} \langle 2\sigma_{z\theta}^{eff} \varepsilon_{z\theta}^{eff} + 2\sigma_{rz}^{eff} \varepsilon_{rz}^{eff} \rangle = 2\mu_{23}^{eff} \varepsilon_0^2$$

Since, $W^{RVE} = W^{eff}$:

$$\mu_{23}^{eff} = \frac{1}{4\varepsilon_0^2 r_N^2} \sum_{i=1}^N \sum_{j=1}^2 \frac{C_{44}^{(i)}}{n_j^{(i)}} (D_j^{(i)})^2 \left(r_i^{2n_j^{(i)}} - r_{i-1}^{2n_j^{(i)}} \right)$$

3.4.3. Axial Young's Modulus; Uniaxial Tension

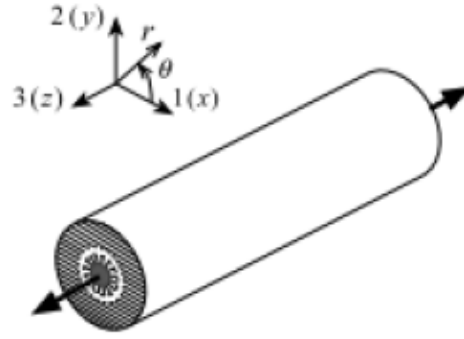


Figure 25: Uniaxial tension along the fiber

The effective axial Young's modulus, E_3^{eff} , through the application of displacement field can be described by:

$$u_r^{(j)} = A^{(i)} r + \sum_{j=1}^2 D_j^{(i)} r^{n_j^{(i)}}; \quad u_\theta^{(i)} = 0, \quad u_z^{(i)} = \varepsilon_0 z, \quad \text{for } i = 1, \dots, N,$$

$$A^{(i)} = \begin{cases} \frac{C_{13}^{(i)} - C_{23}^{(i)}}{C_{22}^{(i)} - C_{11}^{(i)}} \varepsilon_0, & C_{11}^{(i)} \neq C_{22}^{(i)}, \\ 0, & C_{11}^{(i)} = C_{22}^{(i)}, \end{cases}; \quad \left(\frac{C_{rz}^{(i)} - C_{\theta z}^{(i)}}{C_{\theta\theta}^{(i)} - C_{rr}^{(i)}} \right) = \left(\frac{C_{13}^{(i)} - C_{23}^{(i)}}{C_{22}^{(i)} - C_{11}^{(i)}} \right) = 0$$

Where, $n_1^{(i)} = \sqrt{C_{22}^{(i)}/C_{11}^{(i)}} = -n_2^{(i)}$ or $\sqrt{G_{\theta\theta}^{(i)}/G_{rr}^{(i)}} = \sqrt{C_{22}^{(i)}/C_{11}^{(i)}} = 1$

The axis of symmetry is parallel to the fiber axis, $n_1^{(i)} = 1$, for isotropic and transversely isotropic materials. The displacement field thus, reduces to the expression:

$$u_r^i = B_1^i r + \frac{B_2^i}{r}$$

$$u_\theta^i = 0$$

$$u_z^i = \varepsilon_0 z$$

For, $r_{i-1} \leq r \leq r_i$. Where, B_1^i and B_2^i , are constants with $1 \geq i \geq N$. For this document B_1^i and B_2^i are denoted as $D_1^{(i)}$ and $D_2^{(i)}$ and can be determined from the boundary conditions shown in figure 26.

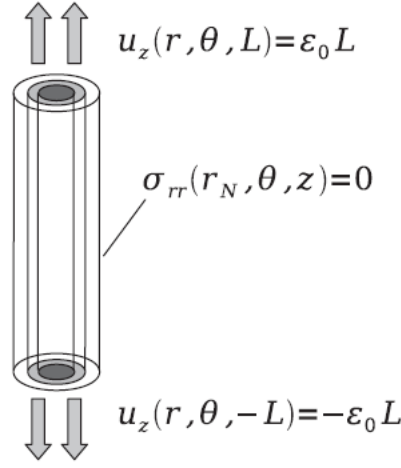


Figure 26: Boundary condition problem for axial Young's modulus

The reinforced interphase has a term, $\Lambda^{(i)}r$, which is required to satisfy the equilibrium equation. $\Lambda^{(i)}$ has a non-zero value only when the material has different radial and circumferential behavior.

The condition for stress non-singularity at fiber center ($r=0$), it ensures that the displacements remain bounded at the origin: $D_2^{(1)} = 0$

$D_1^{(i)}$ and $D_2^{(i)}$, $i = 1, \dots, N$ can be determined by condition of continuity that enforces ideal bonding between phase layers:

$$u_r^{(i)}(r_i) = u_r^{(i+1)}(r_i), \quad \sigma_{rr}^{(i)}(r_i) = \sigma_{rr}^{(i+1)}(r_i),$$

for $i = 1, \dots, N - 1$.

For the axial Young's modulus, the lateral surface is traction free in tension test, the external boundary condition at infinity ($r_N \rightarrow \infty$): $\sigma_{rr}^{(N)}(r_N) = 0$

The volume averaged strain for the composite is:

$$\begin{aligned} W^{\text{RVE}} &= \frac{1}{2} \left\langle \sigma_{rr}^{(i)} \epsilon_{rr}^{(i)} + \sigma_{\theta\theta}^{(i)} \epsilon_{\theta\theta}^{(i)} + \sigma_{zz}^{(i)} \epsilon_{zz}^{(i)} \right\rangle \\ &= \frac{1}{r_N^2} \sum_{i=1}^N \left[\left(A_1^{(i)} \epsilon_0 + 2A_2^{(i)} \Lambda^{(i)} \right) \frac{r_i^2 - r_{i-1}^2}{2} + 2\epsilon_0 \sum_{j=1}^2 D_j^{(i)} \left(C_{23}^{(i)} + n_j^{(i)} C_{13}^{(i)} \right) \right. \\ &\quad \times K_j^{(i)} + \sum_{j=1}^2 \left(\frac{C_{22}^{(i)}}{n_j^{(i)}} + C_{12}^{(i)} \right) D_j^{(i)} \left(D_j^{(i)} \left(r_i^{2n_j^{(i)}} - r_{i-1}^{2n_j^{(i)}} \right) \right. \\ &\quad \left. \left. + 2\Lambda^{(i)} \left(r_i^{1+n_j^{(i)}} - r_{i-1}^{1+n_j^{(i)}} \right) \right) \right] \end{aligned}$$

Here, $A_1^{(i)} = C_{13}^{(i)} \Lambda^{(i)} + C_{23}^{(i)} \Lambda^{(i)} + C_{33}^{(i)} \epsilon_0$, $A_2^{(i)} = C_{11}^{(i)} \Lambda^{(i)} + C_{12}^{(i)} \Lambda^{(i)} + C_{13}^{(i)} \epsilon_0$

$$K_j^{(i)} = \begin{cases} \frac{r_i^{1+n_j^{(i)}} - r_{i-1}^{1+n_j^{(i)}}}{1+n_j^{(i)}}, & n_j^{(i)} \neq -1, \\ \ln(r_i) - \ln(r_{i-1}), & n_j^{(i)} = -1. \end{cases}$$

For a homogenized cylinder, the effective medium, the boundary conditions for axial Young's modulus give: $D_1^{\text{eff}} = -C_{13}^{\text{eff}} \varepsilon_0 / (C_{11}^{\text{eff}} + C_{12}^{\text{eff}})$

Accordingly, the strain energy of the effective homogenous material is:

$$W^{\text{eff}} = \frac{1}{2} \langle \sigma_{rr}^{\text{eff}} \varepsilon_{rr}^{\text{eff}} + \sigma_{\theta\theta}^{\text{eff}} \varepsilon_{\theta\theta}^{\text{eff}} + \sigma_{zz}^{\text{eff}} \varepsilon_{zz}^{\text{eff}} \rangle = \frac{1}{2} E_3^{\text{eff}} \varepsilon_0^2$$

Since, $W^{RVE} = W^{\text{eff}}$:

$$E_3^{\text{eff}} = \frac{2}{\varepsilon_0^2 r_N^2} \sum_{i=1}^N \left[\left(A_1^{(i)} \varepsilon_0 + 2 A_2^{(i)} A^{(i)} \right) \frac{r_i^2 - r_{i-1}^2}{2} + 2 \varepsilon_0 \sum_{j=1}^2 D_j^{(i)} \left(C_{23}^{(i)} + n_j^{(i)} C_{13}^{(i)} \right) K_j^{(i)} + \sum_{j=1}^2 \left(\frac{C_{22}^{(i)}}{n_j^{(i)}} + C_{12}^{(i)} \right) D_j^{(i)} \left(D_j^{(i)} \left(r_i^{2n_j^{(i)}} - r_{i-1}^{2n_j^{(i)}} \right) + 2 A^{(i)} \left(r_i^{1+n_j^{(i)}} - r_{i-1}^{1+n_j^{(i)}} \right) \right) \right].$$

3.4.4. Transverse Shear Modulus; Pure Shear Across the Fiber

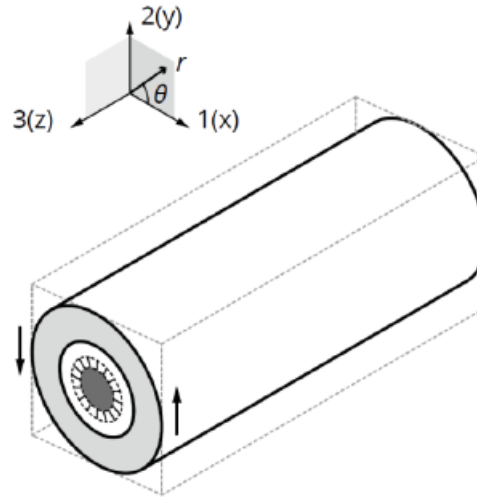


Figure 27: Shear across the fiber

The composite cylinders method that has been used to determine the other effective moduli of the whisker was proposed by Hashin [89] but for the transverse shear modulus they could only provide the bounds, Hashin further discussed transverse shear [93]. Hence, a generalized self-consistent composite cylinder model is used proposed by Christensen [87].

The additional layer $N+1$ represents the effective medium and the external radius is $r_{N+1} \rightarrow \infty$. The transverse shear modulus, μ_{12}^{eff} , through the application of displacement field can be described by:

$$u_r^{(i)} = \sum_{j=1}^4 a_j^{(i)} D_j^{(i)} r^{n_j^{(i)}} \sin(2\theta)$$

$$u_r^{(N+1)} = \varepsilon_0 \frac{r_N}{4\mu_{12}^{\text{eff}}} \left(\frac{2r}{r_N} + D_3^{(N+1)} \frac{r_N^3}{r^3} + 2 \left(1 + \frac{\mu_{12}^{\text{eff}}}{K_{12}^{\text{eff}}} \right) D_4^{(N+1)} \frac{r_N}{r} \right) \sin(2\theta)$$

$$u_\theta^{(i)} = \sum_{j=1}^4 D_j^{(i)} r^{n_j^{(i)}} \cos(2\theta)$$

$$u_\theta^{(N+1)} = \varepsilon_0 \frac{r_N}{4\mu_{12}^{\text{eff}}} \left(\frac{2r}{r_N} - D_3^{(N+1)} \frac{r_N^3}{r^3} + \frac{\mu_{12}^{\text{eff}}}{K_{12}^{\text{eff}}} D_4^{(N+1)} \frac{r_N}{r} \right) \cos(2\theta)$$

For, $u_z^{(j)} = 0$, for $i = 1, \dots, N, j = 1, \dots, 4$,

Where,

$$\alpha_j^{(i)} = 2 \frac{C_{22}^{(i)} + C_{66}^{(i)} - n_j^{(i)} (C_{12}^{(i)} + C_{66}^{(i)})}{C_{22}^{(i)} + 4C_{66}^{(i)} - (n_j^{(i)})^2 C_{11}^{(i)}}$$

For, for $i = 1, \dots, N, j = 1, \dots, 4$.

The exponents $n_j^{(i)}, j = 1, \dots, 4$; are solutions to the polynomial: $An^4 - Bn^2 + \Gamma = 0$

Where,

$$A = C_{11}^{(i)} C_{66}^{(i)},$$

$$B = (C_{11}^{(i)} + C_{22}^{(i)} - 8C_{12}^{(i)}) C_{66}^{(i)} + 4(C_{11}^{(i)} C_{22}^{(i)} - (C_{12}^{(i)})^2),$$

$$\Gamma = 9C_{22}^{(i)} C_{66}^{(i)}.$$

Here; $n_1^{(i)}, n_2^{(i)}$ are positive solutions whereas, $n_3^{(i)}, n_4^{(i)}$ are the negative solutions. The layer symmetry axis is the same as the fiber axis, the displacement field thus, reduces to the expressions:

$$u_r^i = \left(B_1^i r + \left(\frac{\lambda_i}{3\mu_i + 2\lambda_i} \right) B_2^i r^3 - B_3^i r^{-3} \right. \\ \left. + \left(\frac{2\mu_i + \lambda_i}{\mu_i} \right) B_4^i r^{-1} \right) \sin(2\theta)$$

$$u_\theta^i = (B_1^i r + B_2^i r^3 + B_3^i r^{-3} + B_4^i r^{-1}) \\ \times \cos(2\theta)$$

$$u_z^i = 0$$

For, $r_{i-1} \leq r \leq r_i$. Where, B_k^i ($k = 1, \dots, 4$ and $i = 1, \dots, N$), are constants with and λ_i and μ_i are Lamé constants. For this document B_k^i are denoted as $D_k^{(i)}$ and can be determined from the boundary conditions shown in figure 28.

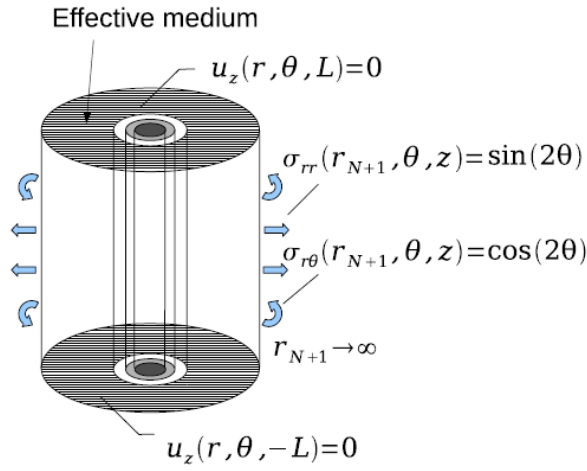


Figure 28: Christensen's generalized self-consistent method for transverse shear modulus

The external boundary condition at infinity ($r_N \rightarrow \infty$):

$$\sigma_{rr}^{N+1}(r_{N+1}, \theta) = \cos(2\theta),$$

$$\sigma_{r\theta}^{N+1}(r_{N+1}, \theta) = \sin(2\theta);$$

The condition for stress non-singularity at fiber center ($r=0$), it ensures that the displacements remain bounded at the origin: $D_3^{(1)} = D_4^{(1)} = 0$

The condition of continuity that enforces ideal bonding between phase layers:

$$u_r^{(i)}(r_i) = u_r^{(i+1)}(r_i), \quad u_\theta^{(i)}(r_i) = u_\theta^{(i+1)}(r_i),$$

$$\sigma_{rr}^{(i)}(r_i) = \sigma_{rr}^{(i+1)}(r_i), \quad \sigma_{r\theta}^{(i)}(r_i) = \sigma_{r\theta}^{(i+1)}(r_i), \quad \text{for } i = 1, \dots, N$$

The strain energy equivalency or surface energy is given by:

$$\int_0^{2\pi} \left[\sigma_{rr}^{(N+1)} u_r^{\text{eff}} + \sigma_{r\theta}^{(N+1)} u_\theta^{\text{eff}} - \left(\sigma_{rr}^{\text{eff}} u_r^{(N+1)} + \sigma_{r\theta}^{\text{eff}} u_\theta^{(N+1)} \right) \right]_{r=r_N} d\theta = 0$$

The displacement field of the effective medium is:

$$u_r^{\text{eff}} = \frac{r \sin(2\theta)}{2\mu_{12}^{\text{eff}}}, \quad u_\theta^{\text{eff}} = \frac{r \cos(2\theta)}{2\mu_{12}^{\text{eff}}}, \quad u_z^{\text{eff}} = 0$$

The strain/surface energy equivalency gives the condition, $D_4^{(N+1)} = 0$, from which μ_{12}^{eff} is solved.

3.4.5. Other Properties

The effective axial Poisson's ratio is given by:

$$\nu_{23}^{\text{eff}} = \sqrt{\frac{C_{33}^{\text{eff}} - E_3^{\text{eff}}}{4K_{12}^{\text{eff}}}}$$

The effective transverse Poisson's ratio is given by:

$$v_{12}^{\text{eff}} = \frac{K_{12}^{\text{eff}} - \mu_{12}^{\text{eff}} - 4(v_{23}^{\text{eff}})^2 \mu_{12}^{\text{eff}} K_{12}^{\text{eff}} / E_3^{\text{eff}}}{K_{12}^{\text{eff}} + \mu_{12}^{\text{eff}} + 4(v_{23}^{\text{eff}})^2 \mu_{12}^{\text{eff}} K_{12}^{\text{eff}} / E_3^{\text{eff}}}$$

The effective transverse Young's modulus is given by:

$$E_1^{\text{eff}} = \frac{4\mu_{12}^{\text{eff}} K_{12}^{\text{eff}}}{K_{12}^{\text{eff}} + \mu_{12}^{\text{eff}} + 4(v_{23}^{\text{eff}})^2 \mu_{12}^{\text{eff}} K_{12}^{\text{eff}} / E_3^{\text{eff}}}$$

4. COMPUTATIONAL MICROMECHANICS MODELING

A crucial issue in computational micromechanics is to take into account the real deformation and failure micromechanics of each phase in the simulations and this is done normally through the implementation of the appropriate constitutive equations.

In this section, a computational micromechanics model based on representative volume element (RVE) modeling is developed to investigate the failure mechanisms, stiffness matrices and elastic moduli of a uni-directional carbon fiber-reinforced polymer compared to a whiskered fiber embedded composite polymer. The whiskered fiber taken into consideration are carbon nanotubes (CNT). To validate the procedure, the findings are compared to those acquired by a numerical method.

4.1. Materials of the RVE

Two constitutive models were developed for the comparative study of a conventional fibrous carbon fiber-reinforced composite polymer (CFRP) and a CNT embedded composite polymer.

For the conventional composite polymer, Hexcel® IM7/8552 uni-directional composite model is used. In this composite, the base fiber is HexTow® IM7 carbon fiber which is continuous and can improve the interlaminar shear properties and structural properties, it is characterized by high tensile strength, shear strength and respective moduli. HexPly® 8552 is used as the epoxy resin for the composite, it is a versatile epoxy matrix that has high service temperatures (~250°F) and is impact tolerant. An IM7/8552 hybrid-ply laminate showed advantageous performance over a standard-ply laminate in compression after impact (CAI) tests [94]. A large increase in compression after impact strength was observed for quasi-isotropic hybrid laminates and there was also 13% weight savings. However, in the notched specimen test, the hybrid laminate had marginal improvements over the standard-ply laminate. The carbon fiber is assumed to be transversely isotropic.

For the CNT embedded composite polymer, the base fiber is HexTow® IM7 carbon fiber with 62% V_{CNT} . CNTs tend to agglomerate CNT volume fraction in HexPly® 8552 epoxy resin. As CNTs have small diameter and elastic modulus in the radial direction, they are easy to agglomerate in a polymer matrix [95] [96], however, the agglomeration of CNTs is not considered and it is assumed that the CNTs are uniformly dispersed. It is considered that the fiber is in the direction of the z -axis, and straight CNTs are aligned in the direction of y -axis where each CNT is modeled as a long fiber with transversely isotropic elastic properties. While the CNTs are considered to be isotropic, straight CNTs behave as transversely isotropic effective media [97].

Material	Poisson's Ratio (ν)	Young's Modulus (E) [GPa]	Coefficient of Thermal Expansion (CTE) [1/K]	Density (ρ) [g/cm ³]
IM7 Carbon Fiber	0.28	276	-6.3x10 ⁻⁷	1.78
Epoxy 8552	0.37	5.1	5.2x10 ⁻⁷	1.301
Carbon CNT	0.14	1100	1x10 ⁻⁵	2.267

Table 1: Mechanical properties of composite constituents [98] [99] [100]

	$E_1=E_2$	E_3	$\nu_{12}=\nu_{21}$	$\nu_{31}=\nu_{32}$	$\nu_{13}=\nu_{23}$	G_{12}	G_{31}
IM7 Carbon Fiber	22.4 GPa	250 GPa	0.35	0.3	0.027	8.3 GPa	22.1 GPa

Table 2: Properties for modeling [101]

The carbon nanotube constructed is an armchair type as shown in figure 30. Armchair CNTs are metallic and are characterized by excellent electrical properties, exhibiting ballistic conduction even at room temperature as they have no gaps between valence and conduction bands [102].

For this work, the following dimensions, volume fraction and packing density of the CNTs have been considered:

Diameter (d_b) [nm]	Length (l_b) [μ m]	Thickness (t_b) [nm]	Chiral Index (n, m)	M_b^{max}	M_b	Packing Density (ρ_b)	Volume Fraction (V_{CNT})
1.2	1	0.38	(9,9)	13607	12246	90%	53%

Table 3: CNT dimensional characteristics

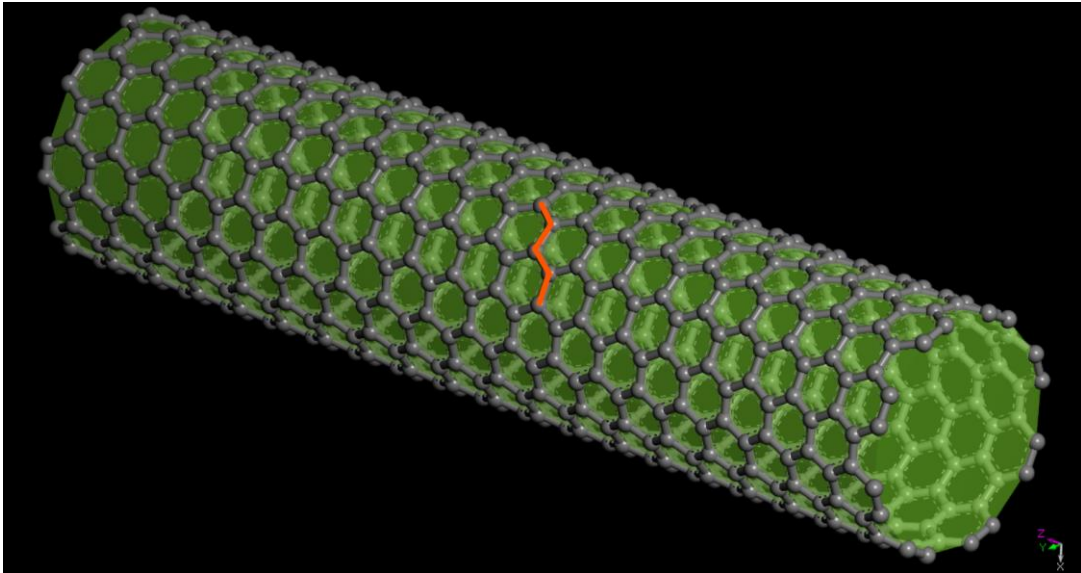


Figure 29: SWCNT isometric view

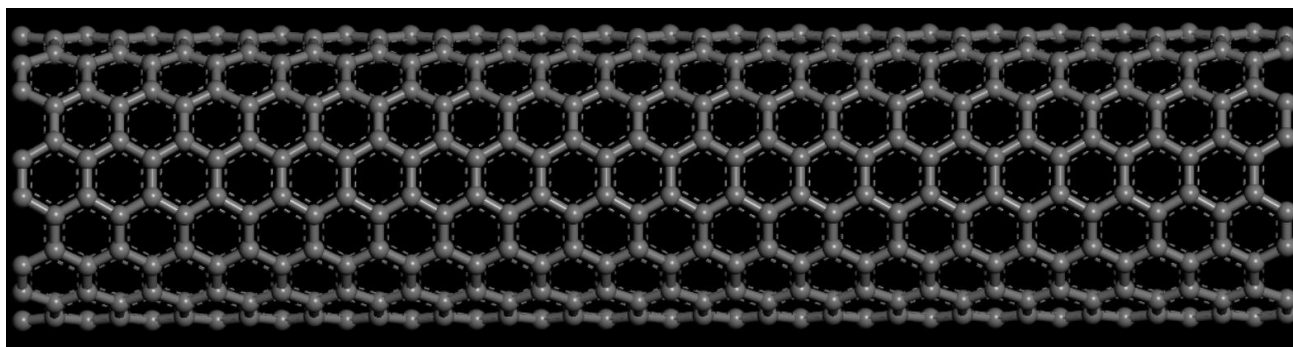


Figure 30: SWCNT length = 1 μm



Figure 31: SWCNT diameter = 1.2 nm

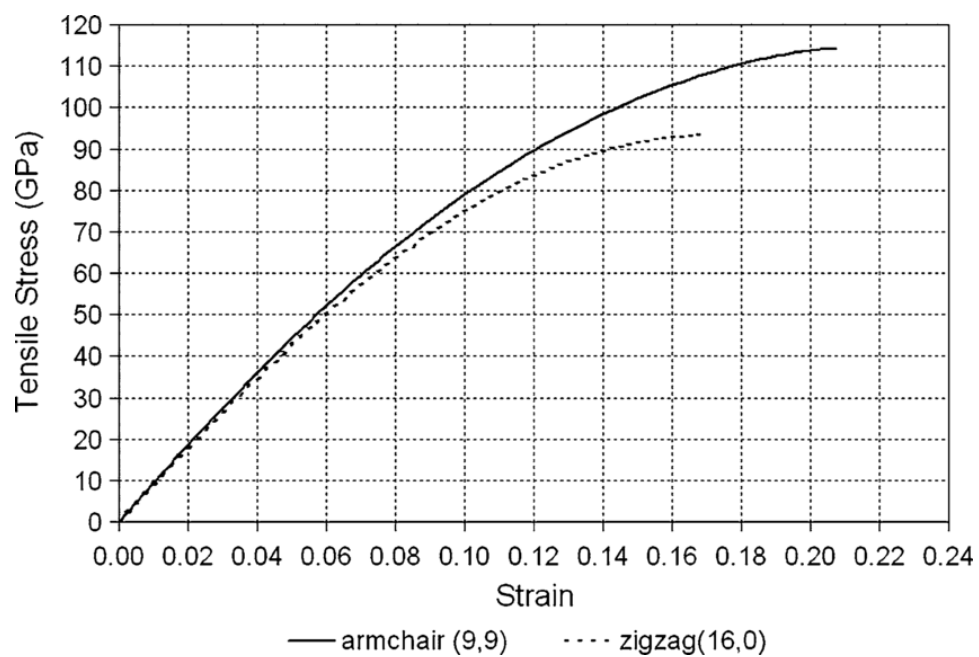


Figure 32: Stress-Strain curve for armchair CNT (9,9) [103]

4.2. Multiscale Approach for Effective Properties

The whiskered fiber embedded composite is considered as a three-phase medium consisting of homogenous fiber, matrix and a heterogenous interphase [97]. The heterogenous interphase consists of the CNTs and the matrix material. A multiscale approach for such a medium has two steps. In the first homogenization step, the interphase layer is treated as a homogenized coating layer attached to the main fibers. In the second step, the mesoscale is taken into consideration which consists of all the three phases and homogenization at this step gives the overall properties of the whiskered fiber embedded composite.

Considering a multiscale approach allows us to assume the unit cell as a mesoscopic material point. Therefore, at the mesoscopic scale, the heterogenous medium is replaced by an equivalent homogenized medium, that has an equivalent mesoscopic response. It is necessary to link the microscale and mesoscale equations. Similar to homogenization, the macroscopic stress and strain fields can be identified through volume averaging of their microscopic counterparts over the unit cell. The Hill-Mendel lemma allows for the verification of compliance of mesoscopic quantities with relevant microscopic quantities in accordance to the micro-meso energy equivalence. The theorem also allows the determination of boundary conditions compatible with the micro-meso energy equivalence [104]. Thus, according to the Hill-Mendel lemma, the mesoscopic mechanical work is equal to the volume average of its microscopic field within the unit cell.

4.2.1. Microscale Modeling of CNT Interphase

A strain driven approach is adopted wherein, a unique solution of the displacement field is found for a given mesoscopic strain in the unit cell while adopting the required boundary conditions, subsequently the mesoscopic stress is determined.

Figure 33 shows the microstructure of a CNT embedded composite. The CNT interphase shows a quasi-cylindrically orthotropic behavior, in a manner that it is transversely isotropic with the axis of symmetry parallel to the radial direction as shown in figure 34.

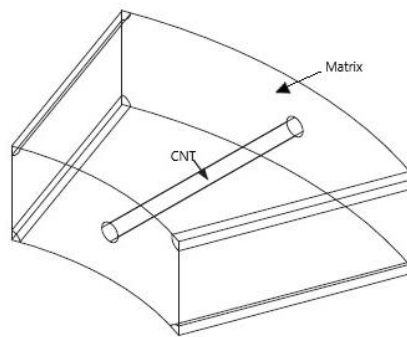


Figure 33: CNT alignment in matrix

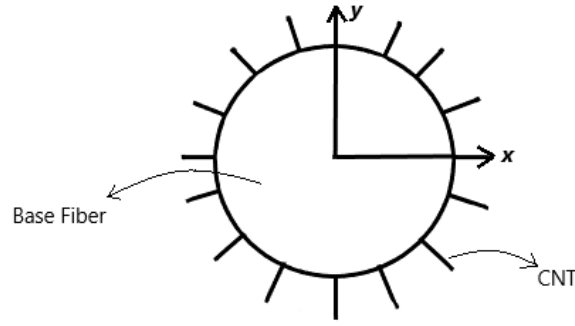


Figure 34: Cartesian orientation of CNTs

The interphase thickness is calculated to be $0.75\mu\text{m}$ (interphase thickness equation). Uneven dispersion of high-volume fraction of CNTs in the interphase has been found to leading to decreased effective properties of the overall composite, therefore, for the present study even dispersion of CNTs in the matrix has been assumed. For computational purposes, a square array packing of CNTs is considered.

The elasticity tensor for the interphase layer is given as:

$$\mathbf{L}_2^{\text{cyl}} = \begin{bmatrix} L_2^{rr} & L_2^{r\theta} & L_2^{rz} & 0 & 0 & 0 \\ L_2^{r\theta} & L_2^{\theta\theta} & L_2^{\theta z} & 0 & 0 & 0 \\ L_2^{rz} & L_2^{\theta z} & L_2^{zz} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu_2^{r\theta} & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu_2^{rz} & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu_2^{\theta z} \end{bmatrix}.$$

4.2.2. Mesoscale Modeling of CNT Composite

The solution of the mesoscopic problem is treated as a mechanical problem in where, for each mesoscopic material point, the stress-strain relationship is obtained from the solution of the unit cell. After homogenization at microscale, the CNTs and the surrounding matrix is substituted as an equivalent nanocomposite.

4.3. Finite Element Model RVE

IM7 carbon fiber for both the models has a diameter of $5.2\mu\text{m}$ with 62% fiber volume fraction [105]. The 8552-epoxy matrix is assumed to be elastic and isotropic with 38% resin volume fraction. For the CNT embedded composite the CNT volume fraction is calculated to be 53% in interphase.

The carbon fibers are coated with radially aligned hollow carbon nanotubes. The SWCNTs have properties of graphite and the interphase is assumed to behave like a reinforced interphase with aligned CNTs and its effective properties are obtained with the aforementioned generalized self-consistent composite cylinders method. It is assumed that there is a perfect bond between CNTs and resin and CNTs and fiber, the imperfect bonding that leads to a different composite behavior

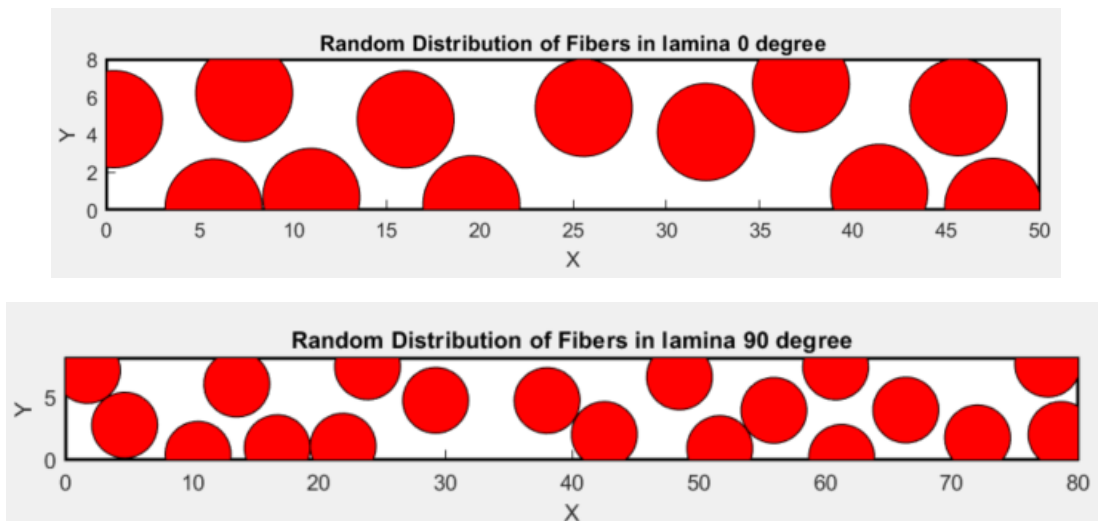
have not been considered. It is presumed that the CNTs contain no defects or residual catalysts and are straight.

4.3.1. Monte Carlo Method for Fiber Distribution

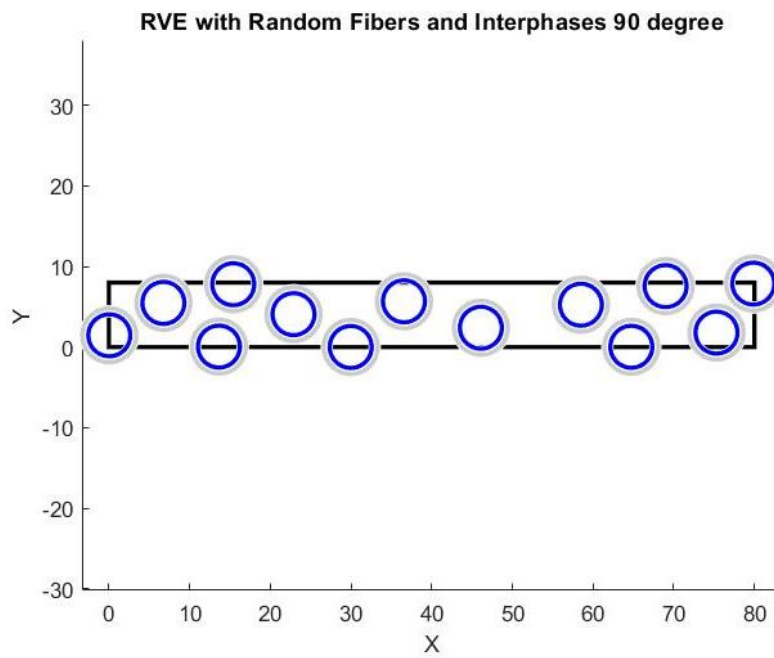
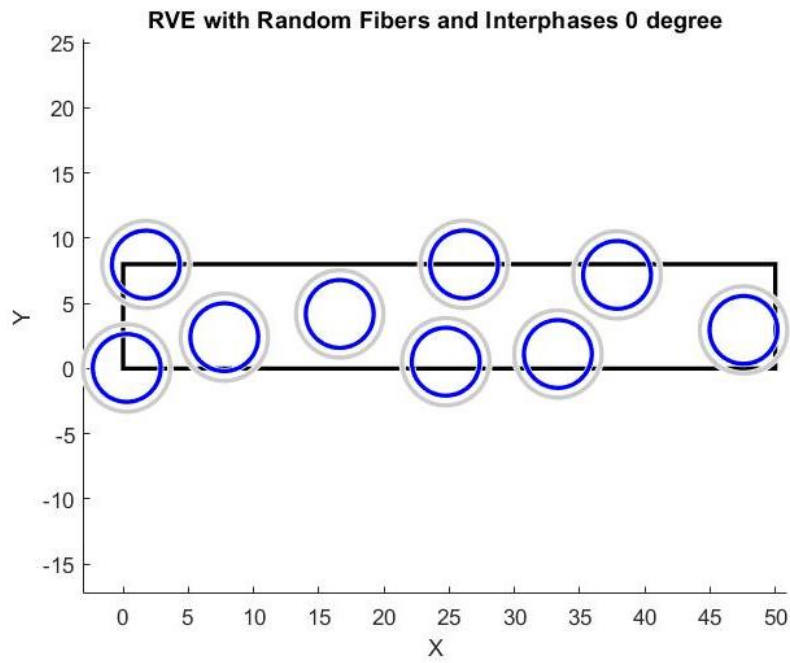
A Monte Carlo method is used to estimate the deterministic outputs of the randomness in the input. This method is effective for estimating the fiber distribution in an RVE. This approach entails creating random samples to replicate the placement and orientation of fibers in the RVE, this allows for the assessment of their spatial distribution and statistical representation of the material's microstructure.

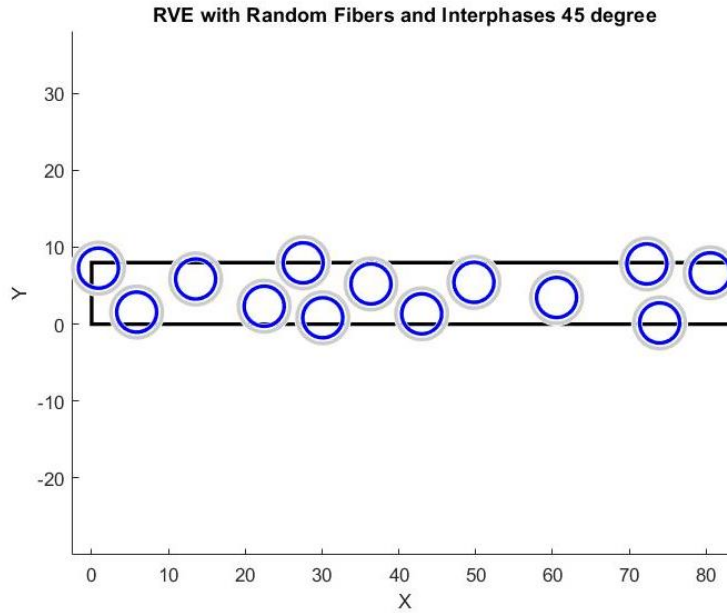
The Monte Carlo styled script was used in MATLAB. For the conventional laminae the fiber radius was taken as $2.6\mu\text{m}$, corresponding to the diameter $5.2\mu\text{m}$ of IM7 carbon fiber. As for the CNT reinforced polymer laminae, the fiber radius was taken as $3.35\mu\text{m}$ corresponding to the $6.69\mu\text{m}$ diameter of fiber and interphase. Volume fraction of carbon fiber for both the cases was 62%.

4.3.1.1. Conventional Lamina



4.3.1.2. Lamina with CNTs





4.3.2. Uni-Directional Unit Cell

In simulations, the minimum size of the RVE is crucial as it should contain all the important statistical information about the microstructure. It must additionally be large enough to ensure that the average properties of the volume element remain independent of its size and position within the material.

The RVE is a cuboidal three-dimensional section of a lamina with homogenous fiber distribution. The dimensions of the conventional fiber unit cell is $8.27 \times 8.27 \times 10 \mu\text{m}$ proportional to the fiber diameter of $5.2 \mu\text{m}$. Whereas, the CNT reinforced carbon fiber unit cell has the dimensions $11.48 \times 11.48 \times 10 \mu\text{m}$ proportional to the fiber diameter $5.2 \mu\text{m}$ and interphase thickness of $0.75 \mu\text{m}$.

Periodicity of unit cell has been implemented, wherein the displacement field of any point in the unit cell is the sum of the affine part and the periodic fluctuation. Periodic boundary conditions have been introduced through constraint equations between periodic nodes, requiring for periodic meshing of the unit cell. In the simulations, macro strains are imposed on the periodic RVEs. The macrostrains are imposed as a linear displacement.

For the conventional fiber unit cell, matrix and fiber both have been meshed as first order hexahedral elements under reduced integration, C3D8R. For CNT reinforced fiber unit cell, the matrix, interphase and fiber have been also meshed as first order hexahedral elements under reduced integration C3D8R.

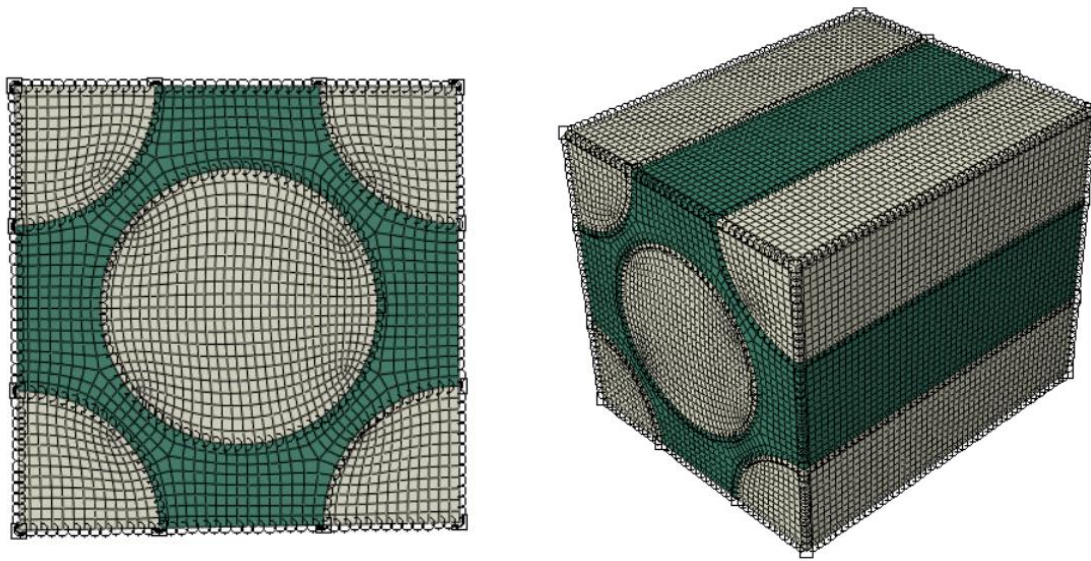


Figure 35: Conventional unit cell

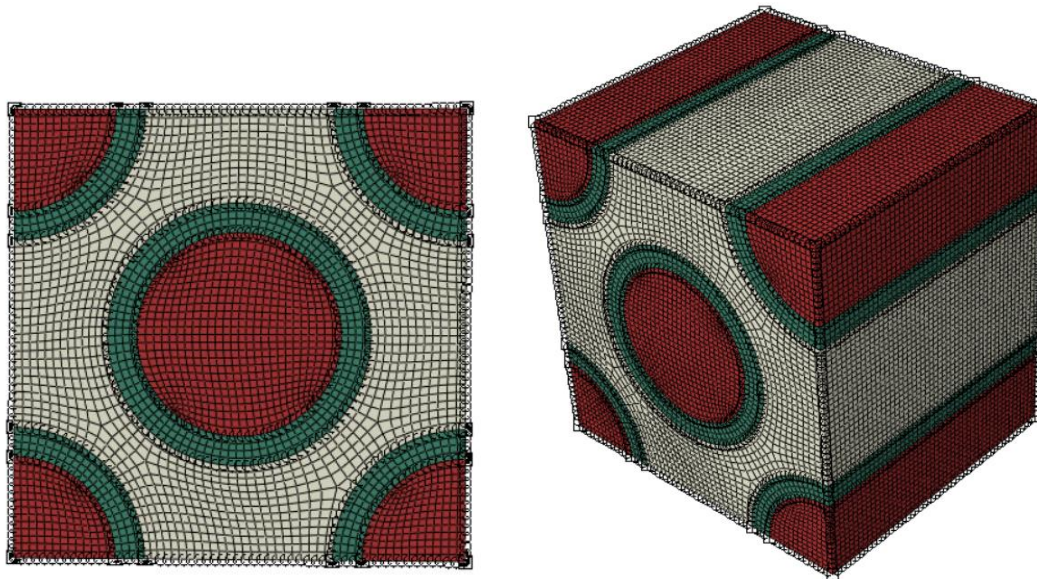


Figure 36: CNT reinforced interphase unit cell

4.3.3. Uni-Directional Lamina

To determine the properties of CNT reinforced composite material, the individual laminae of the quasi-isotropic composite layup were modelled. For application of the composite in wing skin, spars and ribs a layup orientation of $[0^\circ/90^\circ/45^\circ/-45^\circ]$ has been chosen. A quasi-isotropic layup has previously proven to offer several advantages over other configurations. Such a layup provides a good balance stiffness and strength in all in-plane directions. The need for a lamina analysis stems from the fact that the properties of a composite material can significantly change with different lamina orientations in the layup. In order to estimate the properties of the material efficiently it is important to do a lamina-by-lamina analysis. The distribution of fibers is as per the Monte Carlo plot.

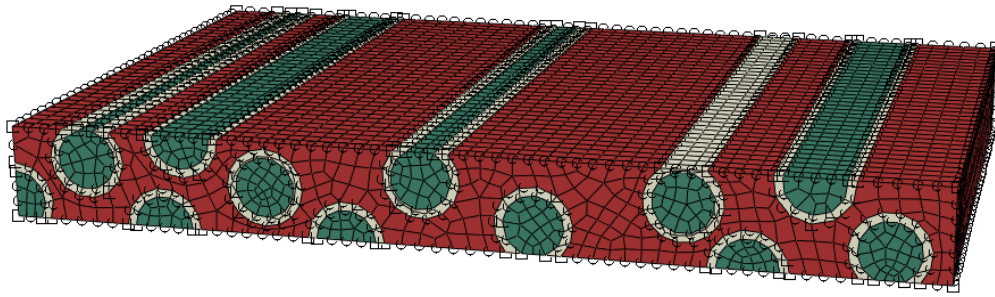


Figure 37: 90-degree lamina with CNT reinforced interphase

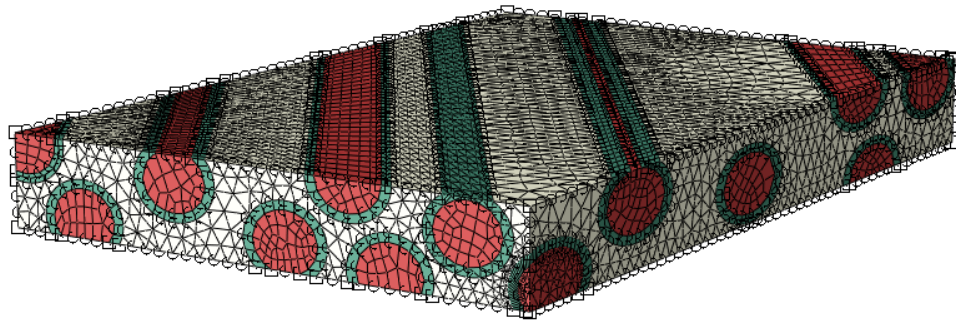


Figure 38: 45-degree lamina with CNT reinforced interphase

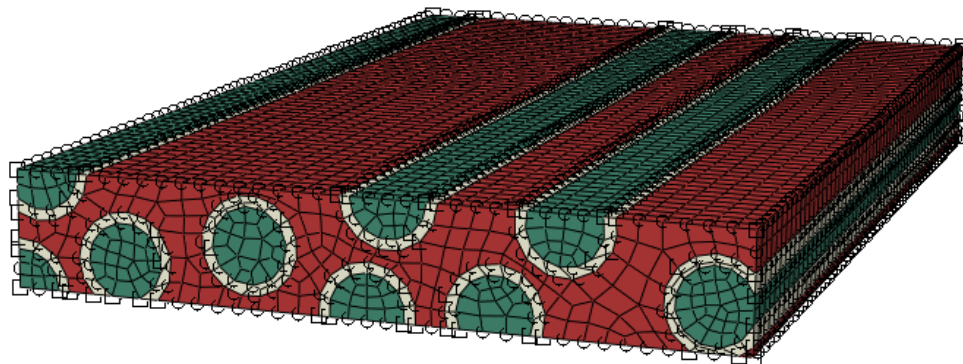


Figure 39: 0-degree lamina with CNT reinforced interphase

4.3.4. Bi-Directional Laminate

Although, theoretically, CNTs strengthen a given composite, the properties of constituent materials suffer when amalgamated in a polymer composite. Hence, to evaluate the effective properties of a composite material a laminate needs to be modelled and analyzed.

For ease computation, a four-ply bi-directional laminate, with $[0^\circ/90^\circ/90^\circ/0^\circ]$ layup, has been modelled with linear placement of fibers following the fiber population generated through the Monte Carlo method. The dimensions of both the RVE, with and without CNT interphase, are

8x50x80 μm . The RVE are subjected to suitable strains for tensile and shear cases with 10% value of the respective dimension.

Periodic boundary conditions through appropriate constraint equations are implemented, resulting in a periodic hex-dominated mesh. The effective elastic properties of the composite material are derived from the global stiffness matrix of the CNT reinforced bi-directional laminate.

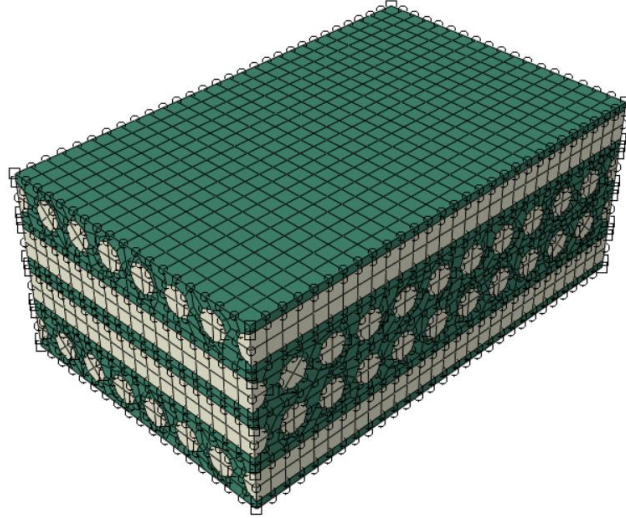


Figure 40: Bi-directional conventional laminate

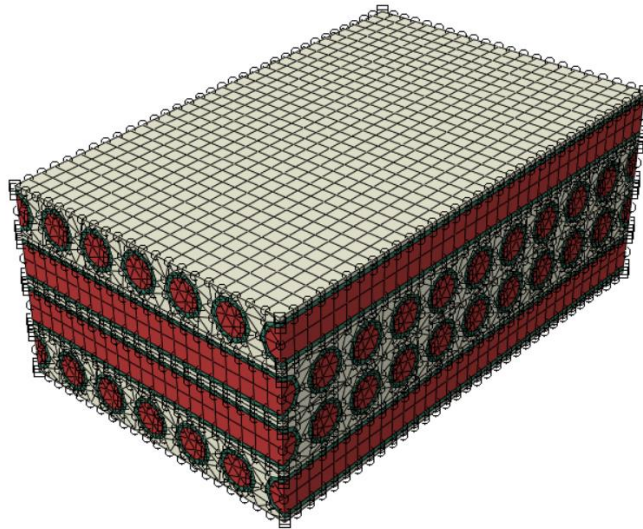


Figure 41: Bi-directional CNT reinforced laminate

5. AIRFRAME COMPONENT MODELING

Several special structural considerations must be considered while designing and manufacturing airplane wings. High strength and lightweight are the two key functional requirements to consider when selecting materials for an airplane wing. distinct materials used to create wings will exhibit distinct structural properties. The lifting surface is an airplane's most crucial component since it is the primary assembly responsible for generating lift. The wing not only ensures flight stability but also serves as a facility to support the strategic operating unit.

The failure of an airplane structural component can have devastating consequences, causing damage to both the lifecycle and the aircraft. The investigation of the structural behavior of aircraft wing components, and hence catastrophes in airliner structures, is of critical importance in preventing future occurrences. Fatal occurrences in the aviation business are nothing new to consider, as there are several examples in aircraft history that demonstrate catastrophic and fatal consequences. In years past, the majority of structural disasters investigated have been in metal components, mirroring the prevalence of metal structures in airplanes. Nonetheless, since the mid-1980s, an increasing number of aircraft manufacturers have used fiber-reinforced polymer amalgams as structural components. Thus, structural failures caused by metallic structures in airplanes can be minimized. As a result, in order to deal with failures and fatal accidents caused by aircraft wing components, the structural behavior and health of three-dimensional wings must be evaluated. While assisting in the investigation of mishaps in civil airplanes, there are numerous instances of wing section catastrophes and faults that were discovered before a misfortune occurred.

5.1. Aircraft Prototype

In the 1980s, Airbus introduced the A320, the most efficient single-aisle aircraft in aviation history. It is the company's third airliner, following the A300 and A310, and the world's most popular narrow-body aircraft. As of 2024, 11,471 A320s have been built. The A320 aircraft family pioneered the use of digital fly-by-wire and side-stick flight controls in airliners. The aircraft employs two CFM International CFM56 engines, which are high-bypass turbofan engines providing a thrust range of 82-150 kN.



Figure 42: Airbus A320

Cockpit Crew	2
Length, [m]	37.57
Wingspan, [m]	34.10
Wingsweep	25°
Height, [m]	11.76
Maximum Take-Off Weight, [Kg]	78,000
Fuel Capacity, [L]	27,200
Cruise Speed, [Km/hr]	829 (Mach 0.78)
Range, [Km]	6,100
Ceiling, [ft]	41,000
Engines (x2)	CFM56

Table 4: A320 Parameters

The aircraft is considered to be a low-wing cantilever monoplanes with a conventional empennage. The airframe of the aircraft includes composite materials and aluminium alloys to save weight and reduce the total number of parts to decrease the maintenance costs. The tail assembly is made almost entirely of composites, as are the elevators.

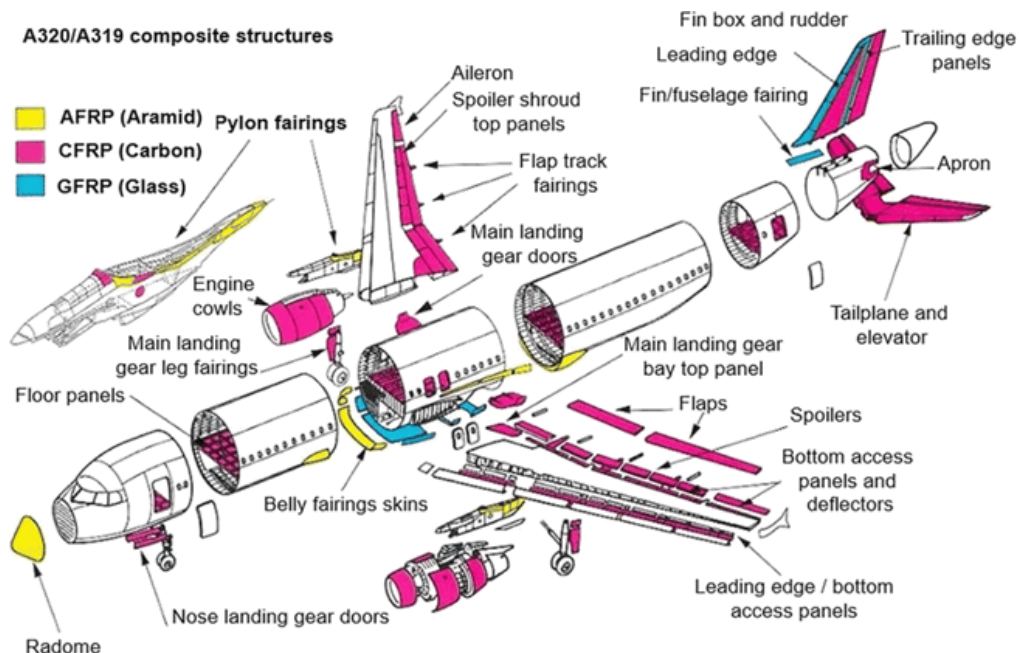


Figure 43: Airbus A320 material distribution

5.2. Establishing Wingbox Model

A wing of Airbus A32 has been considered for the design of the wingbox. Correspondingly, the semi-wingspan has been modelled as the wing component is doubly symmetrical. The planform is defined using root chord length, taper ratio, wingspan, leading-edge sweep angle, kink chord length, tip chord length.

The aircraft employs a NACA 23015 airfoil which has a maximum thickness of 15% at 30% chord and maximum camber of 1.8% at 15% chord length. This airfoil design has high lift characteristics in subsonic speed and is highly suitable for the mission profile of the aircraft.

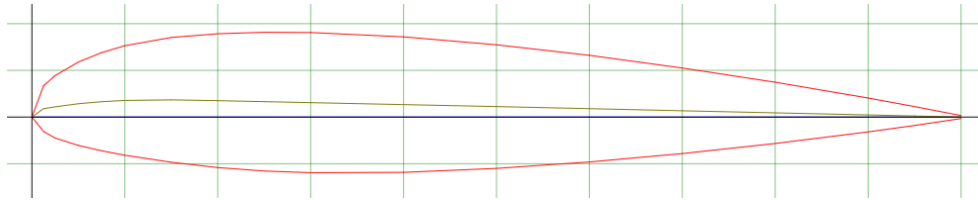


Figure 45: NACA 23015

Following the wing model showcased in figure 44, the wingbox is constructed with 27 ribs and 2 spars. For ease of modelling, the angles between ribs are not considered and the ribs are all modelled parallel.

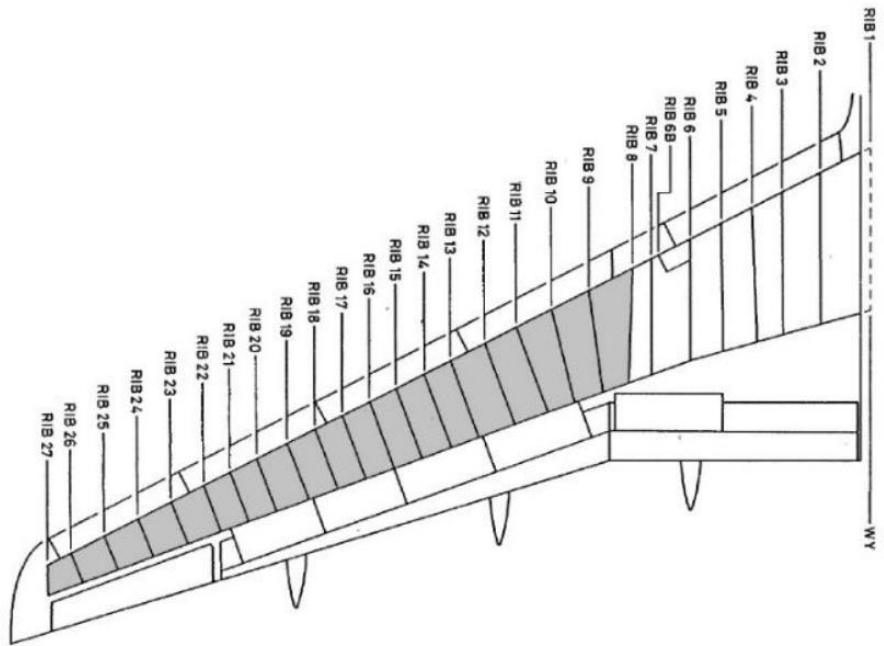


Figure 46: Wing model

$$\text{Wing Loading} = \frac{\text{MTOW [Kg]}}{\text{Wing Area [m}^2\text{]}} = \frac{78,000}{122.6} = 636.22 \text{ kg/m}^2$$

5.2.1. Geometry and Materials

The characteristic dimensions mentioned below have been used to model the wing box. The ribs have been modelled parallel to each other. The root ribs, kink rib, tip rib and rib no.6 have been modelled as reinforced ribs, rest are typical ribs.

Wing semi span, [m]			15.08
Airfoil			NACA 23015
Sweep angle			25°
Skin thickness, [m]			0.0018
Spars		Front	I shape
		Rear	C shape
Ribs	Chord length, [m]	Root	6.07
		Kink	4.53
		Tip	1.5
	Thickness, [m]	Root	0.25
		Kink	0.25
		Tip	0.2

Table 6: Wing components dimensions used for modelling

$$\text{Wing Loading} = \frac{MTOW [Kg]}{\text{Wing Area} [m^2]} = 636.22 \text{ kg/m}^2$$

$$\text{Pressure load} = \frac{L}{A} \cdot (\text{Load Factor}) = 11.99 \text{ kPa}$$

$$\text{Applied Load} = (\text{Pressure load}) \cdot (FoS) = 17.985 \text{ kPa}$$

Wing Area, [m ²]	122.6
Lift Force, [kN]	490
Load Factor	3
Factor of Safety	1.5

Table 7: Considered values for applied load calculation

The coefficient of lift calculated for the aircraft was 0.289. This calculation was based on cruise condition values given in the A320 flight manual. The altitude considered was 29,000ft, thus FL290. Corresponding to the altitude the air density was taken as 0.475 kg/m³ and speed of sound as 304.5 m/s.

CRUISE - M.78										
MAX. CRUISE THRUST LIMITS NORMAL AIR CONDITIONING ANTI-ICING OFF					ISA CG = 33.0%		EPR KG/H/ENG NM/1000KG		MACH IAS (KT) TAS (KT)	
WEIGHT (1000KG)	FL290		FL310		FL330		FL350		FL370	
50	1.217	.780	1.223	.780	1.232	.780	1.246	.780	1.265	.780
	1280	302	1177	289	1087	277	1019	264	969	252
	180.3	462	194.4	458	208.7	454	220.5	450	230.7	447
FL390		1.293	.780	1.293	.780	1.293	.780	1.293	.780	1.293
		930	241	930	241	930	241	930	241	930
		240.5	447	240.5	447	240.5	447	240.5	447	240.5

4.1.1.1. Standard Lift Equation

$$\text{Weight} = mg = \frac{1}{2} \rho S (TAS)^2 C_L \quad (1)$$

With m = Aircraft mass
 g = Gravitational acceleration
 ρ = Air density
 S = Wing area
 C_L = Lift coefficient

The lift coefficient, C_L , is a function of the angle of attack (α), the Mach number (M), and the aircraft configuration.

Figure 47: Excerpt from A320 flight manual

The finite element analysis of the wing box has been done using the below mentioned polymer composites for a comparative study. The wing box has been fixed on the root rib, giving it a degree of freedom of 0, as it would be for a cantilever beam.

The calculated pressure load is 11.99 kPa, multiply this load with a regulated designing factor of safety, 1.5, the applied load is thus obtained. The wing box has been subjected to half of this load, 8.992 kPa, on the bottom skin panel, mimicking the lift load on one wing in flight.

	E_{11} [GPa]	E_{22} [GPa]	E_{33} [GPa]	G_{23} [GPa]	G_{12} [GPa]	G_{13} [GPa]	ν_{13}	ν_{12}	ν_{23}
Conventional IM7/8552	73.1	73.1	10.6	4.1	4.8	4.1	0.04	0.35	0.04
CNT reinforced IM7/8552	52.14	52.88	13.97	6.37	4.06	76.47	0.05	0.42	0.05

Table 8: Effective properties of quasi-isotropic composites for comparative study

5.2.2. Modelling and FEA

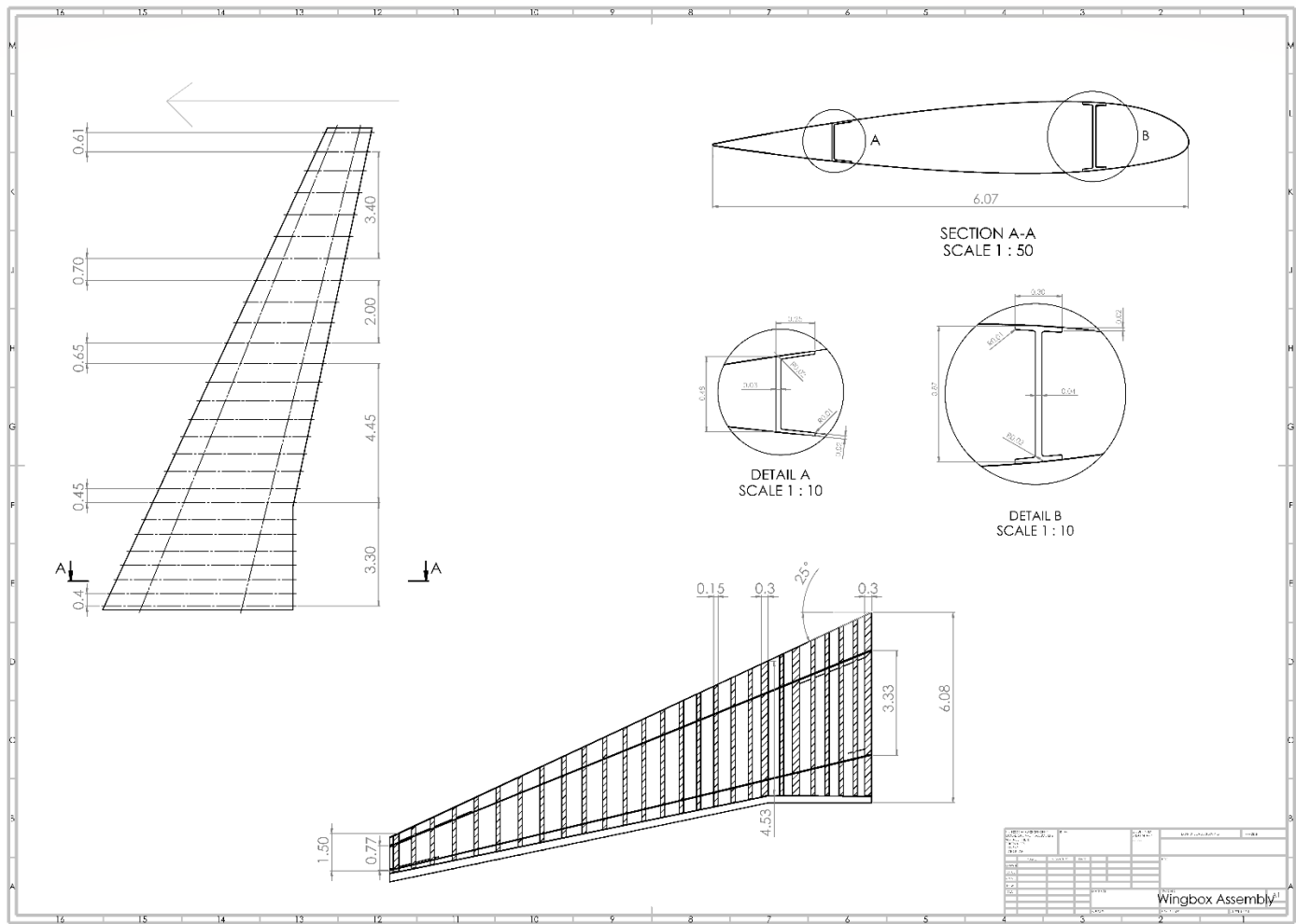


Figure 48: 2D Draft of wing box geometry

Owing to low computerization power, only 6 ribs have been modelled with equivalent distancing to imitate the load bearing capacity of the wing. The dimensions of root rib, kink rib and tip rib have been retained. The semi wingspan has also been maintained.

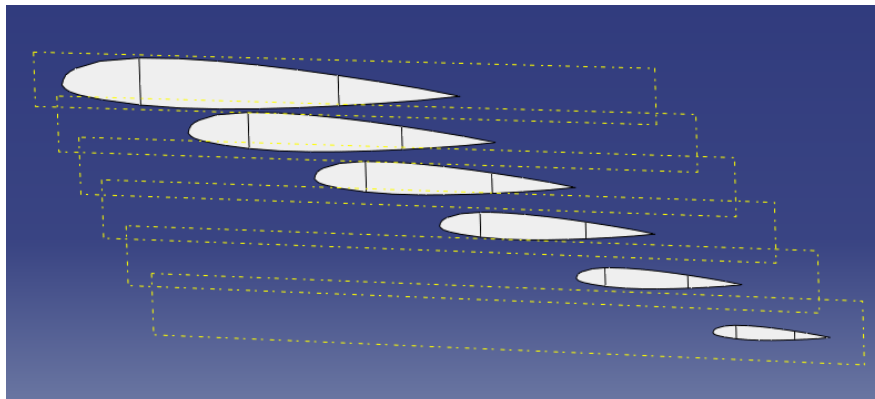


Figure 49: 6 equivalent ribs for FEA

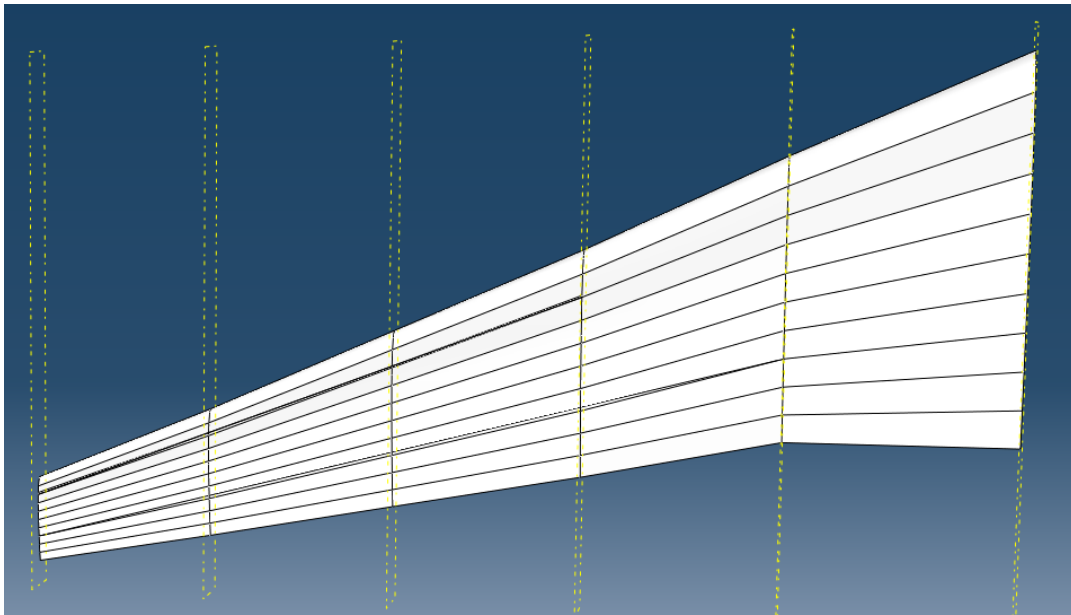


Figure 50: Wingbox model in ABAQUS

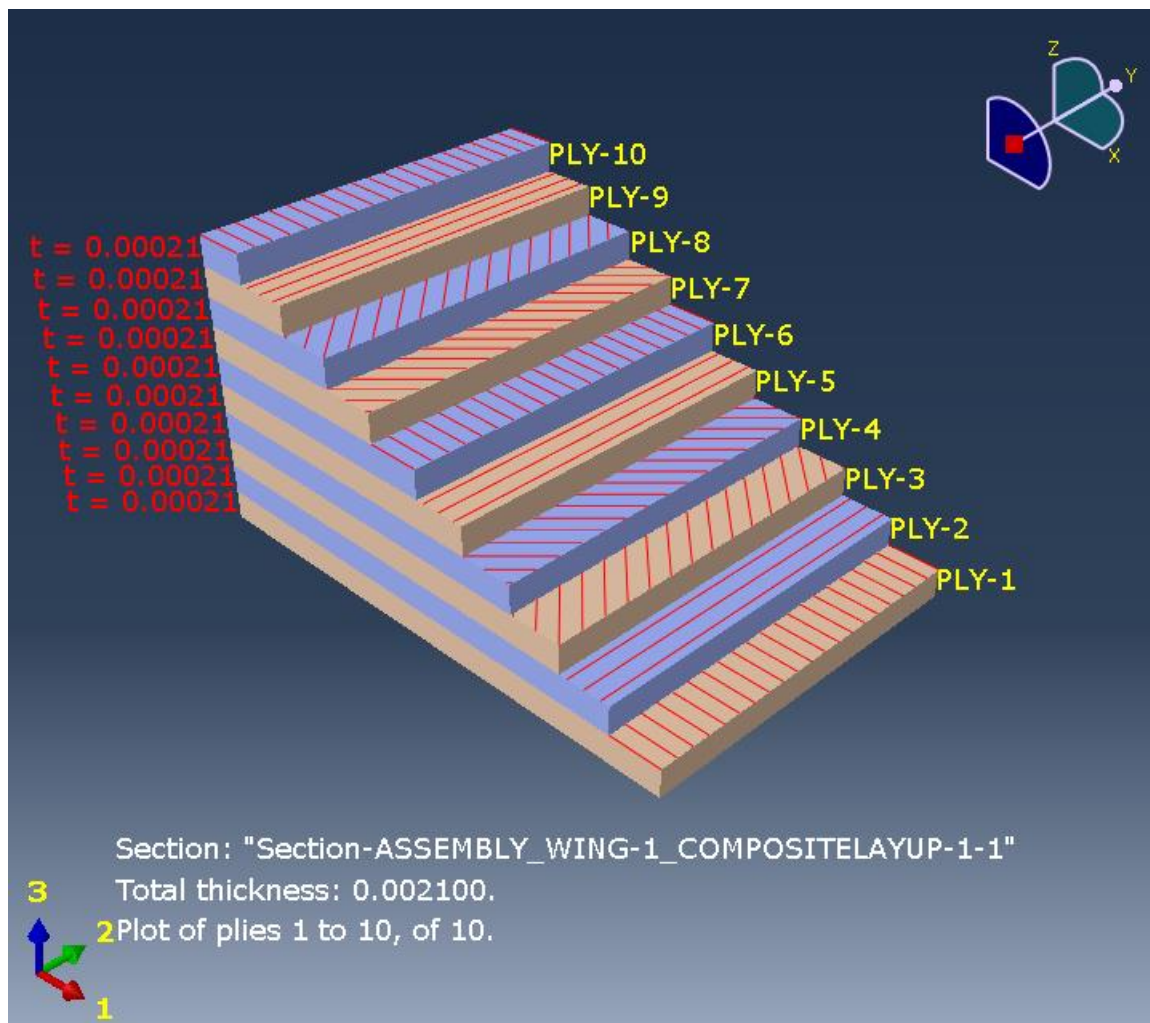


Figure 51: Ply stack plot

6. RESULTS APPENDIX

6.1. Properties Estimation

The element stiffness matrix of the interphase in an RVE under uniaxial loading in z-direction was generated and the interphase compliance matrix was calculated.

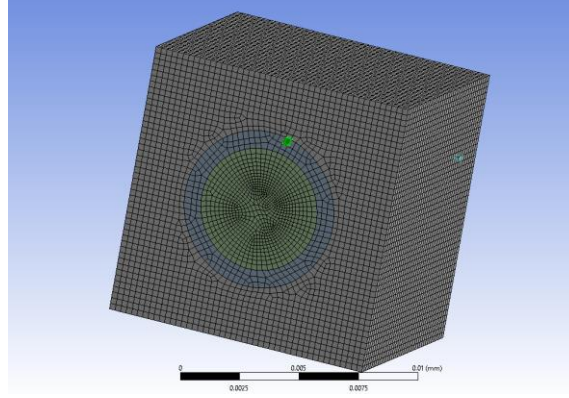


Figure 52: Unit cell RVE for stiffness matrix determination

$$C^{-1} = \begin{pmatrix} 4.625 \times 10^{-12} & -7.397 \times 10^{-13} & -9.639 \times 10^{-13} & 0 & 0 & 0 \\ -7.397 \times 10^{-13} & 4.625 \times 10^{-12} & -9.639 \times 10^{-13} & 0 & 0 & 0 \\ -9.639 \times 10^{-13} & -9.639 \times 10^{-13} & 2.143 \times 10^{-12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 5.681 \times 10^{-12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 5.681 \times 10^{-12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.476 \times 10^{-11} \end{pmatrix}$$

From this compliance matrix, using Christensen's composite cylinders generalized self-consistent method, the effective elastic properties of the interphase were calculated.

CNT reinforced interphase	E_{33}^{eff} [GPa]	$E_{11}^{eff} = E_{22}^{eff}$ [GPa]	$G_{13}^{eff} = G_{23}^{eff}$ [GPa]	G_{12}^{eff} [GPa]	ν_{12}^{eff}	$\nu_{13}^{eff} = \nu_{23}^{eff}$	K_{eff} [GPa]	ρ [g/cm ³]
	467	216	176	67.75	0.16	0.45	14.021	1.813

Table 9: Effective properties of CNT reinforced interphase

Laminae of orientation angles 0°, 90° and 45° were individually analyzed to evaluate the properties of the quasi-isotropic laminate composite.

Lamina 0°	E_{33} [GPa]	$E_{22} = E_{11}$ [GPa]	G_{23} [GPa]	$G_{12} = G_{13}$ [GPa]	ν_{23}	$\nu_{12} = \nu_{13}$
	23.96	14.45	12.32	6.59	-0.028	0.1

Lamina 90°	E ₁₁ [GPa]	E ₂₂ =E ₃₃ [GPa]	G ₁₂ [GPa]	G ₂₃ =G ₁₃ [GPa]	ν ₁₂	ν ₂₃ =ν ₁₃
	28.69	12.19	12.47	6.92	0.15	-0.12

Lamina 45°	E ₃₃ [GPa]	E ₁₁ [GPa]	E ₂₂ [GPa]	G ₂₃ [GPa]	G ₁₂ [GPa]	G ₁₃ [GPa]
	13.25	12	8.13	9.82	8.3	7.86
	ν ₁₂	ν ₂₃	ν ₁₃			
	-0.28	-0.3	-0.4			

The stiffness matrix of a four-ply composite layup made of CNT reinforced carbon fiber reinforced polymer was generated. Inverse of the matrix, or the compliance matrix, was found.

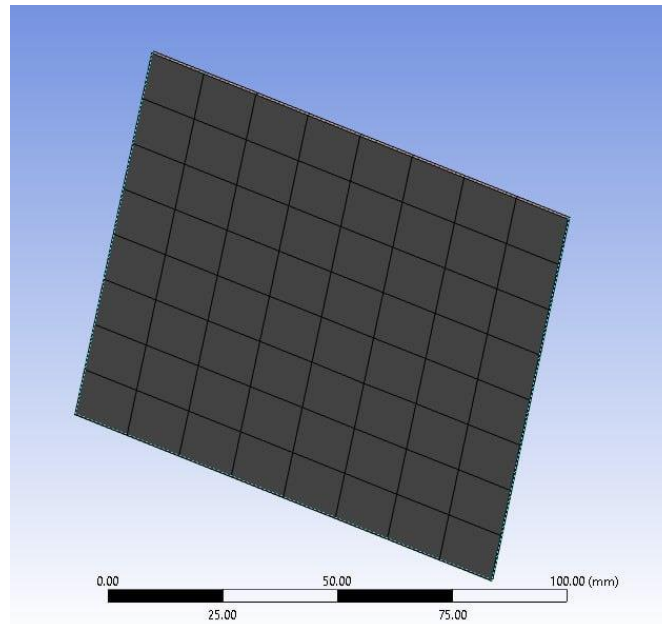


Figure 53: Composite plank for stiffness matrix determination

$$C^{-1} = \begin{pmatrix} 1.918 \times 10^{-5} & -8.421 \times 10^{-6} & -9.517 \times 10^{-7} & 0 & 0 & 0 \\ -8.421 \times 10^{-6} & 1.891 \times 10^{-5} & -9.414 \times 10^{-7} & 0 & 0 & 0 \\ -9.517 \times 10^{-7} & -9.414 \times 10^{-7} & 7.159 \times 10^{-5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 7.351 \times 10^{-5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 7.729 \times 10^{-5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.231 \times 10^{-3} \end{pmatrix}$$

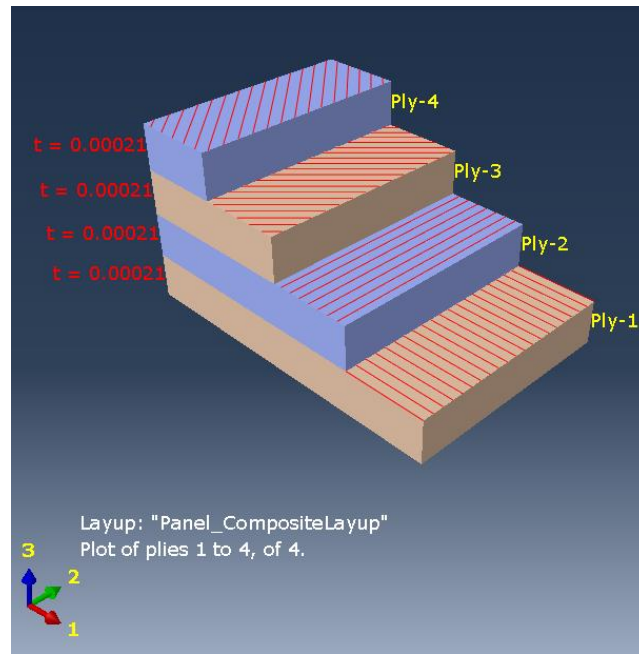


Figure 54 Ply stack plot of plank

Using this matrix and following classical laminate theory, the elastic constants of the composite laminate were calculated.

	E_{11} [GPa]	E_{22} [GPa]	E_{33} [GPa]	G_{23} [GPa]	G_{12} [GPa]	G_{13} [GPa]	ν_{13}	ν_{12}	ν_{23}
CNT reinforced IM7/8552	52.14	52.88	13.97	6.37	4.06	6.47	0.05	0.42	0.05

Table 10: CNT reinforced composite's engineering constants

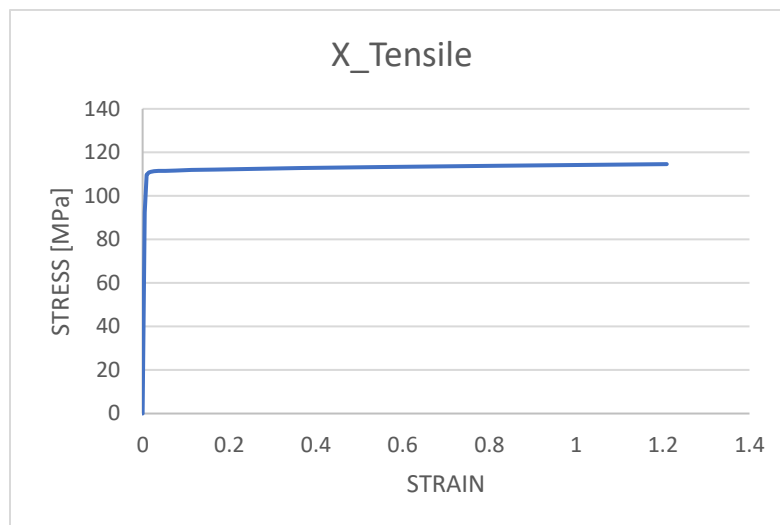
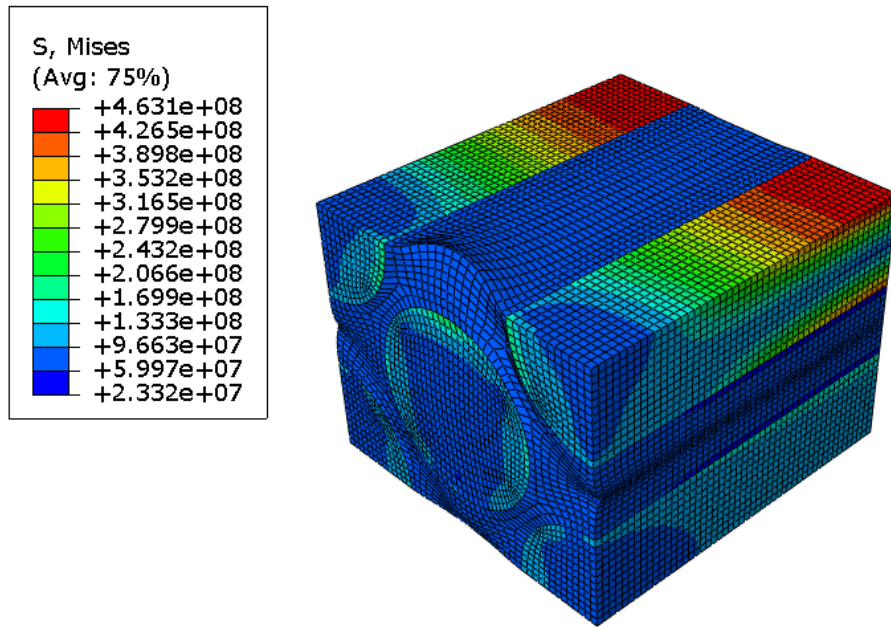
6.2. FEA of Uni-Directional Unit Cell

6.2.1. Unit Cell with Conventional Fiber

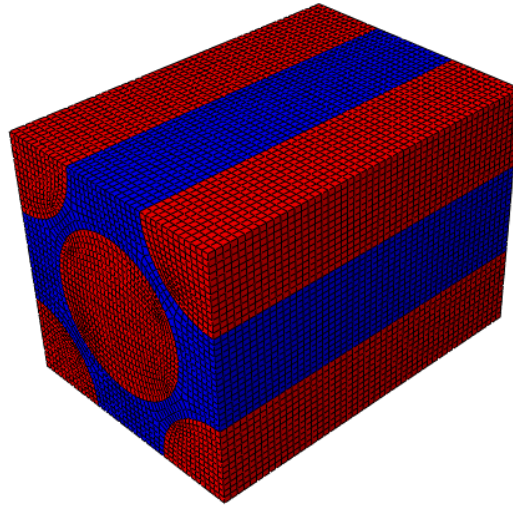
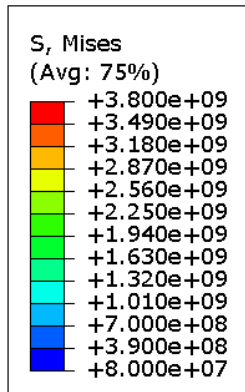
	X_Tensile	Z_Tensile	XY_Shear	YZ_Shear
Elasticity Modulus [GPa]	11.64	22.34	13.76	12.87

Table 11: Calculated Young's and Shear modulus for conventional unit cell

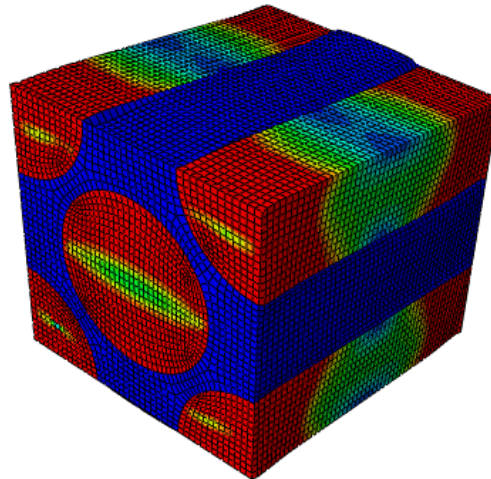
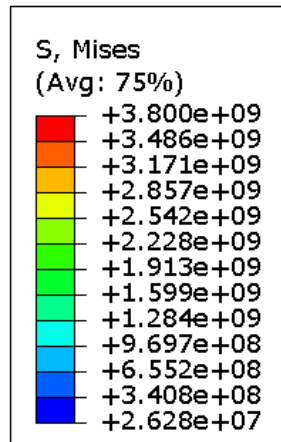
6.2.1.1. Uniaxial Tension in X-axis

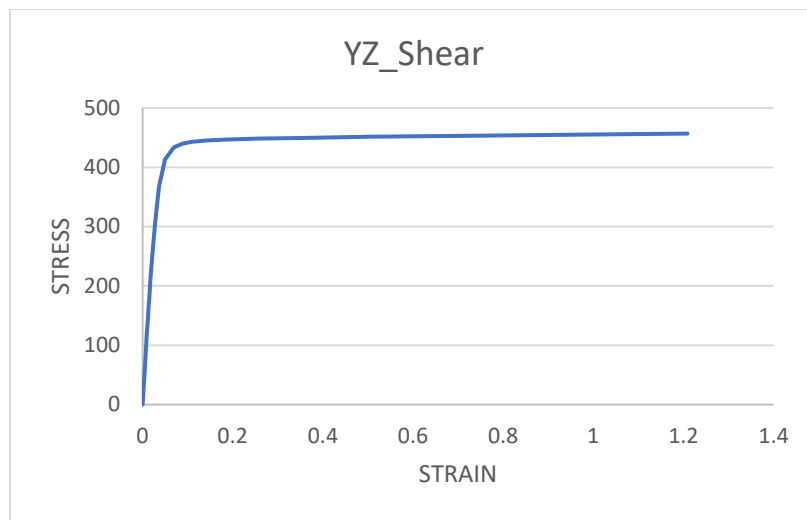


6.2.1.2. Uniaxial Tension in Z-axis

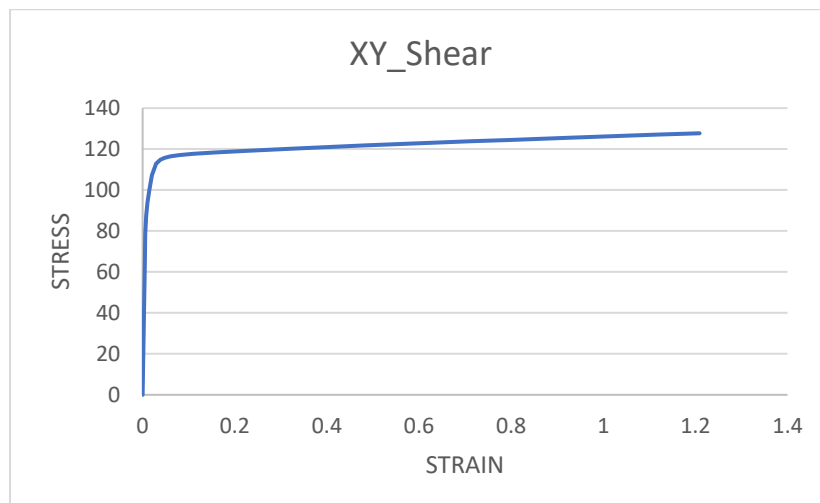
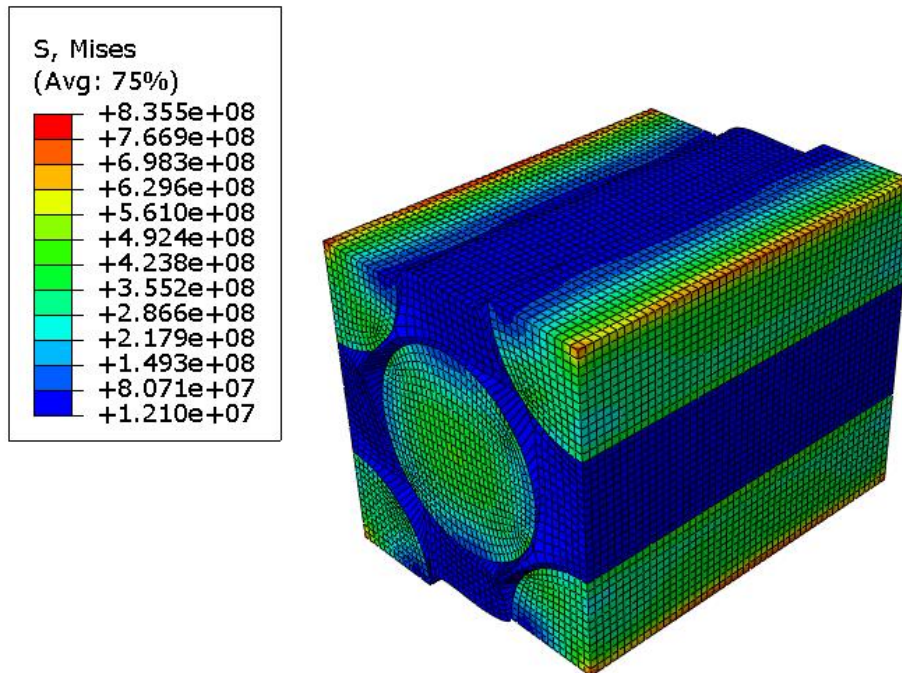


6.2.1.3. Shear in YZ-plane





6.2.1.4. Shear in XY-plane

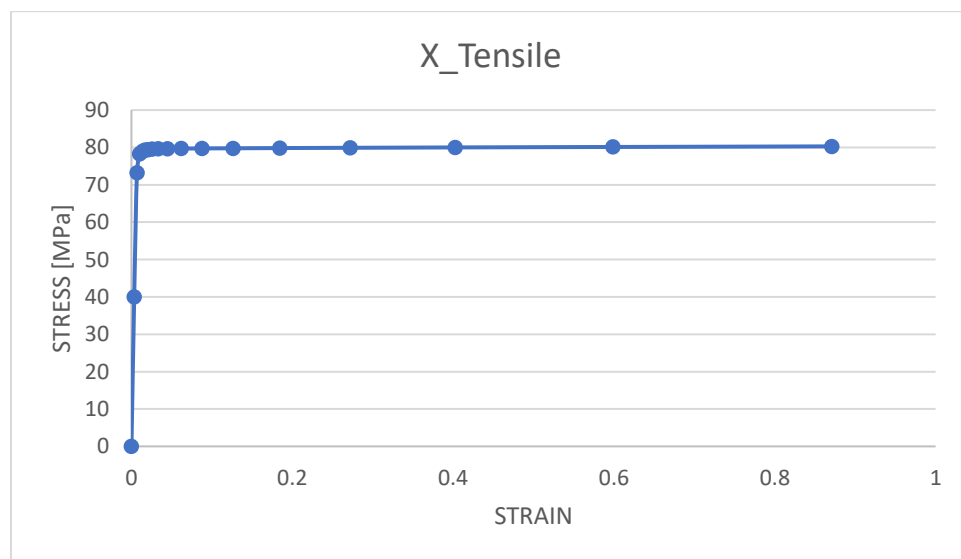
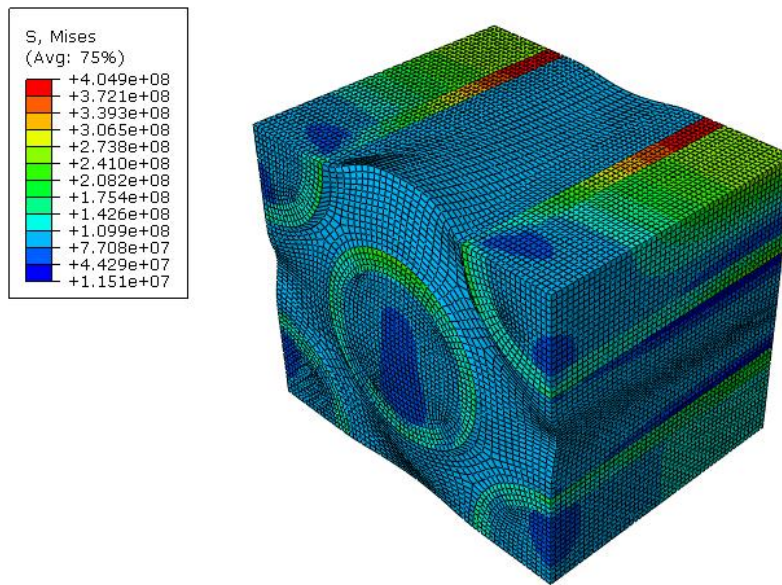


6.2.2. Unit Cell with CNT Reinforced Carbon Fiber

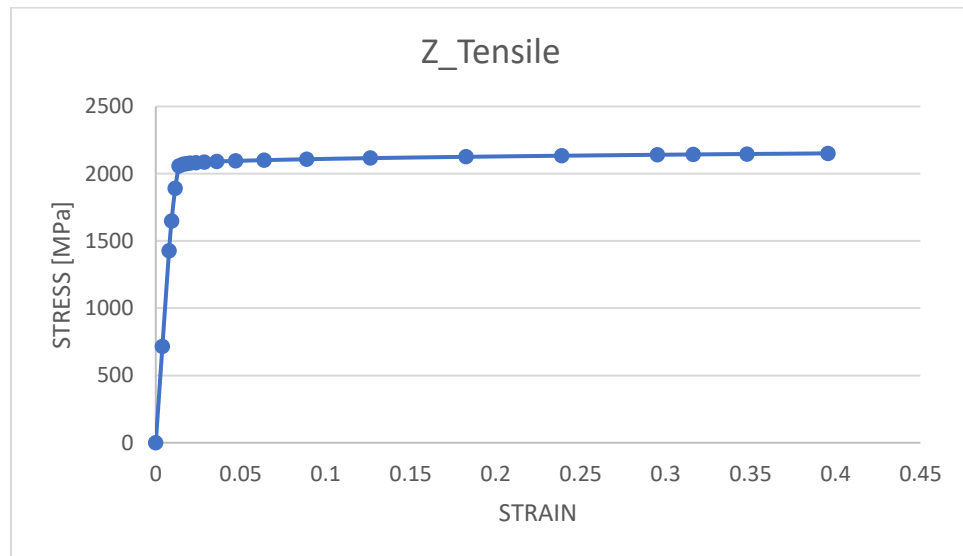
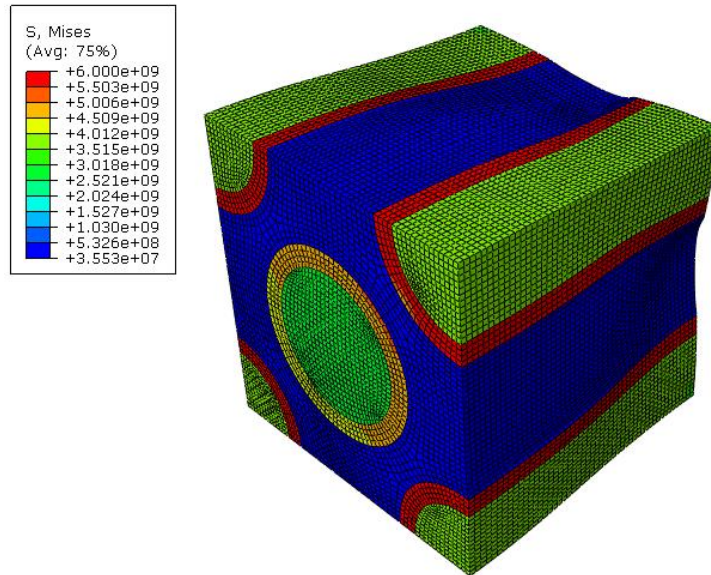
	X_Tensile	Z_Tensile	XY_Shear	YZ_Shear
Elasticity Modulus [GPa]	10.76	168.75	13.77	33.41

Table 12: Calculated Young's and Shear modulus for unit cell with CNT

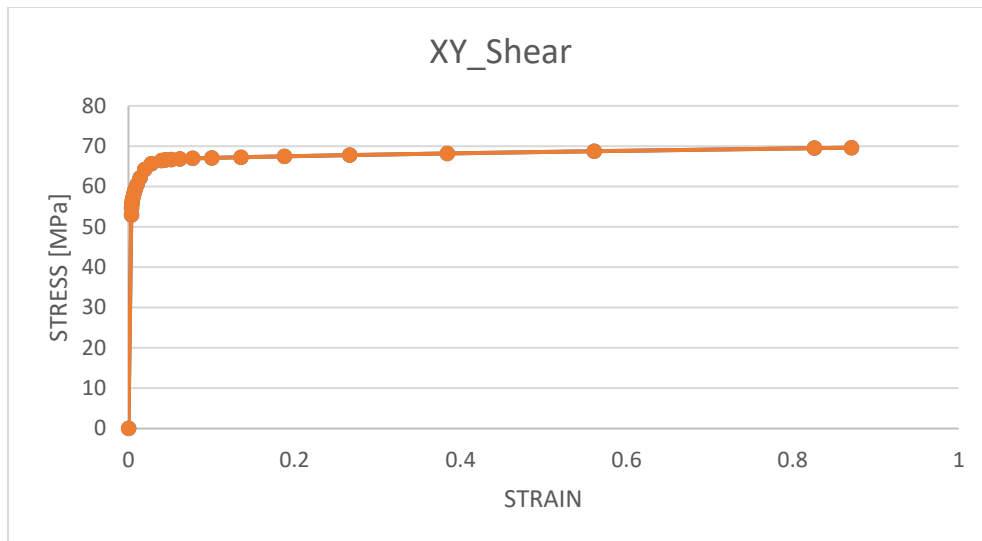
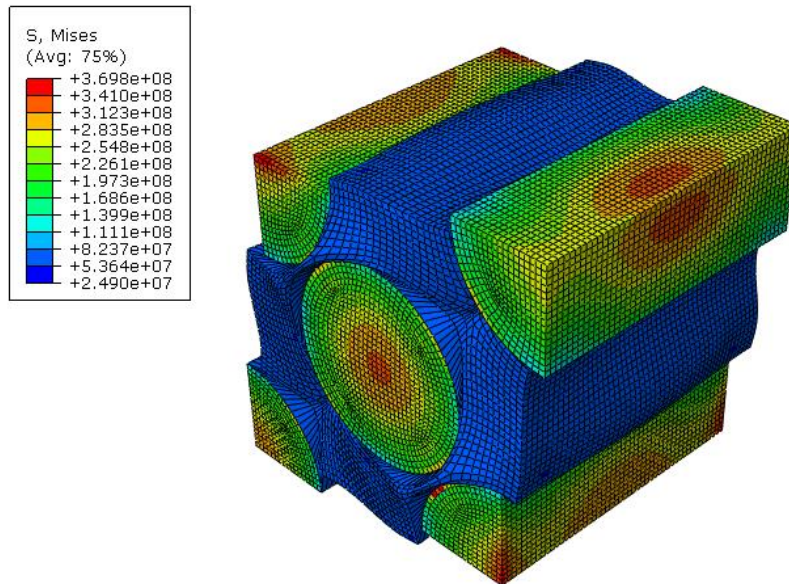
6.2.2.1. Uniaxial Tension in X-axis



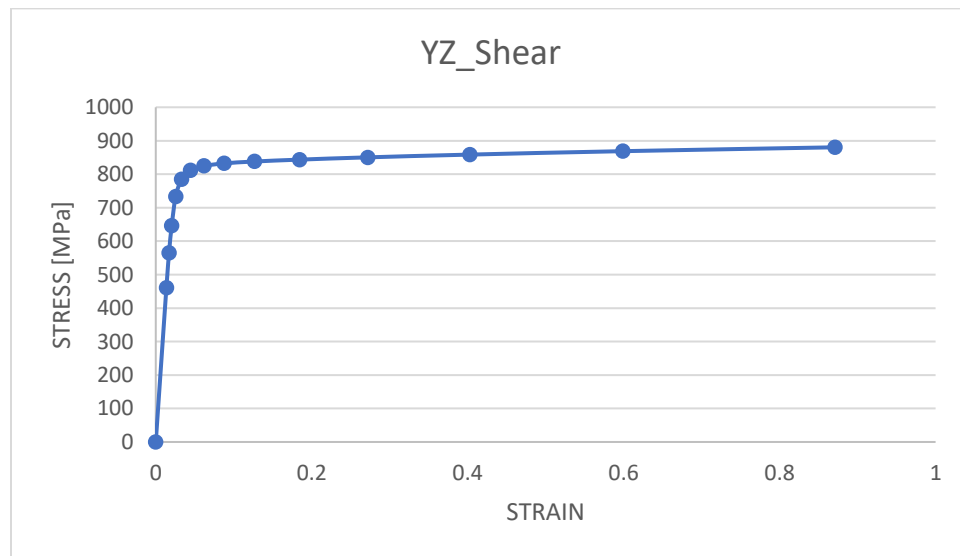
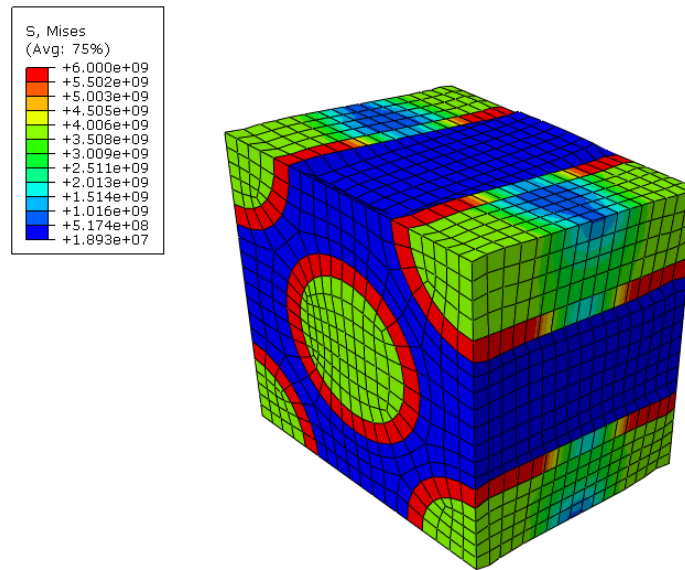
6.2.2.2. Uniaxial Tension in Z-axis



6.2.2.3. Shear in XY-plane



6.2.2.4. Shear in YZ-plane



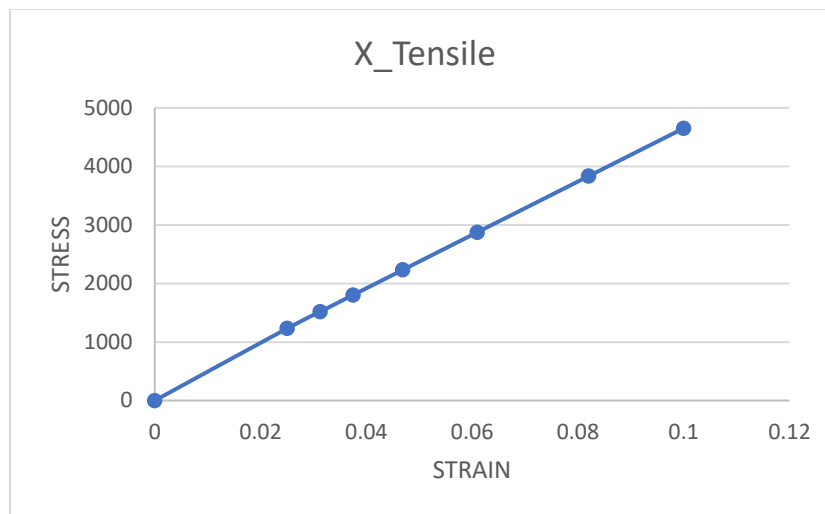
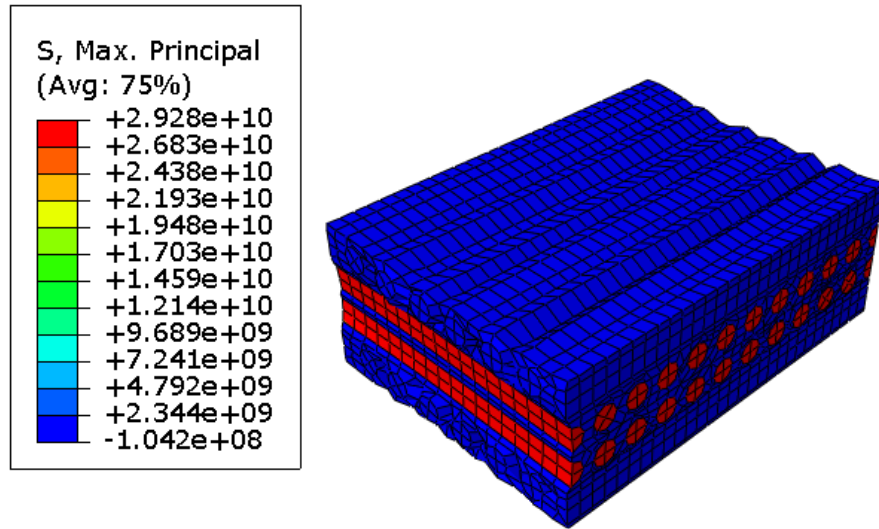
6.3. FEA of Bi-Directional Laminate

6.3.1. Laminate with Conventional Fiber

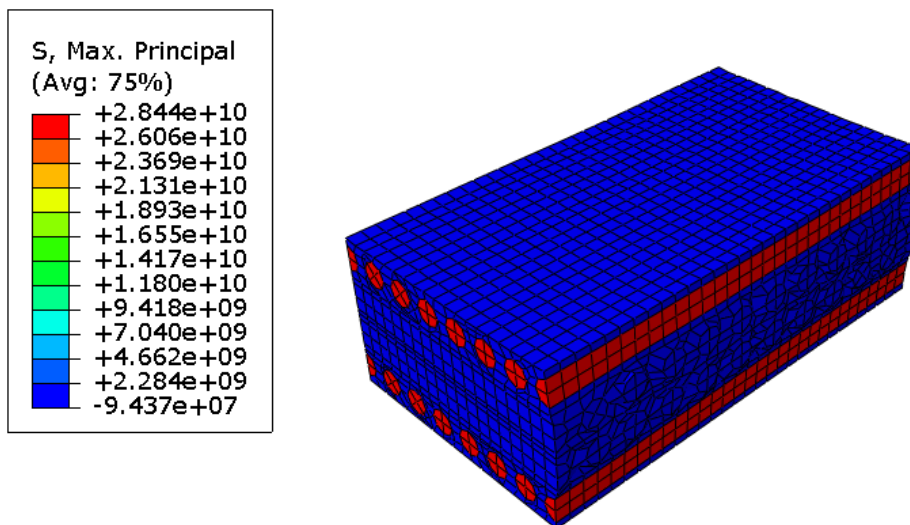
	X_Tensile	Y_Tensile	Z_Tensile	XY_Shear	ZX_Shear
Elasticity Modulus [GPa]	48.74	31.09	52.63	4.57	10.63

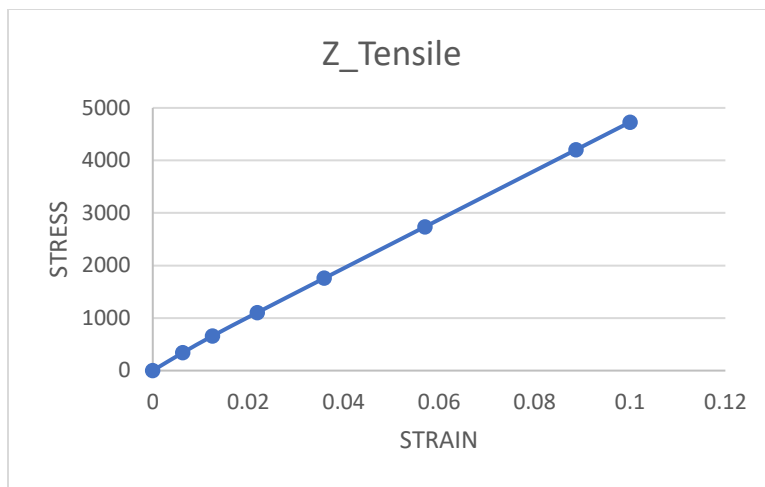
Table 13: Calculated Young's and Shear modulus for conventional composite

6.3.1.1. Uniaxial Tension in X-axis

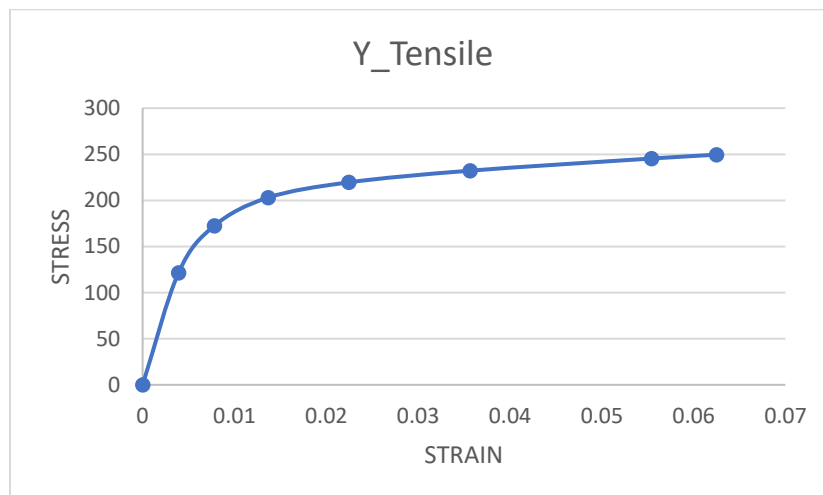
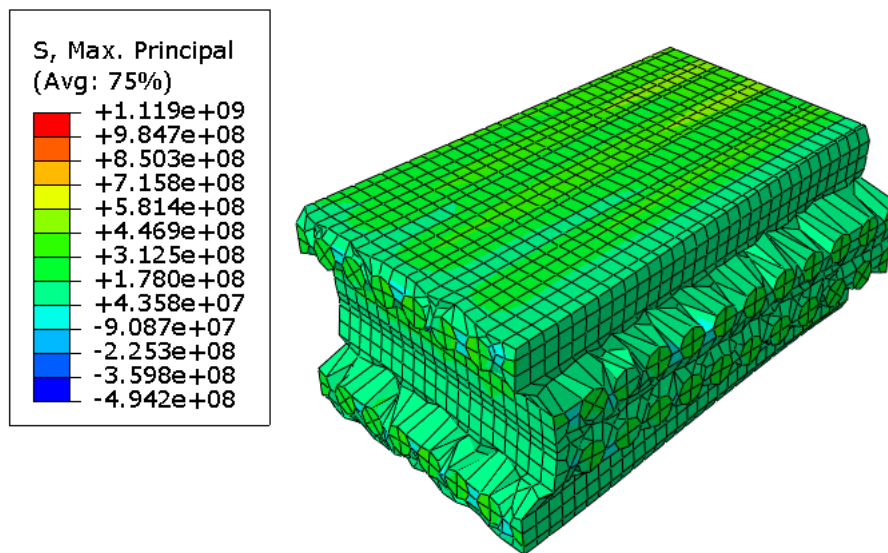


6.3.1.2. Uniaxial Tension in Z-axis

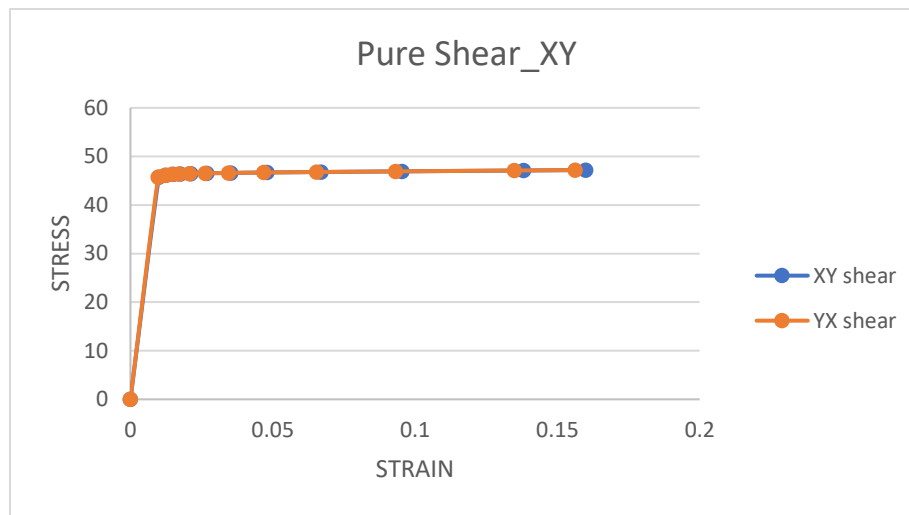
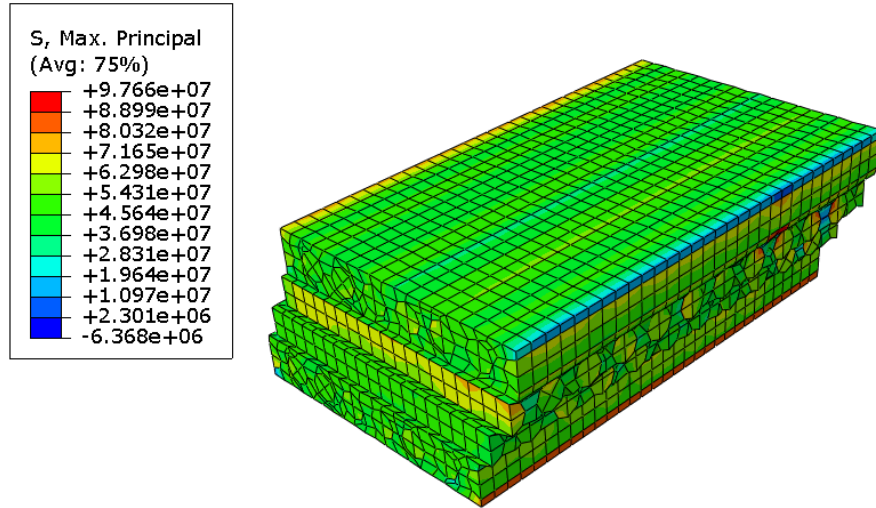




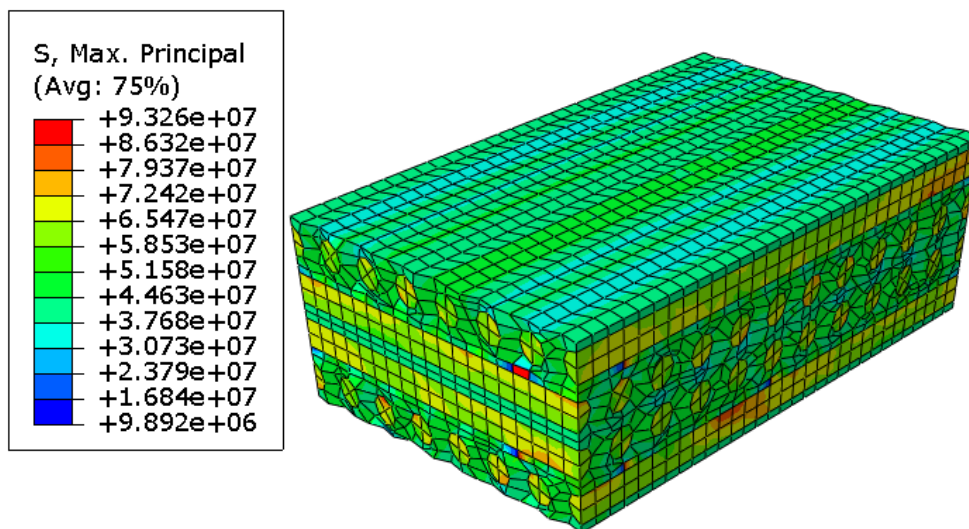
6.3.1.3. Uniaxial Tension in Y-axis

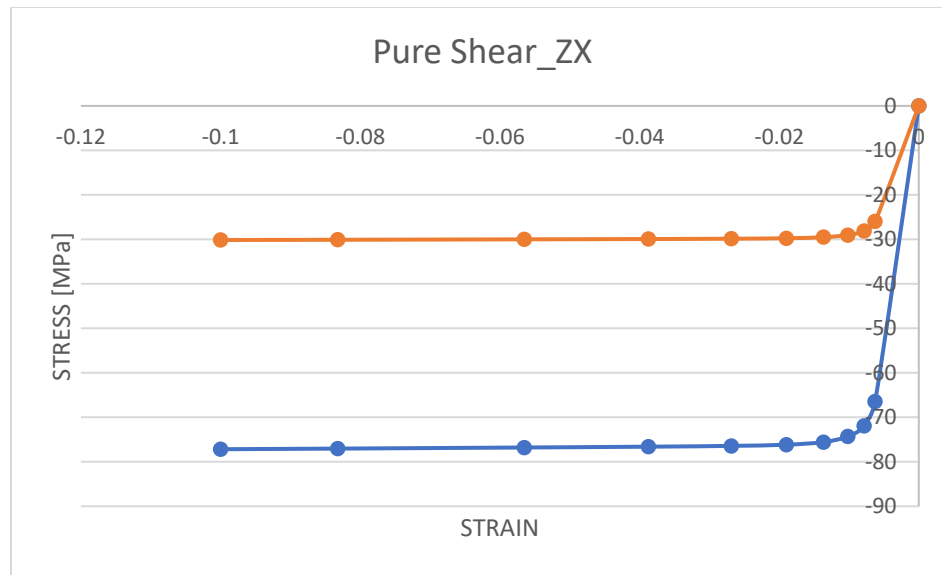


6.3.1.4. Shear in XY-plane



6.3.1.5. Shear in ZX-plane



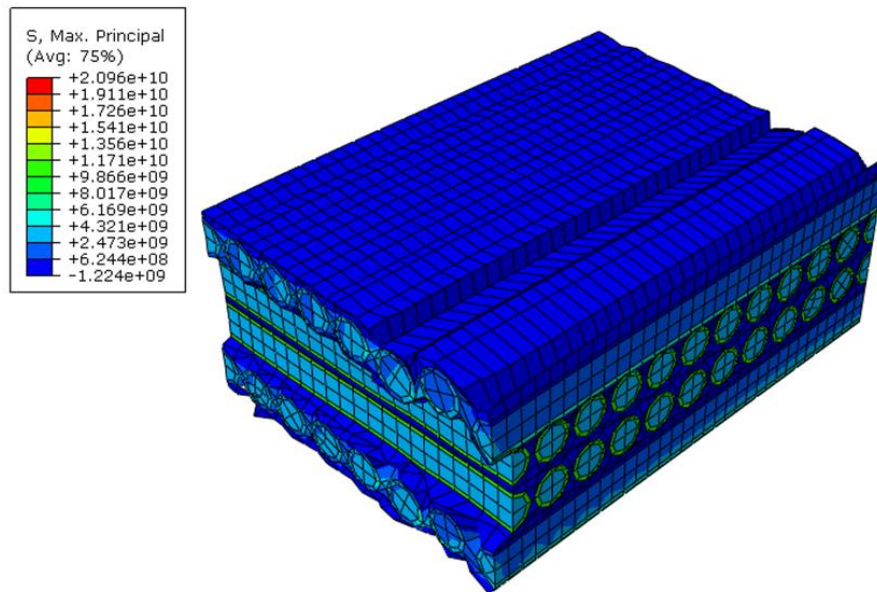


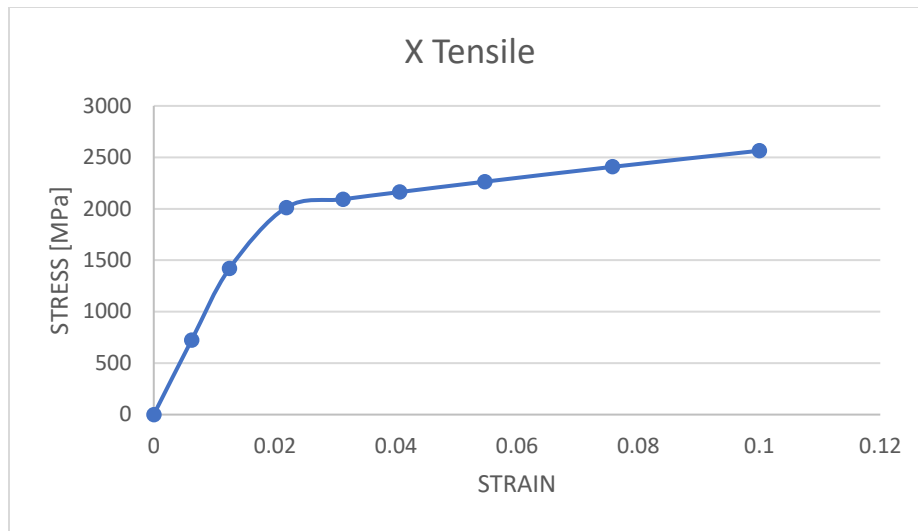
6.3.2. Laminate with CNT Reinforced Fiber

	X_Tensile	Y_Tensile	Z_Tensile	XY_Shear	ZX_Shear
Elasticity Modulus [GPa]	113.54	63.39	31.65	9.03	23.58

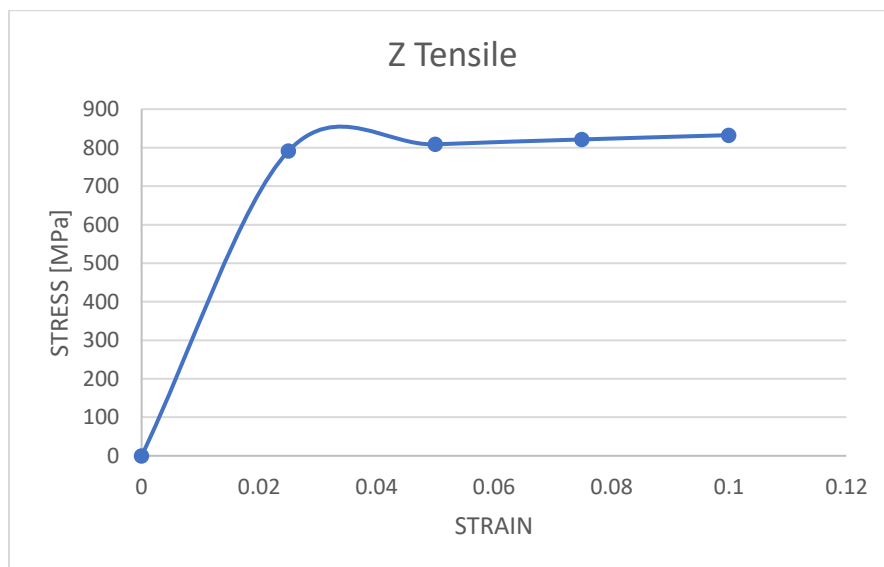
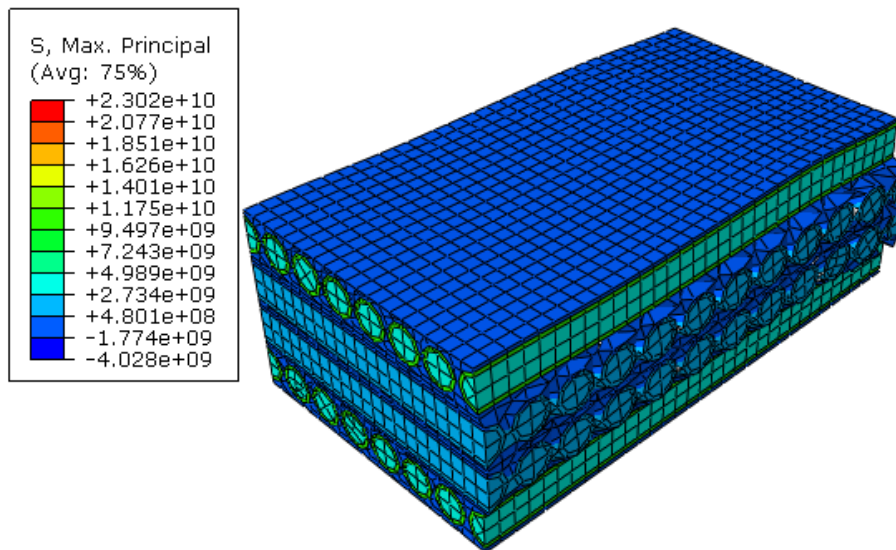
Table 14: Calculated Young's and Shear modulus for laminate with CNT reinforced fiber

6.3.2.1. Uniaxial Tension in X-axis

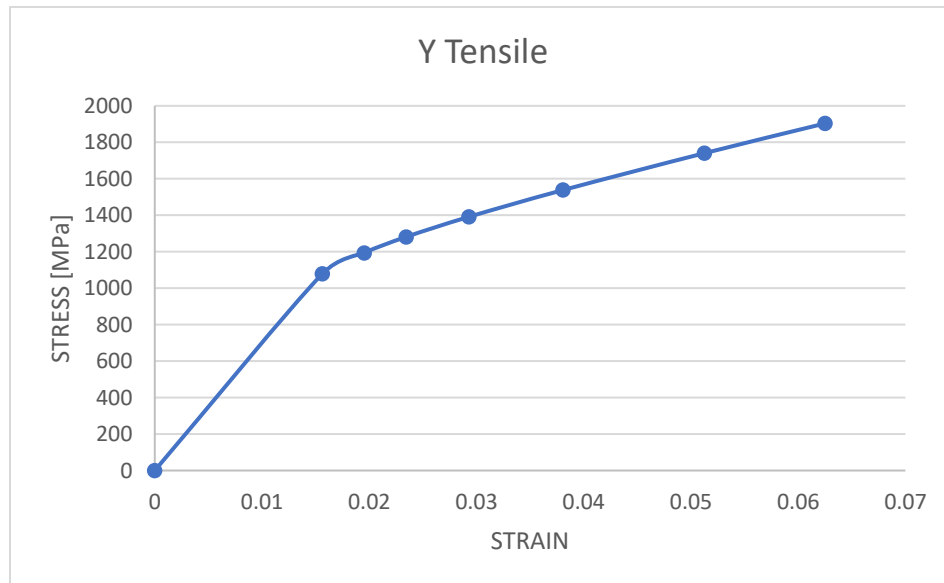
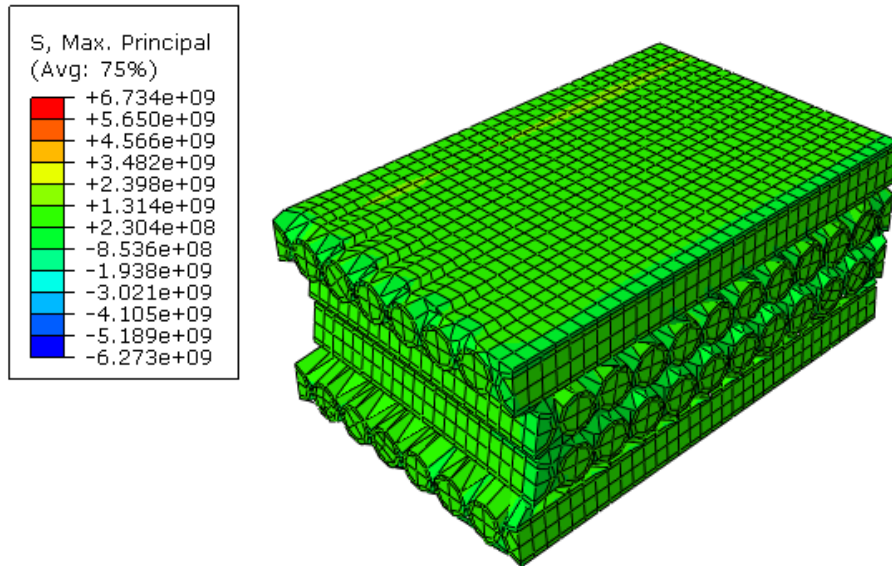




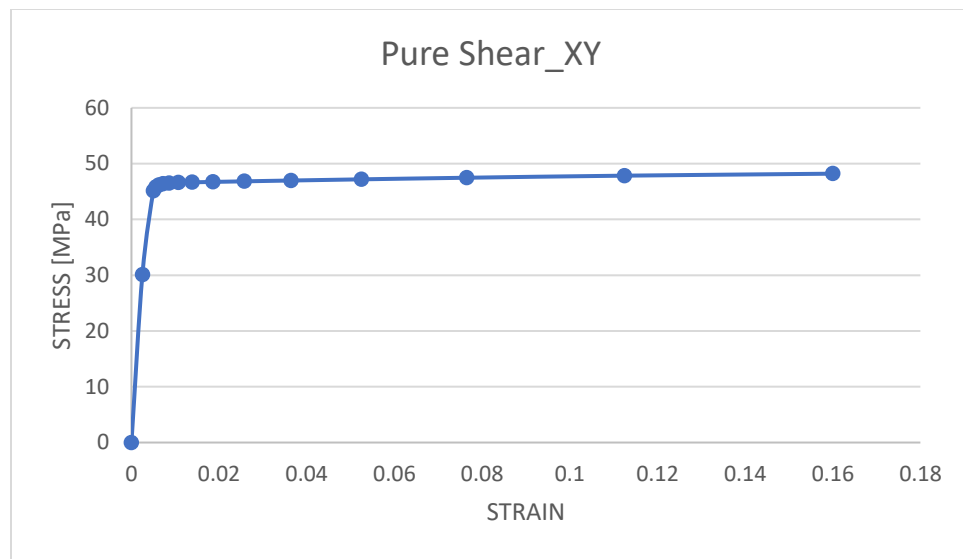
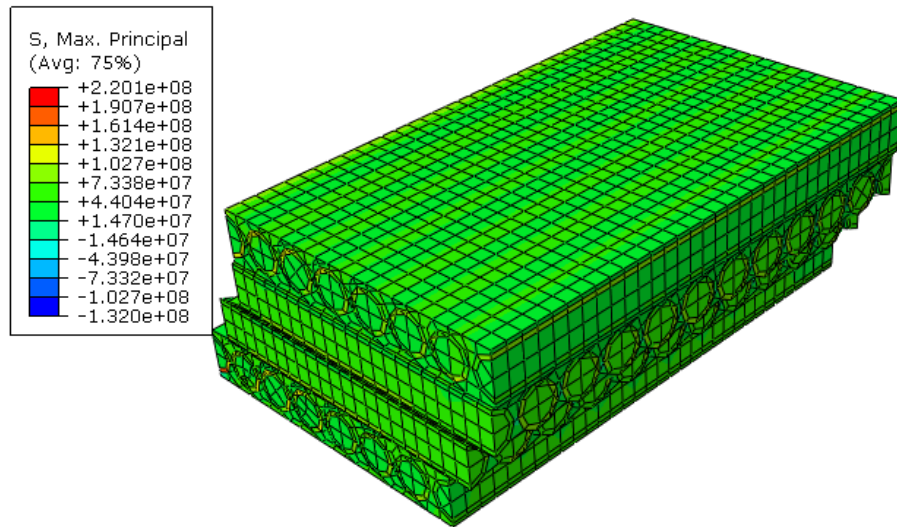
6.3.2.2. Uniaxial Tension in Z-axis



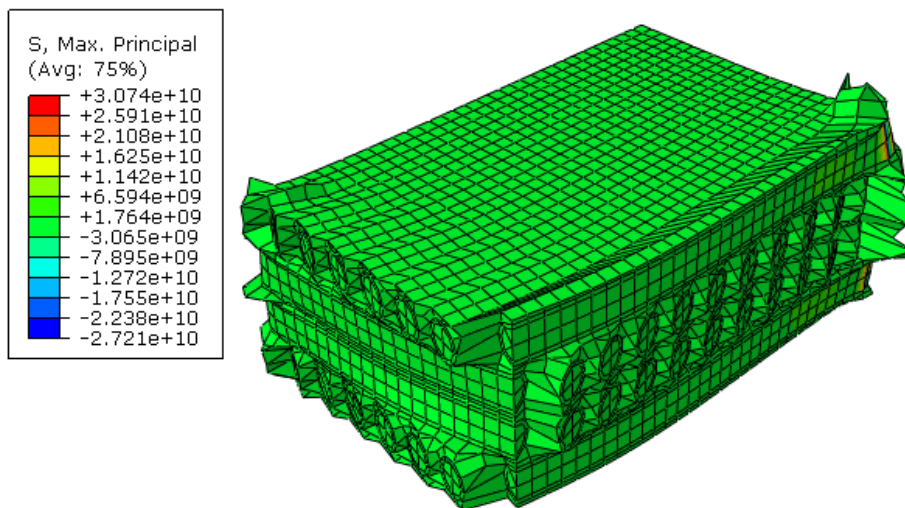
6.3.2.3. Uniaxial Tension in Y-axis

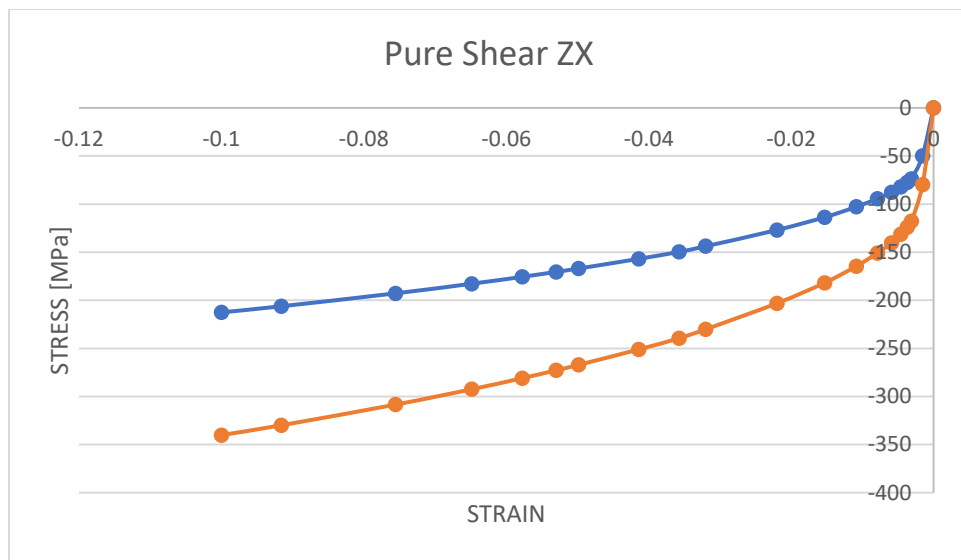


6.3.2.4. Shear in XY-plane



6.3.2.5. Shear in ZX-plane

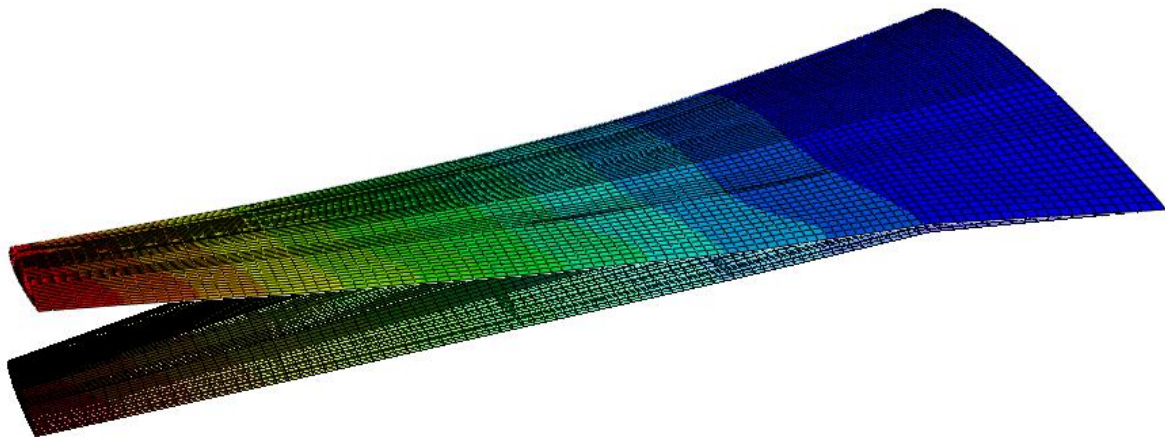




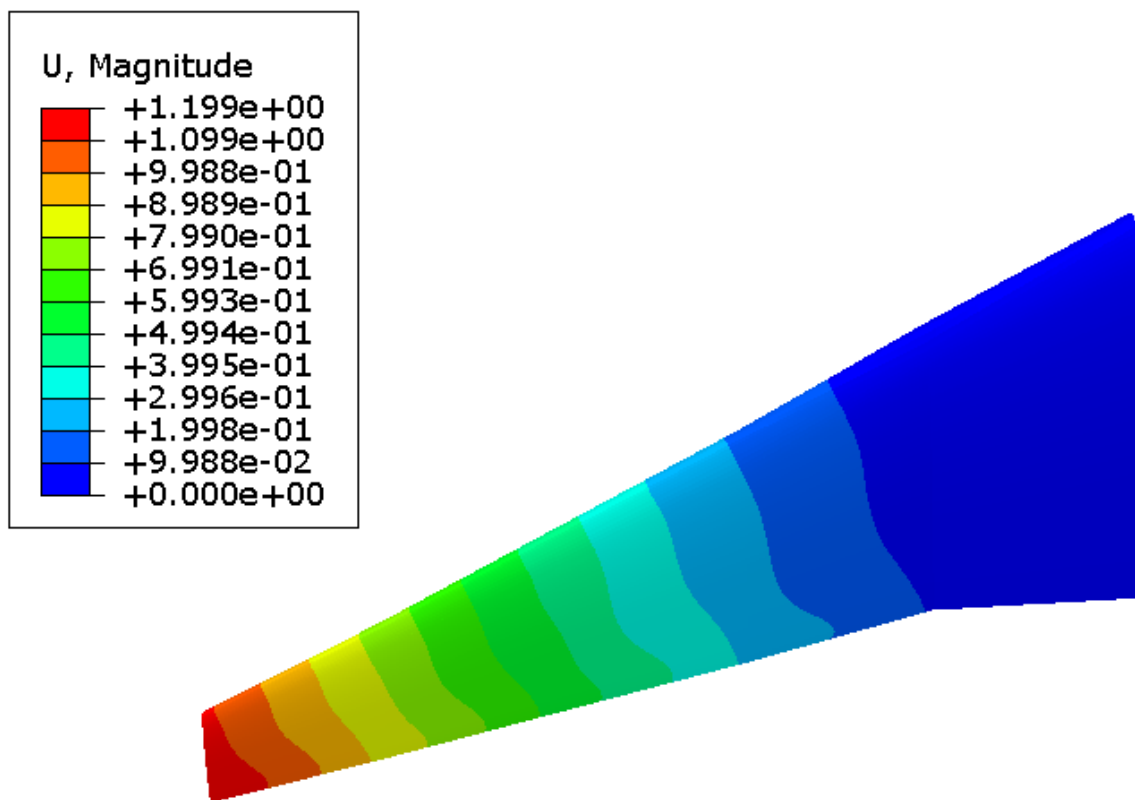
6.4. FEA of Wing Structure

6.4.1. Wing Structure with Conventional Polymer Composite

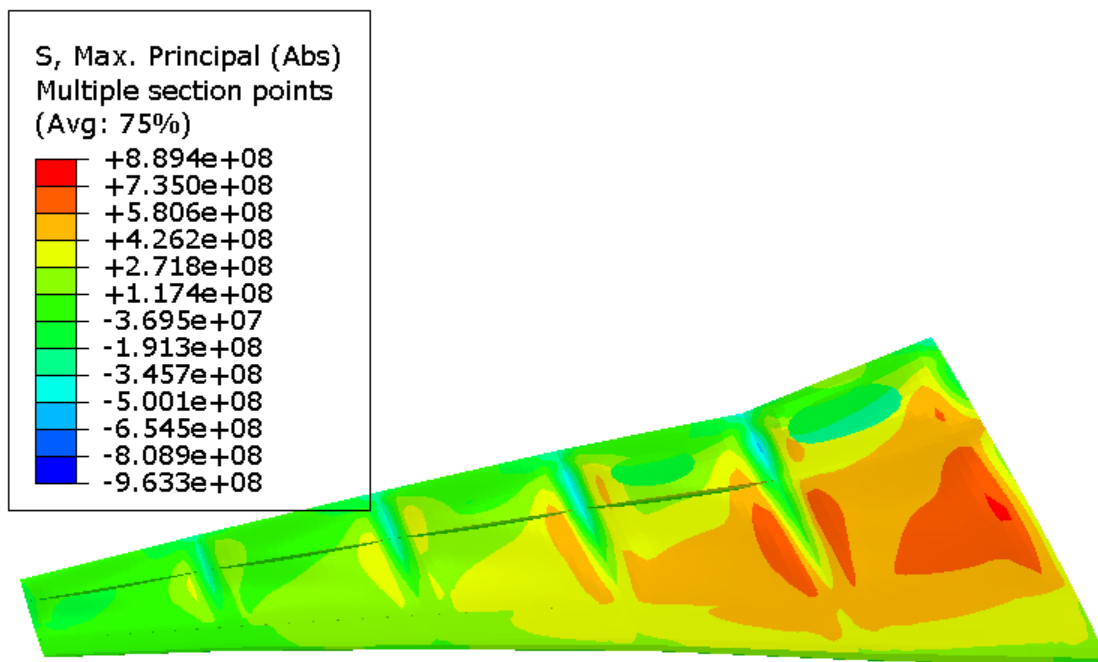
Deformation:



Maximum Displacement:

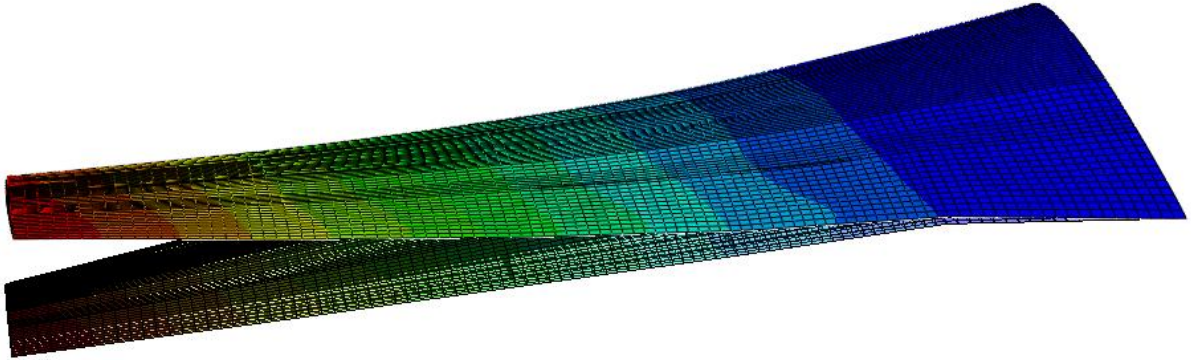


Maximum Principal Stress (Abs):

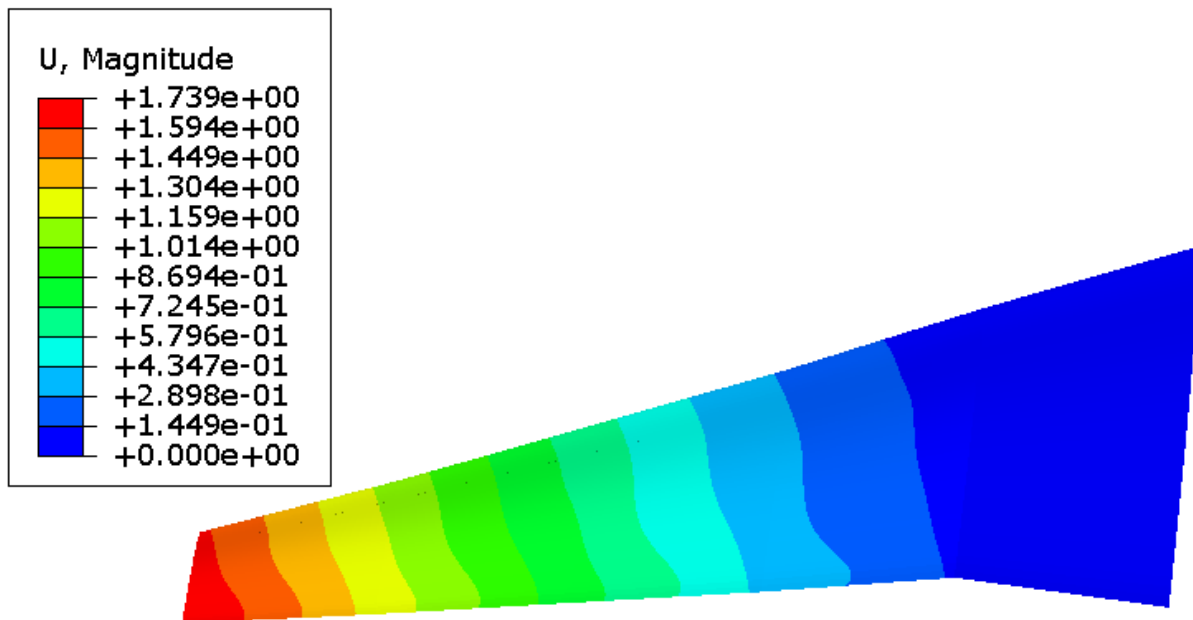


6.4.2. Wing Structure with CNT Reinforced Polymer Composite

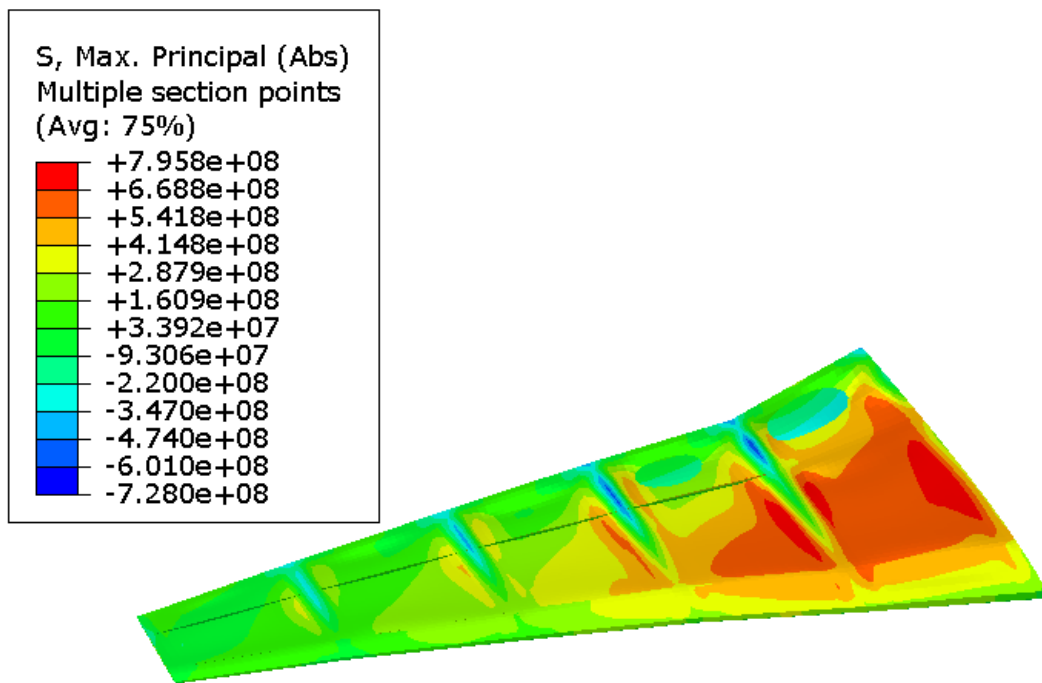
Deformation



Maximum Displacement



Maximum Principal



7. FABRICATION EVALUATION

As discussed earlier in this thesis, polymer composites are being utilized in an increased number of ways in aerospace industry applications. Owing to their high durability and strength, they are on the fast-track to one day replace all metal made aircraft constituent parts. Currently, with the polymer composites available in the market, problems pertaining to electric conduction, impact resistance, temperature sensitive adhesiveness and more have not been curbed. Attempts are being made to alter properties of composites that make them susceptible to such damages. Usage of carbon nanotubes has come out on top as the most effective method.

Even though CNTs have been observed to possess a broad range of properties, they are essentially existent at the nanoscale level. In order to be able to utilize CNTs and their properties in real-world applications, there must exist certain means or ways that would help us realize the properties of these fascinating nanostructures at macroscopic levels. Significant research work is being done in this area, in an effort to establish standard methods that can accomplish this task of utilizing the properties of CNTs at macroscopic levels. Some of the past work done in this area involved the inclusion of CNTs in a variety of polymers by numerous methods, which resulted in the imparting of some of the properties of nanotubes to the nanocomposites thus formed.

Albeit difficult, there exist several methods of CNT dispersion in polymer composites, magnetic stirring, shear mixing, reflux, most commonly used ultra sonication, solution mixing, melt blending, In situ polymerization, latex technology and layer-by-layer deposition [106]. Solution mixing is one of common method for the fabrication of CNT/polymer laminated composites because it can be accepted for small sample size. Blending of solution involves in three methods; dispersion of CNTs in a solvent by mechanical mixing, magnetic agitations or sonication. Carbon nanotubes can also dissolve in polymer resins at room or elevated temperature. Melt blending is commonly used method to fabricate CNT/polymer nano composites. Thermo-plastic polymers, such as polypropylene, polystyrene, poly (ethylene 2, 6-naphthalate) can be processed as matrix materials in this method. In situ polymerization is an efficient method for uniform dispersion of CNTs in a thermos-setting polymers. The advantage of this method is that covalent bonding can be formed between functionalized CNTs and polymer matrix, hence much improved mechanical properties of composite through strong interfacial bonds.

CNT based polymers, for the most part, are fabricated through resin transfer moulding and hand lay-up method. Resin transfer moulding is used for complex shapes at relatively high production rates. The production and manufacturing of CNT reinforced polymer composites is still an emerging field and numerous have been and are being conducted to come up with novel and efficient processing methods.

Four requirements are vital for effectively reinforcing the properties of a CNT-polymer nanocomposite, namely, aspect ratio of the CNTs, method of dispersion used, CNT alignment inside the polymer, and transfer of interfacial stress between the polymer matrix and the CNTs [107]. Among these four points, one of the critical points is the degree of dispersion. A perfectly uniform dispersion helps in the effective transfer of load to the CNTs and reduces the stress concentration spots. In a good dispersion, each CNT has to be coated with the polymer matrix. Poor mixing of CNTs in the polymer composite creates a decline in strength and modulus when the loading is increased beyond a limit. Strong van der Waals interactions present in the CNTs causes agglomeration in making large aggregates. This will lead to the irregular and nonuniform

mixing of CNTs among the polymer medium and adversely affects the different properties of the polymer composite. It is reported that modifying CNTs by chemical methods may destroy the CNT structure and will result in reduced properties.

CNT reinforced polymer composites have excellent properties, but these properties differ depending on the fabrication process used. Shape of CNTs, size, CNT-polymer contact, mixing, and alignment are found to be important in developing CNT incorporated polymer composites. Different methods of fabrication can give rise to optimized properties for the composite material. Characteristics of the filler material used and the synthetic methodology determine the different engineering application of the resulting composites [108].

Solution mixing is one of the simplest and most popular methods for creating CNT-polymer complexes. This process involves preparing CNT-polymer composites in a specific solvent and then evaporating the solvent to produce the composite film. Magnetic stirring, shear mixing, refluxing, ultrasonication, and other agitation techniques are commonly employed to disperse CNTs in solvents. The ideal sonication conditions, such as duration and power, have not been published in the literature, although they can be dictated by the CNT concentration and form. At ambient or increased temperatures, CNTs can be dissolved in polymer resins. This approach is handy for preparing a film-type composite. A different method for synthesizing composites with individual dispersing CNTs was developed [109]. Solution mixing of CNT/(polymethylmethacrylate) PMMA was carried out followed by the formation of a suspension. This suspension obtained was added dropwise into excess of non-solvent so that precipitation of the polymer happens instantly. The precipitated polymer contains entangled CNTs and prevents further agglomeration. The nanocomposites were dried in vacuum to obtain homogeneously distributed CNTs.

Electrospinning yields a fibrous matrix containing micro and nanopores. This approach is extremely valuable since it creates aligned CNTs embedded in a polymer matrix, resulting in an anisotropic scaffold with superior mechanical and electrical characteristics. The potential of incorporating nano-sized carbon materials into fibers makes this process even more attractive for making composite fibers. It is possible to disperse fillers in the electro spun fibers, which allows the manipulation of properties of the composite materials. The alignment of CNTs in the polymer causes increase in the aspect ratio and thereby increases the interfacial bonding and reinforcement of the composite. Different polymers such as polyacrylonitrile, polyvinyl alcohol, polyethylene oxide, PMMA, PU, polycaprolactum, polylactic acid, nylon 6,6, and silk fibroin are used for making CNT-nanocomposite using electrospinning technique.

Layer-by-layer assembly is another way for mixing CNTs with polymers to create CNT-polymer nanocomposites. The preparation process involves layer-by-layer assemblies formed by the adsorption of CNTs and polyelectrolytes on a substrate, which are held together by electrostatic and van der Waals forces. The technique is continued until the required thickness is achieved. The width of the film is determined by the polymer solution's concentration and temperature, as well as its ionic strength, molecular weight, pH, and other parameters. Different materials with opposing charges can be put between these layers.

Epoxy 8552 is a thermoset epoxy resin; it is a type of diglycidyl ether of bisphenol-A (DGEBA) epoxy resin. Thermoset epoxies are characterized by their ability to undergo a curing reaction that creates a cross-linked network of molecules, resulting in a rigid and inflexible polymer matrix. It is imperative to look at composite manufacturing process that might useful for thermoset epoxies.

Stirring, ultrasonication, and hot pressing are common procedures for thermosetting polymers, however the results are unsatisfactory due to weak interfacial bonding between the polymer and the CNT. Chemical functionalization procedures are commonly employed to increase the interfacial connection between CNTs and the polymer matrix. Acid treatments are carried out to create reactive groups on the CNTs. This results in an increase in mechanical characteristics; however, structural faults are discovered, causing property loss. To achieve high-performance composites, several non-covalent functionalization processes must be used.

A novel manufacturing technique for MWCNT/epoxy composite with a loading range of 10-68% has been proposed wherein, hot press infiltration is allowed through a semipermeable membrane so that nanocomposites containing aligned (APNCs), randomly aligned oriented CNTs (RPNCs) with a filler load in a broad range of $\approx 10\text{--}68\text{ wt } \%$ is developed. APNCs are MWCNT arrays with length in the range of mm and developed using chemical vapor deposition method (CVD). RPNCs are a mixture of aligned and millimeter long and randomly oriented commercial micrometer MWCNTs. The process involved the following steps: (i) CVD process for the production of high quality MWCNTs, (ii) pre-densification, (iii) encapsulation of CNTs in the membrane, (iv) CNT wetting using epoxy under vacuum, (v) hot pressing process for the wet samples, and (vi) curing and removal of membranes used for pressing.

Although the scientific community has provided multiple studies giving insights into the properties of CNTs, there still remains some elucidation on the CNT-polymer interaction front. The production of CNTs faces many challenges, MWCNTs are generally entangled in the form of curved agglomerates and SWCNTs are produced as bundles. To achieve optimal results and enhancement of properties, improved dispersion and alignment of CNTs in the polymer matrix is crucial.

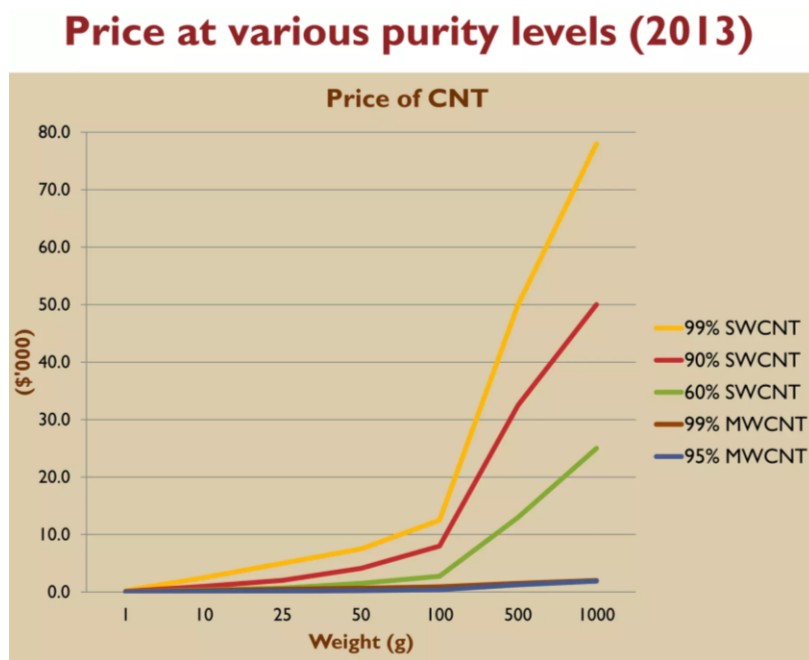


Figure 55: Cost of CNT production based on purity

8. CONCLUSION

In this thesis work, a micromechanics based finite element model has been developed to estimate the effective material properties of a carbon fiber reinforced composite modified with carbon nanotubes as inclusions, and to further investigate whether such a polymer composite material would be suitable to construct the wingbox of a commercial airliner.

This work can be considered as a comparative study between a preexistent IM7/8552 carbon fiber reinforced composite and the same composite but further reinforced with radially grown single walled carbon nanotubes on the carbon fibers forming a strengthened interphase in between the carbon fiber and the epoxy matrix.

Firstly, the elastic constants of the carbon nanotube interphase in a RVE were found by subjecting the RVE to a tensile load in its plane of isotropy, which was the z-axis, while fixing the 3d RVE in the three axes. The displacement and stress strain curve were found as well as the element stiffness matrix of the interphase. The matrix was generated as a 6x6 stiffness matrix for an orthotropic material. The values of the stiffness tensors show that the interphase is transversely isotropic with axis of symmetry parallel to that of the principal fiber. Solving the boundary conditions for the admissible displacement fields using Christensen's composite cylinders model, the effective properties were obtained.

A comparison between a RVE with and a RVE without CNT reinforcements was done where both the types of RVEs were subjected to the same value of displacement load under tensile and shear strains. The results were plotted as stress-strain curves for each load case and the Young's moduli were calculated for comparison purposes. Although the elastic moduli for the case of tensile load in x-axis were almost the same for both the materials, it was observed that the stress concentration shifted from the carbon fiber and matrix to the interphase, a lower plastic strain was also seen in the matrix for the CNT reinforced interphase case. The biggest change and improvement were observed in z tensile and yz shear cases. A shift in stress concentration from fiber to interphase was seen even though the stress values were higher. From the stress-strain curves it was observed that generally the RVE with CNT reinforced interphase had elongated elasticity region as compared to the RVE without reinforcements. They also showcased higher strength.

The next step in homogenization involved mesoscale finite element analysis of a composite laminate with reinforced interphase. A cross-ply layup was chosen for a bi-directional laminate $[0^\circ/90^\circ/90^\circ/0^\circ]$. Such a stacking sequence was chosen to avoid the complexity of a woven composite for ease of simulation. Again, the two laminates were subjected to five load cases, z tensile, y tensile, x tensile, xy shear and xz shear. While a similar trend of increment in Young's modulus was observed in all the cases, it was also observed that the matrix was not being subjected to high plastic strains as was in the conventional composite RVE. The adopted ply stacking sequence lacked in the z direction but had high strength in the x direction. An exponential increase in yz shear was observed owing to the increased stiffness in y and z directions due to CNT reinforcements.

To sum up the RVE study, it was concluded that the CNT interphase provided high stiffness in principal fiber axis. The axial Young's modulus of the laminates was CNT interphase dominated while the transverse Young's modulus was matrix dominated.

Considering the effective properties of the laminate, it is observed that due to the particular stack plot used, the laminate might be weaker in the z-axis. Thus, a different ply orientations plot must be used. Nevertheless, in the x-axis it gives a much higher elastic modulus than the conventional carbon fiber reinforced polymer. The shear moduli of both the laminates was comparable, the interphase next to no additional stiffness.

The essence of this study lies in using the effective properties of the modified composite laminate to make the wingbox and compare the stress and deformation results. Hence, the next step was to model a wingbox. For this purpose, wing of Airbus A320 was modelled. The wingbox was fixed on the root while a lifting pressure was applied on the suction surface, the value of which was corresponding to the lifting force experienced by the wing in cruise flight.

The wing analysis for the two composite polymers yielded similar results. The wing tip deflection for the CNT based composite was more, yet in the allowable range for A320, showcasing that the wing exhibits better bending properties, it also has good bending stiffness. The maximum principal stress, however, was comparable but slightly higher for a conventional CFRP. CFRP experiences slightly higher stress for the same wing loading, which is not desirable. Though the results of the two analyses did not differ much, the high mechanical properties of the interphase and laminate did not translate well to providing exponentially high stiffness to the wing. It can still be concluded that a CNT based composite can provide better rigidity to the wing structure and may be used to produce airframe structures. It can also be seen that the CNT based wing has low ductility which can make it prone to fatigue cracking.

There are still a multitude of limitations and obstacles. It is evident that a different volume fraction incorporation of CNTs is required, also functionalization of CNTs might be necessary to enable the high mechanical values of CNTs to translate to composite laminates.

8.1. Limitations of the Study

Although this work succeeded in finding the effective elastic constants of a CNT modified carbon polymer composite and thus determining the stresses and deformations induced in a wingbox of such a material, there were certain assumptions made and conditions not considered that would otherwise be present in the real world.

Multiple assumptions and considerations were made to accommodate for the low computerization power available. Firstly, the interphase was considered a homogenous medium and CNT agglomeration, which is highly common in production, is not considered. For the laminate, a simple quasi-isotropic laminate was chosen for ease of meshing. When modelling the wing, instead of 27 ribs, only 6 equivalent ribs were constructed so that there are a smaller number of elements.

While analyzing the wing's structural integrity, all the components of the wingbox have been given the same material and CFRP and metal interactions have not been considered. The structural analysis was done only for cruise flight while only considering gravity load and lifting pressure.

In terms of material properties, only mechanical properties have been estimated without including the thermal properties. Structure and material damage and failure have not been analyzed.

8.2. Future Scope

When determining whether a given material can be used to make an airframe component, it is detrimental to do a holistic analysis of not only the material but the airframe structure as well. This comprehensive analysis would require one to evaluate the mechanical, physical and thermal properties.

Apart from the elastic moduli and ultimate tensile strength, it is important to estimate the material's thermal conductivity, specific heat capacity and coefficient of thermal expansion. It is also essential to evaluate the chosen material's fatigue properties. In addition to numerical and simulation models, the material should also undergo extensive experimentation.

Different volume fractions of CNTs should be analyzed to see which provides the best properties. Also, functionalization of CNTs might be imperative for the interface to have better interfacial stress transfer and bonding to the matrix and base fiber.

Apart from simulation and testing, when it comes to production of CNTs and CNT based nanocomposites, many challenges need to be dealt with. The biggest obstacles being homogenous dispersion.

To efficiently compete with existing carbon fiber-based composites, further efforts are needed to make CNT based composites stand out. Producing CNT based composites is complex and energy-intensive. Pure carbon is a very costly raw material and CNT yield is usually very low leading to significant wastage of material.

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